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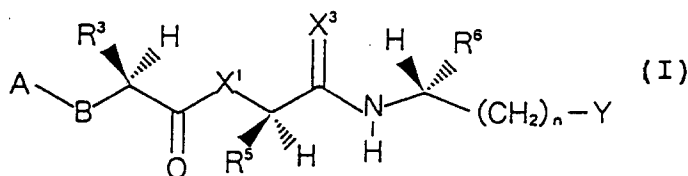
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(54) **Endothelin antagonistic substance.**

(57) A peptide derivative of the formula:



wherein A is a group of the formula $R^{11}O-CO-$ (wherein R^{11} is a lower alkyl group or a phenyl group), or a group of the formula $R^{12}R^{13}N-C(=O)-$ (wherein R^{12} is a lower alkyl group, a cycloalkyl group, a 1-adamantyl group, a phenyl group wherein one or two optional hydrogen atoms on the benzene ring may independently be replaced by a halogen atom, a trifluoromethyl group, a nitro group, an amino group or a formylamino group, a pyridyl group, or a thienyl group, R^{13} is a hydrogen atom, a lower alkyl group or a cycloalkyl group, or R^{12} and R^{13} form, together with the adjacent nitrogen atom, a 5- to 9-membered nitrogen-containing saturated heterocyclic ring

having 4 to 8 carbon atoms, wherein among methylene groups forming the ring, one optional methylene group not adjacent to the above nitrogen atom may be replaced by a thio group, and one to four optional hydrogen atoms on the carbon atoms of the heterocyclic ring may independently be replaced by a lower alkyl group, and further two adjacent carbon atoms in the heterocyclic ring may form a benzo-fused ring); B is an oxygen atom or a group of the formula $-NR^2$ - (wherein R^2 is a hydrogen atom or a lower alkyl group); R^3 is a lower alkyl group, a cycloalkyl group, an aryl group, a heterocyclic group, a cycloalkyl lower alkyl group, an aryl lower alkyl group or a heterocyclic lower alkyl group; X^1 is an oxygen atom or a group of the formula $-NR^4$ - (wherein R^4 is a hydrogen atom or a lower alkyl group); R^5 is a 3-indolylmethyl, 3-benzothienylmethyl, 1-naphthylmethyl or benzyl group wherein one or two optional hydrogen atoms on the ring may be replaced by a hydroxyl group, a halogen atom, a formyl group, a lower alkyl group, a lower alkoxy group, a lower alkylthio group, a lower alkylsulfinyl group, a lower alkylsulfonyl group, a lower alkoxycarbonyl group, a nitro group or a group of the formula $R^{51}-CO-X^2$ {wherein R^{51} is a lower alkyl group, a lower alkoxy group, or an amino group which may be substituted by a lower alkyl group, and X^2 is an oxygen atom or a group of the formula $-NR^{52}$ - (wherein R^{52} is a hydrogen atom or a lower alkyl group)}; X^3 is an oxygen atom or a sulfur atom; R^6 is a hydrogen atom, or a lower alkyl or lower alkenyl group which may have a substituent selected from the group consisting of a hydroxyl group, a lower alkoxy group, a lower alkylthio group and a heterocyclic group; n is 0 or 1; Y is a hydroxymethyl group, a group of the formula CO_2R^{71} (wherein R^{71} is a hydrogen atom or a lower alkyl group), a group of the formula $CONHR^{72}$ (wherein R^{72} is a hydrogen atom or a lower alkyl group which may have a substituent selected from the group consisting of a hydroxyl group, a carboxyl group and a sulfo group), a 1H-tetrazol-5-yl group, a sulfo group and a phosphono group; or a pharmaceutically acceptable salt thereof.

The present invention relates to novel compounds having antagonism against three endothelin isopeptides (endothelin-1, endothelin-2 and endothelin-3), which are physiologically highly active endogenous peptides, processes for their preparation and their use as a drug.

Two endothelin receptor subtypes ET_A and ET_B are known so far. The compounds of the present invention possess high affinity to at least the ET_B receptors, thereby inhibiting vasoconstriction and bronchoconstriction induced by the endothelins. The compounds of the present invention provide a new therapeutic potential, particularly for the treatment of hypertension, pulmonary hypertension, Raynaud's disease, acute renal failure, myocardial infarction, angina pectoris, cerebral infarction, cerebral vasospasm, arteriosclerosis, asthma, gastric ulcer, diabetes, endotoxin shock, endotoxin-induced multiple organ failure or disseminated intravascular coagulation, and/or cyclosporin-induced renal failure or hypertension.

Endothelin is a polypeptide composed of 21 amino acids, and it is produced by vascular endothelial cells of human or pig. Endothelin has a potent vasoconstrictor effect and a sustained and potent pressor action (Nature, 332, 411-415 (1988)).

Three endothelin isopeptides (endothelin-1, endothelin-2 and endothelin-3), which resemble one another in structure, exist in the bodies of animals including human, and these peptides have vasoconstriction and pressor effects (Proc. Natl. Acad. Sci. USA, 86, 2863-2867 (1989)).

As reported, the endothelin levels are clearly elevated in the blood of patients with essential hypertension, acute myocardial infarction, pulmonary hypertension, Raynaud's disease, diabetes or atherosclerosis, or in the washing fluids of the respiratory tract or the blood of patients with asthmaticus as compared with normal levels (Japan. J. Hypertension, 12, 79, (1989), J. Vascular Medicine Biology, 2, 207 (1990), Diabetologia, 33, 306-310 (1990), J. Am. Med. Association, 264, 2868 (1990), and The Lancet, ii, 747-748 (1989) and ii, 1144-1147 (1990)).

Further, an increased sensitivity of the cerebral blood vessel to endothelin in an experimental model of cerebral vasospasm (Japan. Soc. Cereb. Blood Flow & Metabol., 1, 73 (1989)), an improved renal function by the endothelin antibody in an acute renal failure model (J. Clin. Invest., 83, 1762-1767 (1989), and inhibition of gastric ulcer development with an endothelin antibody in a gastric ulcer model (Extract of Japanese Society of Experimental Gastric Ulcer, 50 (1991)) have been reported. Therefore, endothelin is assumed to be one of mediators causing acute renal failure or cerebral vasospasm following subarachnoid hemorrhage.

Further, endothelin is secreted not only by endothelial cells but also by tracheal epithelial cells or from kidney cells (FEBS Letters, 255, 129-132 (1989), and FEBS Letters, 249, 42-46 (1989)).

Endothelin was also found to control the release of physiologically active endogenous substances such as renin, atrial natriuretic peptide, endothelium-derived relaxing factor (EDRF), thromboxane A₂, prostacyclin, noradrenaline, angiotensin II and substance P (Biochem. Biophys. Res. Commun., 157, 1164-1168 (1988); Biochem. Biophys. Res. Commun., 155, 167-172 (1989); Proc. Natl. Acad. Sci. USA, 85, 9797-9800 (1989); J. Cardiovasc. Pharmacol., 13, S89-S92 (1989); Japan. J. Hypertension, 12, 76 (1989) and Neuroscience Letters, 102, 179-184 (1989)). Further, endothelin causes contraction of the smooth muscle of gastrointestinal tract and the uterine smooth muscle (FEBS Letters, 247, 337-340 (1989); Eur. J. Pharmacol., 154, 227-228 (1988); and Biochem. Biophys. Res. Commun., 159, 317-323 (1989)). Further, endothelin was found to promote proliferation of rat vascular smooth muscle cells, suggesting a possible relevance to the arterial hypertrophy (Atherosclerosis, 78, 225-228 (1989)). Furthermore, since the endothelin receptors are present in a high density not only in the peripheral tissues but also in the central nervous system, and the cerebral administration of endothelin induces a behavioral change in animals, endothelin is likely to play an important role for controlling nervous functions (Neuroscience Letters, 97, 276-279 (1989)). Particularly, endothelin is suggested to be one of mediators for pain (Life Sciences, 49, PL61-PL65 (1991)).

On the other hand, endotoxin is one of potential candidates to promote the release of endothelin. Remarkable elevation of the endothelin levels in the blood or in the culture supernatant of endothelial cells was observed when endotoxin was exogenously administered to animals or added to the culture endothelial cells, respectively. These findings suggest that endothelin is one of important mediators for endotoxin-induced diseases (Biochem. Biophys. Res. Commun., 161, 1220-1227 (1989); and Acta Physiol. Scand., 137, 317-318 (1989)).

Further, it was reported that cyclosporin remarkably increased endothelin secretion in the renal cell culture (LLC-PK1 cells) (Eur. J. Pharmacol., 180, 191-192 (1990)). Further, dosing of cyclosporin to rats reduced the glomerular filtration rate and increased the blood pressure in association with a remarkable increase in the circulating endothelin level. This cyclosporin-induced renal failure can be suppressed by the administration of endothelin antibody (Kidney Int., 37, 1487-1491 (1990)). Thus, it is assumed that endothelin is significantly involved in the pathogenesis of the cyclosporin-induced diseases.

Such various effects of endothelin are caused by the binding of endothelin to endothelin receptors widely distributed in many tissues (Am. J. Physiol., 256, R856-R866 (1989)).

It is known that vasoconstriction by the endothelins is caused via at least two subtypes of endothelin receptors (J. Cardiovasc. Pharmacol., 17(Suppl.7), S119-S121 (1991)). One of endothelin receptors is ET_A receptor selective to ET-1 rather than ET-3, and the other is ET_B receptor equally active to ET-1 and ET-3. These receptor proteins are reported to be different from each other (Nature, 348, 730-735 (1990)).

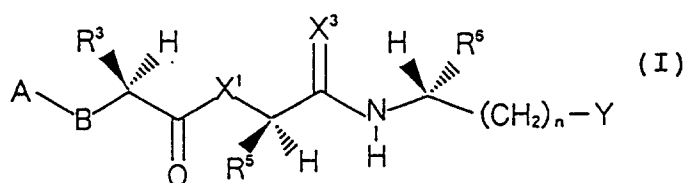
These two subtypes of endothelin receptors are differently distributed in tissues. It is known that the ET_A receptor is present mainly in cardiovascular tissues, whereas the ET_B receptor is widely distributed in various tissues such as brain, kidney, lung, heart and vascular tissues.

Substances which specifically inhibit the binding of endothelin to the endothelin receptors are believed to antagonize various pharmacological activities of endothelin and to be useful as a drug in a wide field. The present inventors already disclosed potent endothelin ET_A receptor antagonists in EP 0460679A2. Since, the action of the endothelins is caused via not only the ET_A receptor but also the ET_B receptor, novel substances with ET_B receptor antagonistic activity are desired to block activities of the endothelins in various diseases.

Endothelin is an endogenous substance which directly or indirectly (by controlling liberation of various endogenous substances) induces sustained contraction of vascular or non-vascular smooth muscles, and its excess production or excess secretion is believed to be one of pathogeneses for hypertension, pulmonary hypertension, Raynaud's disease, bronchial asthma, gastric ulcer, diabetes, arteriosclerosis, acute renal failure, myocardial infarction, angina pectoris, cerebral vasospasm and cerebral infarction. Further, it is suggested that endothelin serves as an important mediator involved in disease such as endotoxin shock, endotoxin-induced multiple organ failure or disseminated intravascular coagulation, and/or cyclosporin-induced renal failure or hypertension. Two endothelin receptors ET_A and ET_B are known so far. An antagonistic agent against the ET_B receptor as well as the ET_A receptor is useful as a drug. Accordingly, it is an object of the present invention to provide a novel therapeutics for the treatment of the above-mentioned various diseases by an invention of a potent ET_B receptor antagonist.

In order to accomplish the above object, the present inventors have synthesized various peptide derivatives and have investigated their endothelin antagonistic activities, and as a result have found that novel peptide derivatives represented by the following formula (I) and their pharmaceutically acceptable salts have strong potent ET_B receptor antagonistic activities. The present invention has been accomplished on the basis of this discovery.

Thus, the present invention provides a peptide derivative of the formula:



wherein A is a group of the formula R¹¹O-CO- (wherein R¹¹ is a lower alkyl group or a phenyl group), or a group of the formula R¹²R¹³N-C(=O)- (wherein R¹² is a lower alkyl group, a cycloalkyl group, a 1-adamantyl group, a phenyl group wherein one or two optional hydrogen atoms on the benzene ring may independently be replaced by a halogen atom, a trifluoromethyl group, a nitro group, an amino group or a formylamino group, a pyridyl group, or a thienyl group, R¹³ is a hydrogen atom, a lower alkyl group or a cycloalkyl group, or R¹² and R¹³ form, together with the adjacent nitrogen atom, a 5- to 9-membered nitrogen-containing saturated heterocyclic ring having 4 to 8 carbon atoms, wherein among methylene groups forming the ring, one optional methylene group not adjacent to the above nitrogen atom may be replaced by a thio group, and one to four optional hydrogen atoms on the carbon atoms of the heterocyclic ring may independently be replaced by a lower alkyl group, and further two adjacent carbon atoms in the heterocyclic ring may form a benzo-fused ring); B is an oxygen atom or a group of the formula -NR²- (wherein R² is a hydrogen atom or a lower alkyl group); R³ is a lower alkyl group, a cycloalkyl group, an aryl group, a heterocyclic group, a cycloalkyl lower alkyl group, an aryl lower alkyl group or a heterocyclic lower alkyl group; X¹ is an oxygen atom or a group of the formula -NR⁴- (wherein R⁴ is a hydrogen atom or a lower alkyl group); R⁵ is a 3-indolylmethyl, 3-benzothienylmethyl, 1-naphthylmethyl or benzyl group wherein one or two optional hydrogen atoms on the ring may be replaced by a hydroxyl group, a halogen atom, a formyl group, a lower alkyl group, a lower alkoxy group, a lower alkylthio group, a lower alkylsulfanyl

group, a lower alkylsulfonyl group, a lower alkoxy carbonyl group, a nitro group or a group of the formula $R^{51}-CO-X^2$ {wherein R^{51} is a lower alkyl group, a lower alkoxy group, or an amino group which may be substituted by a lower alkyl group, and X^2 is an oxygen atom or a group of the formula $-NR^{52}$ - (wherein R^{52} is a hydrogen atom or a lower alkyl group)}; X^3 is an oxygen atom or a sulfur atom; R^6 is a hydrogen atom, or a lower alkyl or lower alkenyl group which may have a substituent selected from the group consisting of a hydroxyl group, a lower alkoxy group, a lower alkylthio group and a heterocyclic group; n is 0 or 1; Y is a hydroxymethyl group, a group of the formula CO_2R^{71} (wherein R^{71} is a hydrogen atom or a lower alkyl group), a group of the formula $CONHR^{72}$ (wherein R^{72} is a hydrogen atom or a lower alkyl group which may have a substituent selected from the group consisting of a hydroxyl group, a carboxyl group and a sulfo group), a 1H-tetrazol-5-yl group, a sulfo group and a phosphono group; or a pharmaceutically acceptable salt thereof.

In the accompanying drawings:

Figure 1 shows the activities of 10 μ M of Compound 4 (○) against endothelin-3-induced contraction of isolated guinea pig bronchial tube as compared with the case in which no drug is present (●).

Figure 2 shows the activities of 10 μ M of Compound 39 (○) against endothelin-3-induced contraction of isolated guinea pig bronchial tube as compared with the case in which no drug is present (●).

Figure 3 shows the activities of 10 μ M of Compound 40 (○) against endothelin-3-induced contraction of isolated guinea pig bronchial tube as compared with the case in which no drug is present (●).

Now, the present invention will be described in further detail with reference to the preferred embodiments.

Now, the meanings of various abbreviations used in this specification will be given.

<u>Abbreviation</u>	<u>Meaning of Abbreviation</u>
D β Abu	D-3-aminobutyric acid
DBal	D-3-(3-benzothienyl)alanine
Ile	L-isoleucine
DIle	D-isoleucine
Leu	L-leucine
γ MeLeu	γ -methyl-L-leucine
DMet	D-methionine
DNal	D-3-(1-naphthyl)alanine
Nle	L-norleucine
DNle	D-norleucine
Nva	L-norvaline
DNva	D-norvaline
DPhe(m-NO ₂)	D-3-(3-nitrophenyl)alanine
DPhe(m-NH ₂)	D-3-(3-aminophenyl)alanine
DLPhe(3-COOEt)	DL-3-(3-ethoxycarbonylphenyl)alanine
DLPhe(3-COOMe)	DL-3-(3-methoxycarbonylphenyl)alanine
DSer	D-serine
DSer(Me)	O-methyl-D-serine
D-mTyr	D-methatyrine
DL-mTyr	DL-methatyrine
DTrp	D-tryptophan

	DTrp(7-OBzl)	D-(7-benzyloxy)tryptophan
	DTrp(7-OH)	D-(7-hydroxy)tryptophan
5	DTrp(COOEt)	D-(N ⁱⁿ -ethoxycarbonyl)tryptophan
	DTrp(COOMe)	D-(N ⁱⁿ -methoxycarbonyl)tryptophan
10	DTrp(OH)	D-(N ⁱⁿ -hydroxy)tryptophan
	DTrp(OMe)	D-(N ⁱⁿ -methoxy)tryptophan
	DLTrp(Me)	DL-(N ⁱⁿ -methyl)tryptophan
15	DTrp(2-SMe)	D-(2-methylthio)tryptophan
	DTrp(2-Br)	D-(2-bromo)tryptophan
	DTrp(2-Cl)	D-(2-chloro)tryptophan
20	DTrp(1-Me, 2-Cl)	D-(2-chloro-1-methyl)tryptophan
	Ac	acetyl
25	Boc	tert-butoxycarbonyl
	Et	ethyl
	Me	methyl
30	ⁿ Pr	n-propyl
	ⁱ Pr	isopropyl
	ⁿ Bu	n-butyl
35	^t Bu	tert-butyl
	Ph	phenyl
40	Bzl	benzyl
	c-Pent	cyclopentyl
	Cpeg	L-cyclopentylglycine
45	Cprg	L-cyclopropylglycine
	CDI	1,1'-carbonyldiimidazole
50	DCC	N,N'-dicyclohexylcarbodiimide
	DMAP	4-(dimethylamino)pyridine

	DMF	N,N-dimethylformamide
	DMSO	dimethylsulfoxide
5	NMP	N-methylpyrrolidone
	NMM	N-methylmorpholine
10	EDCI·HCl	1-ethyl-3-(3-dimethylaminopropyl)- carbodiimide·hydrochloride
15	HOBT·H ₂ O	1-hydroxy-1H-benzotriazole· monohydrate
	HOSu	N-hydroxysuccinimide
20	TBAHS	tetra-n-butylammonium hydrogensulfate
	TEA	triethylamine
25	TFA	trifluoro acetic acid
	THF	tetrahydrofuran
	TsOH	p-toluene sulfonic acid
30	Ts	p-toluenesulfonyl
	Z	benzyloxycarbonyl
35	MOPS	3-morpholinopropane sulfonic acid
	HEPES	2-[4-(2-hydroxyethyl)-1- piperazinyl]-ethane sulfonic acid
40	Tris	tris(hydroxymethyl)aminomethane
	PMSF	phenylmethanesulfonyl=fluoride

Now, the definitions of the various terms mentioned in this specification will be explained.

45 In this specification, the lower alkyl group means a linear or branched alkyl group having 1 to 6 carbon atoms such as a methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, tert-pentyl, 1-methylbutyl, 2-methylbutyl, 1,2-dimethylpropyl, 1-ethylpropyl, hexyl, isohexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 2,2-dimethylbutyl, 1-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-2-methylpropyl or 1-ethyl-1-methylpropyl
50 group.

The cycloalkyl group may, for example, be a cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl or cyclononyl group.

The aryl group may, for example, be a phenyl, 1-naphthyl or 2-naphthyl group.

55 The heterocyclic group means a heterocyclic or fused heterocyclic group containing at least one hetero atom such as an oxygen, nitrogen or sulfur atom, for example, a thienyl, furyl, thiazolyl, imidazolyl, pyridyl, indolyl or benzothienyl group.

The lower alkenyl group means a linear or branched alkenyl group having 1 to 6 carbon atoms such as a vinyl, allyl, 2-propenyl, isopropenyl, 3-butenyl, 2-butenyl, 1-butenyl, 1-methyl-2-propenyl, 1-methyl-1-

propenyl, 1-ethyl-1-ethenyl, 2-methyl-2-propenyl, 2-methyl-1-propenyl, 4-pentenyl or 5-pentenyl group.

The lower alkoxy group means a linear or branched alkoxy group having 1 to 6 carbon atoms such as a methoxy, ethoxy, propoxy or isopropoxy group.

The lower alkanoyloxy group means a linear or branched alkanoyloxy group having 2 to 6 carbon atoms such as an acetoxo or propionyloxy group.

The halogen atom means a fluorine, chlorine, bromine or iodine atom.

The lower alkylthio group means a linear or branched alkylthio group having 1 to 6 carbon atoms such as a methylthio, ethylthio, propylthio or isopropyl group. The lower alkylsulfinyl group means a linear or branched alkylsulfinyl group having 1 to 6 carbon atoms such as a methylsulfinyl, ethylsulfinyl, propylsulfinyl or isopropylsulfinyl group. The lower alkylsulfonyl group means a linear or branched alkylsulfonyl group having 1 to 6 carbon atoms such as a methylsulfonyl, ethylsulfonyl, propylsulfonyl or isopropylsulfonyl group.

Now, this invention will be described in more detail with reference to specific examples for the various symbols used in the formula (I).

In A, R^{11} means a lower alkyl group or a phenyl group. Examples of the lower alkyl group are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, tert-pentyl, 1,1-dimethylbutyl, 1-ethyl-1-methylpropyl and 1,1,2-trimethylpropyl groups.

In A, R^{12} means a lower alkyl group, a cycloalkyl group, a 1-adamantyl group, a phenyl group wherein 1 or 2 optional hydrogen atoms on the benzene ring may be replaced by an optional group selected from the group consisting of a halogen atom, a trifluoromethyl group, a nitro group, an amino group and a formylamino group, a pyridyl group or a thienyl group. Otherwise, R^{12} and R^{13} may form, together with the adjacent nitrogen atom, a heterocyclic group as indicated below. Examples of the lower alkyl group are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, tert-pentyl, 1,1-dimethylbutyl, 1-ethyl-1-methylpropyl and 1,1,2-trimethylpropyl groups. Examples of the cycloalkyl group are cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl and cycloheptyl groups. Examples of the phenyl group wherein 1 or 2 optional hydrogen atoms on the benzene ring may be replaced by an optional group selected from the group consisting of a halogen atom, a trifluoromethyl group, a nitro group, an amino group and a formylamino group, are phenyl, 2-fluorophenyl, 2-chlorophenyl, 2-bromophenyl, 3-fluorophenyl, 3-chlorophenyl, 3-bromophenyl, 4-fluorophenyl, 4-chlorophenyl, 4-bromophenyl, 2,6-difluorophenyl, 2,6-dichlorophenyl, 2,6-dibromophenyl, 2-aminophenyl, 2-formylaminophenyl, 2-trifluoromethylphenyl, 2-nitrophenyl, 3-aminophenyl and 3-formylaminophenyl groups. Examples of the pyridyl group are 2-pyridyl, 3-pyridyl and 4-pyridyl groups. Examples of the thienyl group are 2-thienyl and 3-thienyl groups.

In A, R^{13} means a hydrogen atom, a lower alkyl group or a cycloalkyl group. Otherwise, R^{13} and R^{12} may form, together with the adjacent nitrogen atom, a heterocyclic group as indicated below. Examples of the lower alkyl group are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl and tert-pentyl groups. Examples of the cycloalkyl group are cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl and cycloheptyl groups.

In A, R^{12} and R^{13} may also form, together with the adjacent nitrogen atom, a 5- to 9-membered nitrogen-containing saturated heterocyclic group having 4 to 8 carbon atoms. Among methylene groups forming the heterocycle, one optional methylene group not adjacent to the above nitrogen atom may be replaced by a thio group, and one to four optional hydrogen atoms on the carbon atoms of the heterocycle may independently be replaced by a lower alkyl group, and further two adjacent carbon atoms in the heterocycle may form a fused-benzene ring. Examples of the heterocyclic group are pyrrolidino, piperidino, perhydroazepin-1-yl, perhydroazocin-1-yl, perhydroazonin-1-yl, 1,3-thiazolidin-1-yl, indolin-1-yl, isoindolin-2-yl, 3-pyrolin-1-yl, 1,5-dihydro-2H-pyrrol-1-yl, perhydro-1,4-thiadin-4-yl, 1,2,3,4-tetrahydroquinolin-1-yl, 1,2,3,4-tetrahydroisoquinolin-2-yl, 1,2,3,4-tetrahydropyridin-1-yl, 1,2,3,6-tetrahydropyridin-1-yl, perhydro-1,4-thiazepin-4-yl, 2,3,4,5-tetrahydro-1-benzazepin-1-yl, 2,3,4,5-tetrahydro-2-benzazepin-2-yl, 1,2,4,5-tetrahydro-3-benzazepin-3-yl, 2,3,4,5-tetrahydro-1H-azepin-1-yl, 2,3,6,7-tetrahydro-1H-azepin-1-yl, 1,3,4,7-tetrahydro-2H-azepin-1-yl, perhydro-1,4-thiazocin-4-yl, 1,2,3,4,5,6-hexahydro-1-benzazocin-1-yl, 1,2,3,4,5,6-hexahydro-2-benzazocin-2-yl, 1,2,3,4,5,6-hexahydro-3-benzazocin-3-yl, 1,2,3,4,5,6-hexahydroazocin-1-yl, 1,2,3,4,7,8-hexahydroazocin-1-yl and 1,2,3,4,5,8-hexahydroazocin-1-yl groups, or the above-mentioned heterocyclic groups wherein one to four optional hydrogen atoms on the carbon atoms of the heterocycle may independently be replaced by a lower alkyl group. Examples of the lower alkyl group are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl and tert-butyl groups.

In B, R^2 means a hydrogen atom or a lower alkyl group. Examples of the lower alkyl group are methyl and ethyl groups.

R^3 means a lower alkyl group, a cycloalkyl group, an aryl group, a heterocyclic group, a cycloalkyl lower alkyl group, an aryl lower alkyl group or a heterocyclic lower alkyl group. Examples of the lower alkyl

group are propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl and tert-pentyl groups. Examples of the cycloalkyl group are cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl groups. An example of the aryl group is a phenyl group. Examples of the heterocyclic group are 2-thienyl, 3-thienyl, 2-thiazolyl, 4-thiazolyl, 5-thiazolyl, 2-furyl and 3-furyl groups. Examples of the cycloalkyl lower alkyl group are cyclopropylmethyl, 1-cyclopropylethyl, 2-cyclopropylethyl, 1-cyclopropyl-1-methylethyl, cyclobutylmethyl, 1-cyclobutylethyl, 2-cyclobutylethyl, 1-cyclobutyl-1-methylethyl, cyclopentylmethyl, 1-cyclopentylethyl, 2-cyclopentylethyl, 1-cyclopentyl-1-methylethyl, 1-cyclohexylmethyl, 1-cyclohexylethyl, 1-cyclohexyl-1-methylethyl, 1-cycloheptylmethyl, 1-cycloheptylethyl, 1-cyclooctylmethyl and 1-cyclooctylethyl groups. Examples of the aryl lower alkyl group are benzyl and phenylethyl groups. Examples of the heterocyclic lower alkyl group are 2-thienylmethyl, 3-thienylmethyl, 2-thienylethyl, 2-thiazolylmethyl, 2-furylmethyl, 3-furylmethyl, 2-pyridylmethyl, 3-pyridylmethyl, 4-pyridylmethyl, 2-indolylmethyl, 3-indolylmethyl, 2-benzothienylmethyl and 3-benzothienylmethyl groups.

In X¹, R⁴ means a hydrogen atom or a lower alkyl group. Examples of the lower alkyl group are methyl and ethyl groups.

R⁵ is a 3-indolylmethyl, 3-benzothienylmethyl, 1-naphthylmethyl group or benzyl group wherein one or two optional hydrogen atoms on the ring may be replaced by a hydroxyl group, a formyl group, a lower alkyl group, a lower alkoxy group, a lower alkoxycarbonyl group, a nitro group or a group of the formula R⁵¹-CO-X²- {wherein R⁵¹ is a lower alkyl group, a lower alkoxy group, or an amino group which may be replaced by a lower alkyl group, and X² is an oxygen atom or a group of the formula -NR⁵²- (wherein R⁵² is a hydrogen atom or a lower alkyl group)}. Examples of the substituted or unsubstituted 3-indolylmethyl, 3-benzothienylmethyl, 1-naphthylmethyl or benzyl group, are 3-indolylmethyl, (1-methoxycarbonyl-3-indolyl)methyl, (1-ethoxycarbonyl-3-indolyl)methyl, (1-propoxycarbonyl-3-indolyl)methyl, (1-isopropoxycarbonyl-3-indolyl)methyl, (1-butoxycarbonyl-3-indolyl)methyl, (1-isobutoxycarbonyl-3-indolyl)methyl, (1-sec-butoxycarbonyl-3-indolyl)methyl, (1-tert-butoxycarbonyl-3-indolyl)methyl, (1-formyl-3-indolyl)methyl, (1-hydroxyl-3-indolyl)methyl, (7-hydroxy-3-indolyl)methyl, (1-methyl-3-indolyl)methyl, (1-methoxy-3-indolyl)methyl, (7-methoxy-3-indolyl)methyl, (1-acetoxy-3-indolyl)methyl, (2-acetoxy-3-indolyl)methyl, (7-acetoxy-3-indolyl)methyl, (7-hydroxy-1-methoxycarbonyl-3-indolyl)methyl, (7-methoxy-1-methoxycarbonyl-3-indolyl)methyl, (2-bromo-3-indolyl)methyl, (2-chloro-3-indolyl)methyl, (2-fluoro-3-indolyl)methyl, (2-chloro-1-methyl-3-indolyl)methyl, (2-methylthio-3-indolyl)methyl, (2-methylsulfinyl-3-indolyl)methyl, (2-methylsulfonyl-3-indolyl)methyl, 2-nitrophenylmethyl, 3-nitrophenylmethyl, 4-nitrophenylmethyl, 2-methoxycarbonylphenylmethyl, 3-methoxycarbonylphenylmethyl, 4-methoxycarbonylphenylmethyl, 2-ethoxycarbonylphenylmethyl, 3-ethoxycarbonylphenylmethyl, 4-ethoxycarbonylphenylmethyl, 2-acetoxyphenylmethyl, 3-acetoxyphenylmethyl, 4-acetoxyphenylmethyl, 2-methoxycarbonyloxyphenylmethyl, 3-methoxycarbonyloxyphenylmethyl, 4-methoxycarbonyloxyphenylmethyl, 2-carbamoyloxyphenylmethyl, 3-carbamoyloxyphenylmethyl, 4-carbamoyloxyphenylmethyl, 2-methylcarbamoyloxyphenylmethyl, 3-methylcarbamoyloxyphenylmethyl, 4-methylcarbamoyloxyphenylmethyl, 2-dimethylcarbamoyloxyphenylmethyl, 3-dimethylcarbamoyloxyphenylmethyl, 4-dimethylcarbamoyloxyphenylmethyl, 2-acetamidophenylmethyl, 3-acetamidophenylmethyl, 4-acetamidophenylmethyl, 2-methoxycarbonylaminophenylmethyl, 3-methoxycarbonylaminophenylmethyl, 4-methoxycarbonylaminophenylmethyl, 2-ureidophenylmethyl, 3-ureidophenylmethyl, 4-ureidophenylmethyl, 2-(3-methylureido)phenylmethyl, 3-(3-methylureido)phenylmethyl, 4-(3-methylureido)phenylmethyl, 2-(3,3-dimethylureido)phenylmethyl, 3-(3,3-dimethylureido)phenylmethyl, 4-(3,3-dimethylureido)phenylmethyl, 2-hydroxyphenylmethyl, 3-hydroxyphenylmethyl, 4-hydroxyphenylmethyl, 2-methoxyphenylmethyl, 3-methoxyphenylmethyl and 4-methoxyphenylmethyl groups.

Among them, preferred are 3-indolylmethyl groups wherein the 1- and/or 2-position of the indole ring may be substituted by 1 to 2 substituents selected from the group consisting of a halogen atom, a lower alkyl group, a lower alkoxy group, a lower alkanoyloxy group, a lower alkylthio group, a lower alkylsulfinyl group, a lower alkylsulfonyl group, and a lower alkoxycarbonyl group, such as, 3-indolylmethyl, (1-methoxycarbonyl-3-indolyl)methyl, (1-ethoxycarbonyl-3-indolyl)methyl, (1-propoxycarbonyl-3-indolyl)methyl, (1-isopropoxycarbonyl-3-indolyl)methyl, (1-butoxycarbonyl-3-indolyl)methyl, (1-isobutoxycarbonyl-3-indolyl)methyl, (1-sec-butoxycarbonyl-3-indolyl)methyl, 1-tert-butoxycarbonyl-3-indolyl)methyl, (1-acetoxy-3-indolyl)methyl, (2-acetoxy-3-indolyl)methyl, (1-methoxy-3-indolyl)methyl, (2-bromo-3-indolyl)methyl, (2-chloro-3-indolyl)methyl, (2-fluoro-3-indolyl)methyl, (2-chloro-1-methyl-3-indolyl)methyl, (2-methylthio-3-indolyl)methyl, (2-methylsulfinyl-3-indolyl)methyl and (2-methylsulfonyl-3-indolyl)methyl groups.

R⁶ means a hydrogen atom, or a lower alkyl or lower alkenyl group which may have 1 to 3 substituents selected from the group consisting of hydroxyl, lower alkoxy, lower alkylthio and heterocyclic groups. Examples of the lower alkyl or lower alkenyl group which may be substituted by 1 to 3 substituents selected from the group consisting of hydroxyl, lower alkoxy, lower alkylthio and heterocyclic groups, are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, tert-pentyl, hexyl,

hydroxymethyl, 1-hydroxyethyl, methoxymethyl, 1-methoxyethyl, methylthiomethyl, ethylthiomethyl, propylthiomethyl, butylthiomethyl, 2-methylthioethyl, 2-ethylthioethyl, 3-indolylmethyl, 4-imidazolylmethyl, vinyl, allyl, 2-propenyl, isopropenyl, 3-butenyl, 2-butenyl, 1-butenyl, 1-methyl-2-propenyl, 1-methyl-1-propenyl, 1-ethyl-1-ethenyl, 2-methyl-2-propenyl, 2-methyl-1-propenyl, 4-pentenyl, 5-pentenyl, 1-hydroxy-2-propenyl, 1-methoxy-2-propenyl, 1-methylthio-2-propenyl, 2-hydroxy-3-butenyl and 2-ethylthio-3-butenyl groups. Among them, preferred are a lower alkyl or lower alkenyl group which may have a substituent selected from the group consisting of lower alkoxy and lower alkylthio groups, such as, methyl, ethyl, propyl, isopropyl, butyl, isobutyl, neopentyl, tert-pentyl, hexyl, methoxymethyl, 1-methoxyethyl, methylthiomethyl, ethylthiomethyl, propylthiomethyl, butylthiomethyl, 2-methylthioethyl, 2-ethylthioethyl, vinyl, allyl, 2-propenyl, isopropenyl, 3-butenyl, 2-butenyl, 1-butenyl, 1-methyl-2-propenyl, 1-methyl-1-propenyl, 1-ethyl-1-ethenyl, 2-methyl-2-propenyl, 2-methyl-1-propenyl, 4-pentenyl, 1-hydroxy-2-propenyl, 1-methoxy-2-propenyl, 1-methylthio-2-propenyl, 2-hydroxy-3-butenyl and 2-ethylthio-3-butenyl groups

R⁷¹ means a hydrogen atom or a lower alkyl group. Examples of the lower alkyl group are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl and hexyl groups

R⁷² means a hydrogen atom or a lower alkyl group which may be substituted by a substituent selected from the group consisting of hydroxyl, carboxyl, and sulfo groups. Examples of the lower alkyl group which may be substituted by a substituent selected from the group consisting of hydroxyl, carboxyl, and sulfo groups, are 2-hydroxyethyl, carboxymethyl, 1-carboxyethyl, 2-carboxyethyl, sulfomethyl and 2-sulfoethyl groups.

Now, the process for producing the novel peptide derivatives of the present invention will be described.

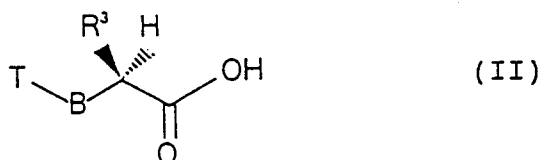
A peptide derivative of the present invention can be prepared by a method wherein amino acids composing the target peptide derivative are condensed one by one, and then, if necessary, removing a C-terminal protective group (Method 1). Peptide derivatives prepared by Method 1 can be converted into other peptide derivatives of the present invention, as the case requires, by a combination of the following reactions: (1) alkoxycarbonylation, aryloxycarbonylation or carbamoylation of the N-terminal α -amino group after removal of an N-terminal amino-protecting group (Methods 2 and 3), (2) alkoxycarbonylation (Method 4) or hydroxylation or alkoxylation (Method 5) at the 1-position of the indole ring of tryptophan, and (3) halogenation (Method 6), or lower alkylthiolation, lower alkylsulfinylation or lower alkylsulfonation (Method 7) at the 2-position of the indole ring of tryptophan. Furthermore, these peptide derivatives can, if necessary, be converted to pharmaceutically acceptable salts.

Each method will be detailed as follows.

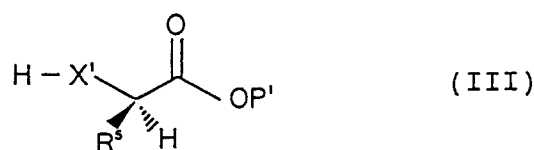
[Method 1]

Method 1 is a conventional synthetic method for peptides, that is, a method wherein amino acids or amino acid derivatives are condensed one by one to prepare a desired peptide derivative. Furthermore, after condensation, a C-terminal protective group can be removed by alkaline hydrolysis, catalytic hydrogenation or acid decomposition. Condensation can be conducted according to known methods such as a DCC method, an azide method, an active ester method and a mixed acid anhydride method (described, for example, by M. Bodansky and M. A. Ondetti in *Peptide Synthesis*, Interscience, New York, 1966; by F. M. Finn and K. Hofmann in *The Proteins*, Vol. 2, ed. by H. Nenrath and R. L. Hill, Academic Press Inc., New York, 1976; by Nobuo Izumiya et al. in *Peptide Synthesis*, Maruzen, 1975).

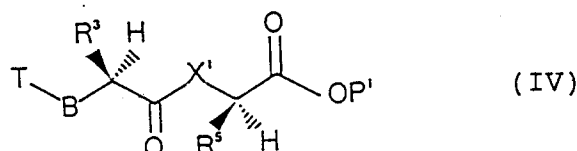
For example, in the case wherein condensation is carried out by a DCC method, an α -amino acid derivative or an α -hydroxycarboxylic acid derivative of the formula



wherein T is A or an α -amino-protecting group, and A, B and R³ are as defined before, is treated with a condensing reagent such as DCC(or EDCI·HCl)-HOBT·H₂O in a suitable solvent such as DMSO, NMP, DMF, THF, 1,4-dioxane, acetonitrile, dichloromethane or chloroform at around -40 °C to room temperature, then condensed with an α -amino acid derivative or an α -hydroxycarboxylic acid derivative having an α -carboxyl-protecting group, of the formula:



wherein P^1 is an α -carboxyl-protecting group, and X^1 and R^5 are as defined before, to afford a dipeptide derivative of the formula:



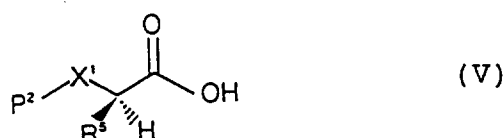
wherein T , B , R^3 , R^5 , X^1 and P^1 are as defined before. An N-terminal α -amino-protecting group is usually selected from the groups well-known to those skilled in the art, for example, from urethane type protective groups such as a Z group, a Boc group, a p-methoxybenzyloxycarbonyl group and a p-nitrobenzyloxycarbonyl group, while the C-terminal α -carboxyl group is usually protected as, for example, a methyl ester, an ethyl ester, a benzyl ester or a tert-butyl ester. Each protective group should be selected so that it can be selectively deprotected after condensation. For example, in the case that a Boc group is selected as an N-terminal protective group, it is preferable to protect the C-terminus as a methyl group, an ethyl group or a benzyl group. A Boc group will be readily removed by use of a mild acid such as TFA, while the carboxyl-protecting groups described above will be usually intact under these conditions. On the other hand, a methyl, ethyl or benzyl ester will be easily deprotected by alkaline hydrolysis and a benzyl ester will be also deprotected by catalytic hydrogenation, while a Boc group will be intact under these conditions.

In the case that T is an α -amino-protecting group, the group T will be formally converted to A by removal of T from the dipeptide derivative (IV) followed by N-alkoxycarbonylation, N-aryloxycarbonylation or N-carbamoylation which will be carried out under the reaction conditions described later in Method 2.

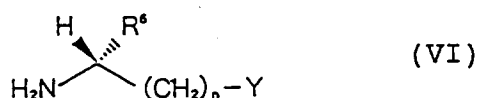
A C-terminal protective group of the dipeptide derivative (IV) prepared in the above-mentioned manner is now removed, and the resulting deprotected dipeptide is treated with a condensation reagent (for example, $\text{EDCI} \cdot \text{HCl} \cdot \text{HOBT} \cdot \text{H}_2\text{O}$) in the same manner described above and then with an amino acid whose C-terminal carboxyl group is protected, to afford a desired peptide derivative.

In the case that each of B and X^1 is $-\text{NR}^2-$ or $-\text{NR}^4-$, and P^1 is a lower alkyl group or a benzyl group, the dipeptide derivative (IV) may be treated with an excess amount of hydrazine in a solvent such as methanol or DMF at room temperature to afford the corresponding hydrazide, which can be converted to a desired peptide derivative by an azide method. Namely, the hydrazide is first converted to the corresponding azide on treatment with a reagent such as a lower alkyl ester of nitrous acid (for example, tert-butyl nitrite or isoamyl nitrite) or an alkaline metal salt of nitrous acid (for example, sodium nitrite or potassium nitrite) in the presence of a strong acid such as hydrochloric acid or sulfuric acid (this reaction can be performed in a solvent such as water, and/or DMF, THF or 1,4-dioxane at around -60°C to -15°C). Then, the azide is mixed with a tertiary amine such as TEA, and a C-terminal ester derivative of an amino acid at -70°C to -60°C , and then allowed to react at -20°C to room temperature to afford a desired peptide derivative. A tetrabutylammonium-, triethylammonium-, sodium- or potassium-salt of an amino acid can also be used instead of the C-terminal ester derivative.

In the process so far described, a C-terminal amino acid or a C-terminal dipeptide is lastly condensed to give a target peptide derivative. The alternative process wherein an N-terminal amino acid is lastly condensed to give a target product is also available. Namely, a compound of the formula:



wherein P² is a hydrogen atom or an α -amino-protecting group, and R⁵ and X¹ are as defined before, is condensed with a compound of the formula:



wherein n, R⁶ and Y are as defined before, by a DCC method or an azide method to afford an N-terminal protected peptide derivative or α -hydroxycarboxylic acid derivative. A suitable α -amino-protecting group can be selected from the urethane type protective groups described before, and a C-terminal carboxyl group can be protected as an ester. In the case that a C-terminal carboxyl group is protected as, for example, a tert-butyl ester, a Z group or a Boc group is preferable for a N-terminal amino-protecting group. A Z group will be readily removed by catalytic hydrogenation and a Boc group will be readily removed by treatment with a mild acid such as formic acid or TFA under ice-cooling, while under these conditions these C-terminal carboxylprotecting groups will be intact.

An α -hydroxycarboxylic acid derivative or a dipeptide derivative (in the case of a dipeptide derivative, an N-terminal amino protecting group is removed prior to following treatments) prepared in the above-mentioned manner is treated with a condensation reagent (for example, EDCI•HCl-HOBT•H₂O) in the same manner described above and then with an N^α-substituted amino acid or an O^α-substituted hydroxycarboxylic acid, to afford a desired peptide derivative. A peptide derivative of the formula (I) wherein X³ is a sulfur atom, can be prepared by condensation of a compound of the formula (V) with a compound of the formula (VI) whose C-terminal carboxyl group is protected, followed by conversion of the resulting amide bond to the thioamide bond on treatment with, for example, the Lawesson's reagent, then condensed with a compound of the formula (II) by, for example, a DCC method or an azide method to afford a target peptide derivative elongated toward the N-terminus.

A C-terminal protective group of a peptide derivative prepared by the method so far described, can be deprotected by a suitable method, if necessary. For example, in the case that a carboxyl group is protected as a tert-butyl ester, the protective group can be readily removed by acid decomposition, that is, by treatment with a mild acid such as formic acid or TFA. In the case that a carboxylic acid is protected as a benzyl ester, the protective group can be readily removed by catalytic hydrogenation, that is, by hydrogenation under 1 to 4 atmospheric pressures of hydrogen in the presence of a catalyst such as Pd-C or palladium black in a solvent such as methanol, ethanol, DMF, THF, 1,4-dioxane or acetic acid.

[Method 2]

Method 2 is a process for producing a peptide derivative which possesses an alkoxycarbonyl, aryloxycarbonyl or carbamoyl group at the N-terminus, wherein an N-terminal amino protective group of an amino acid derivative or a peptide derivative prepared by Method 1, is removed, and the resulting deprotected amino group is reacted with an acid chloride such as a chloroformate (R¹¹OCOCI) or a carbamoyl chloride (R¹²R¹³NCOCI) in the presence of a base, or reacted with an isocyanate (R¹²NCO) (wherein R¹¹, R¹² and R¹³ are as defined before), furthermore, a C-terminal protective group is optionally removed by alkaline hydrolysis, catalytic hydrogenation or acid decomposition.

An N-terminal protective group of the precursor can be readily removed by a conventional method such as catalytic hydrogenation (a Z group) or by treatment with a mild acid such as TFA (a Boc group). The reaction of the resulting deprotected peptide derivative from which the N-terminal amino protective group is removed by the method described above, with an acid chloride such as a chloroformate (R¹¹OCOCI) or a carbamoyl chloride (R¹²R¹³NCOCI) can be performed in a suitable solvent such as chloroform, dichloromethane, THF, 1,4-dioxane, toluene or pyridine in the presence of a base such as TEA, DMAP, N-methylmorpholine or pyridine at 0 °C to the boiling point of the solvent. The reaction with an isocyanate (R¹²NCO) can be performed in a solvent such as chloroform, dichloromethane, THF, 1,4-dioxane or toluene at 0 °C to the boiling point of the solvent.

A C-terminal protective group of peptide derivatives prepared by the above-mentioned method can be removed by alkaline hydrolysis, catalytic hydrogenation or acid decomposition in the same manner described in Method 1, if necessary.

[Method 3]

Method 3 is a process for producing an amino acid or a peptide derivative which possesses a carbamoyl group at the N-terminus, by treatment of an amino acid derivative or a peptide derivative (prepared by Method 1 or 2) having an aryloxycarbonyl group at the N-terminus, with a primary or secondary amine $R^{12}NHR^{13}$ wherein R^{12} and R^{13} are as defined before, furthermore optionally removing a C-terminal protective group by alkaline hydrolysis, catalytic hydrogenation or acid decomposition.

That is, a peptide derivative possessing a carbamoyl group at the N-terminus can be prepared by dissolving a peptide derivative possessing an aryloxycarbonyl group at the N-terminus in a solvent such as chloroform, dichloromethane, THF, 1,4-dioxane, toluene or pyridine, followed by addition of the primary or secondary amine described above, optional addition of a tertiary amine such as TEA or DMAP, and allowing them to react at room temperature to the boiling point of the solvent. A C-terminal protective group of the product can be removed, if necessary, by alkaline hydrolysis, catalytic hydrogenation or acid decomposition in the same manner described in Method 1.

[Method 4]

Method 4 is a process for alkoxylation at the 1-position of the indole ring of a tryptophanyl residue when the peptide derivative prepared in the present invention contains the tryptophanyl residue.

The alkoxylation of the indole ring at the 1-position can be performed by treatment of a tryptophan-containing dipeptide or tripeptide derivative prepared by optionally combining Methods 1, 2 and 3, or an optionally protected tryptophan derivative as the starting material thereof, with a base (such as sodium hydroxide) and alkyl chloroformate in a solvent such as dichloromethane in the presence of TBAHS.

[Method 5]

Method 5 is a process for hydroxylation or lower alkoxylation at the 1-position of the indole ring when the peptide derivative prepared in the present invention contains a tryptophanyl residue.

The hydroxylation of the indole ring at the 1-position can be performed by treatment of a tryptophan-containing dipeptide or tripeptide derivative prepared by optionally combining Methods 1, 2, and 3, or an optionally protected tryptophan derivative as the starting material thereof, with sodium cyanoborohydride in a solvent such as acetic acid, and treatment of the resulting indoline derivative with hydrogen peroxide in methanol, water or a mixed solvent thereof. The lower alkoxylation of the indole ring at the 1-position can be performed by treatment of the hydroxylated derivative with, for example, diazomethane or diazoethane.

[Method 6]

Method 6 is a process for halogenation at the 2-position of the indole ring of a tryptophanyl residue when the peptide derivative prepared in the present invention contains the tryptophanyl residue.

The halogenation of the indole ring at the 2-position can be performed by treatment of a tryptophan-containing dipeptide or tripeptide derivative prepared by optionally combining methods 1, 2 and 3, or an optionally protected tryptophan derivative as the starting material thereof, with an N-halosuccinimide such as N-bromosuccinimide or N-chlorosuccinimide in a solvent such as acetic acid or carbon tetrachloride, optionally in the presence of 2,2'-azobis(isobutyronitrile) or the like.

[Method 7]

Method 7 is a process for lower alkylthiolation, lower alkylsulfinylation or lower alkylsulfonylation at the 2-position of the indole ring of a tryptophanyl residue when the peptide derivative prepared in the present invention contains the tryptophanyl residue.

The lower alkylthiolation of the indole ring at the 2-position can be performed by treatment of a tryptophan-containing dipeptide or tripeptide derivative prepared by optionally combining methods 1, 2 and 3, or an optionally protected tryptophan derivative as the starting material thereof, with a lower alkylsulfonyl chloride in a solvent such as acetic acid. The lower alkylsulfinylation or lower alkylsulfonylation at the 2-position of the indole ring can be performed by treatment of the sulfide derivative thus obtained, with an oxidizing agent such as hydrogen peroxide or m-chloroperbenzoic acid in a solvent such as acetic acid or dichloromethane.

All reaction intermediates and products so far described can be purified by well-known methods such as recrystallization, reprecipitation, partition procedures, normal- or reverse-phase chromatography and ion-exchange chromatography.

Starting materials used in the methods so far described are commercially available except for the following materials, which are prepared by the known method in the literature.

DL-3-(3-ethoxycarbonylphenyl)alanine and DL-3-(4-methoxycarbonylphenyl)alanine: Synthesis, 53 (1984).

D-3-(3-benzothienyl)alanine: Chem. Pharm. Bull., 24, 3149 (1976).

D-S-methylcysteine, D-S-ethylcysteine, D-S-n-propylcysteine: by Nobuo Izumiya et al in Peptide Synthesis, Maruzen, 1975.

D-norleucinol: synthesized in the same manner described in J. Am. Chem. Soc., 107, 7974 (1985).

D-3-cyclopropylalanine: J. Am. Chem. Soc., 111, 6354 (1989).

D-cyclopropylglycine: J. Am. Chem. Soc., 111, 6354 (1989).

D-cyclopentylglycine: J. Org. Chem., 30, 1320 (1965).

D-3-(3-nitrophenyl)alanine: Pept. Res., 3, 176 (1990).

DL-1-aminopentanesulfonic acid: European Laid-open Patent publication No. 33504.

D-1-aminopentylphosphonic acid diethyl ester: synthesized in the same manner described in Liebigs Ann. Chem., 45 (1987).

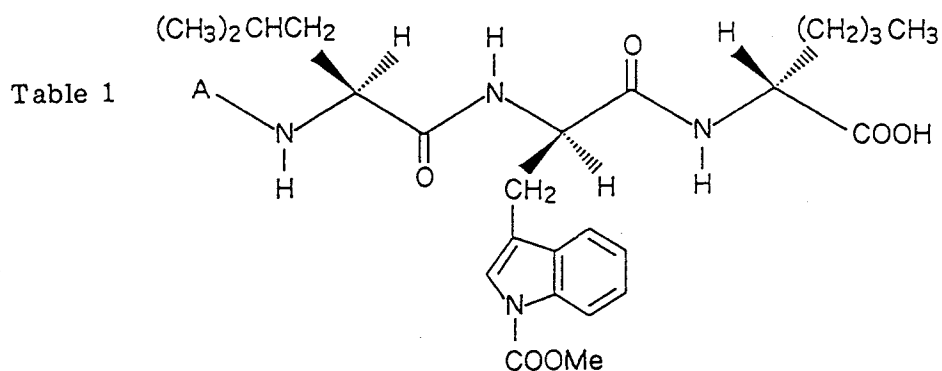
D-(7-benzyloxy)tryptophan: synthesized in the same manner described in Tetrahedron Letters, 30, 4073 (1989).

(R)-5-(1-aminopentyl)-1H-tetrazole hydrochloride: synthesized in the same manner described in J. Org. Chem., 56, 2395 (1991).

α -N-trifluoroacetyl-2-chloro-D-tryptophan methyl ester and

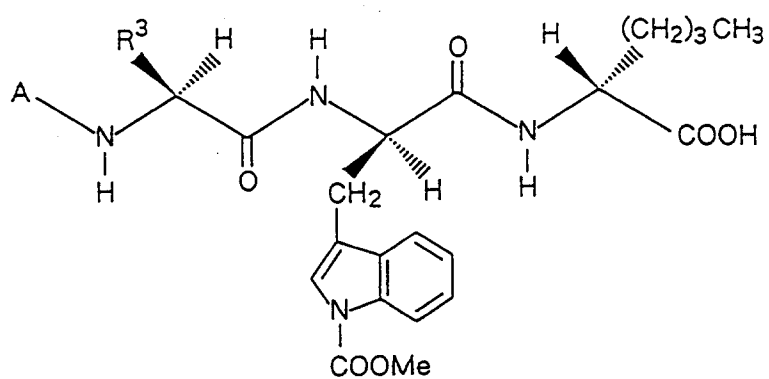
α -N-trifluoroacetyl-1-methyl-2-chloro-D-tryptophan: synthesized in the same manner described in J. Am. Chem. Soc., 108, 2023 (1986).

The chemical structures, Example Nos. and compound Nos. of the prepared peptide derivatives in the present invention are shown in Tables 1 to 7.



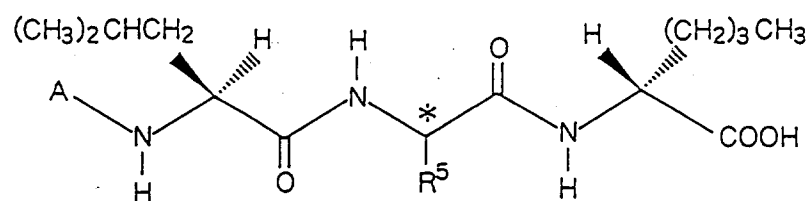
Exp. No.	Compd No.	A	Exp. No.	Compd No.	A
1	1	Boc	31	31	
4	4		32	32	
21	21		33	33	
22	22		34	34	
23	23		35	35	
24	24		36	36	
25	25		37	37	
26	26		38	38	
27	27		39	39	
28	28		44	44	
29	29		46	46	
30	30		47	47	

Table 2



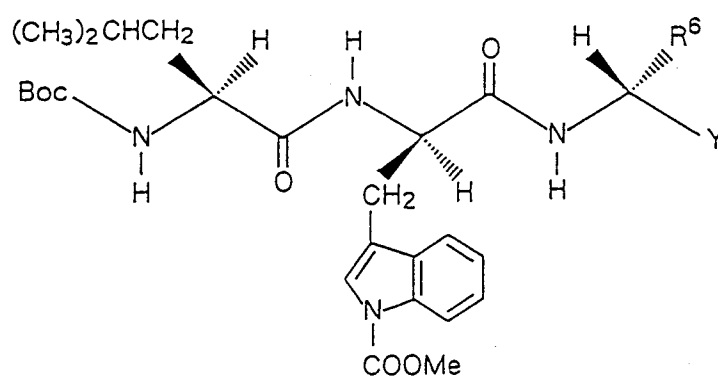
Exp. No.	Compd. No.	A	R ³
14	14	Boc	-(CH ₂) ₃ CH ₃
15	15	Boc	-CH ₂ C(CH ₃) ₃
40	40		-CH ₂ C(CH ₃) ₃
41	41		-CH ₂ Ph
42	42		-CH ₂
43	43		-CH(CH ₃) ₂
62	67		-CH(CH ₃) ₂
63	68		-CH ₂
64	69	=	
65	70		

Table 3



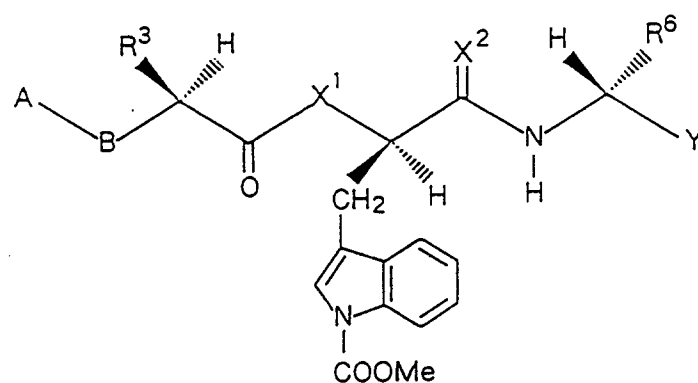
Exp. No.	Compd No.	A	*	R ⁵
3	3	Boc	(R)	
5	5		(R)	
16	16	Boc	(R)	
17	17	Boc	(RS)	
18	18		(R)	
19	19		(RS)	
20	20		(RS)	
97	112		(R)	
98	113		(R)	

Table 4



Exp. No.	Compd No.	R^6	Y
2	2	$-(\text{CH}_2)_2\text{CH}_3$	COOH
6	6	$-(\text{CH}_2)_2\text{SCH}_3$	COOH
7	7	$-\text{CH}_2\text{SCH}_3$	COOH
8	8	$-\text{CH}_2\text{SCH}_2\text{CH}_3$	COOH
9	9	$-\text{CH}_2\text{S}(\text{CH}_2)_2\text{CH}_3$	COOH
10	10	$-\text{CH}_2\text{CH}=\text{CH}_2$	COOH
11	11	$-\text{CH}_2\text{CH}(\text{CH}_3)_2$	COOH
12	12	$(\text{R}) - \overset{*}{\underset{\text{CH}_3}{\text{CH}}}\text{CH}_2\text{CH}_3$	COOH
13	13	$-(\text{CH}_2)_3\text{CH}_3$	CH_2OH
45	45	$-\text{CH}_2\text{CH}_3$	COOH
60	65	$-\text{CH}_2\text{OH}$	COOH
66	71	$-\text{CH}_2\text{OCH}_3$	COOH

Table 5



Exp. No.	Comp. No.	A	B	R ³	X ¹	X ²	R ⁶	Y
48	48	 NCO-	-NH-	-CH ₂ CH(CH ₃) ₂	NH	O	-(CH ₂) ₃ CH ₃	COO ^t Bu
48	49	 NCO-	-NH-	-CH ₂ CH(CH ₃) ₂	NH	O	-(CH ₂) ₃ CH ₃	COOH
49	50	 NCO-	-NH-	-CH ₂ CH(CH ₃) ₂	NH	O	-(CH ₂) ₂ CH ₃	COOH
50	51	 NCO-	-NH-	-CH ₂ C(CH ₃) ₃	NH	O	-(CH ₂) ₂ CH ₃	COOH
51	52	 NCO-	-NH-	(S) - [*] CH(CH ₃)CH ₂ CH ₃	NH	O	-(CH ₂) ₃ CH ₃	COOH
52	53	 NCO-	-NH-	(R) - [*] CH(CH ₃)CH ₂ CH ₃	NH	O	-(CH ₂) ₃ CH ₃	COOH
53	54	 NCO-	-NH-	-C(CH ₃) ₃	NH	O	-(CH ₂) ₃ CH ₃	COOH
54	55	 NHCO-	-O-	-CH ₂ CH(CH ₃) ₂	NH	O	-(CH ₂) ₃ CH ₃	COOH
55	56	Boc	-NH-	-CH ₂ CH(CH ₃) ₂	NH	O	-CH(CH ₃) ₂	COOH

Table 5 (continued)

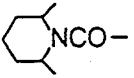
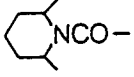
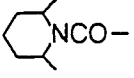
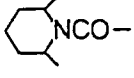
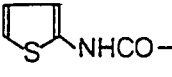
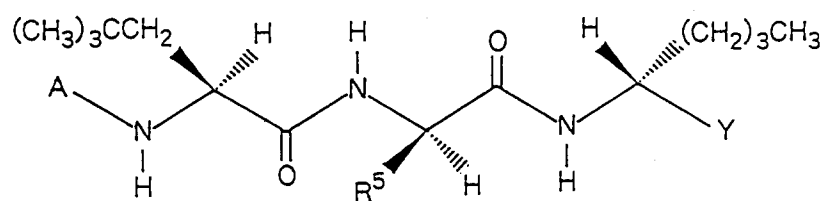
Exp. No.	Comp. No.	A	B	R ³	X ¹	X ²	R ⁶	Y
56	57		-NH-	-CH ₂ C(CH ₃) ₃	NH	S	-(CH ₂) ₃ CH ₃	COO ^t Bu
56	58		-NH-	-CH ₂ C(CH ₃) ₃	NH	S	-(CH ₂) ₃ CH ₃	COOH
57	59		-NH-	-CH ₂ C(CH ₃) ₃	O	O	-(CH ₂) ₃ CH ₃	COOH
58	60		-NH-	-CH ₂ C(CH ₃) ₃	NH	O	-(CH ₂) ₃ CH ₃	COO ^t Bu
61	66		-NH-	-CH ₂ C(CH ₃) ₃	NH	O	-(CH ₂) ₃ CH ₃	COOH

Table 6



Exp. No.	Compd No.	A	R ⁵	Y
59	61			COOH
59	62			COOH
59	63			COOH
59	64			COOH
67	72			COOH
67	73			COOH
67	74			COOH
67	75			COOH
68	76			COO ^t Bu
68	77			COOH

Table 6 (continued)

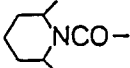
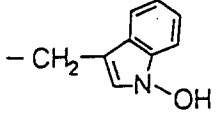
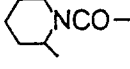
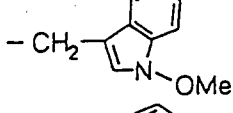
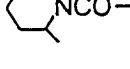
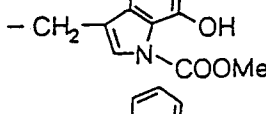
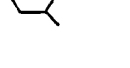
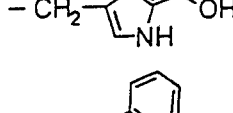
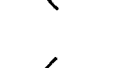
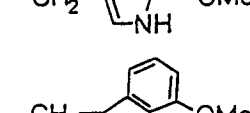
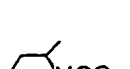
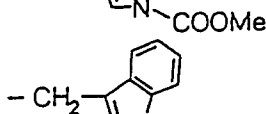
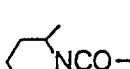
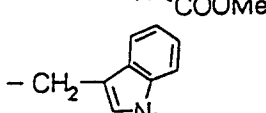
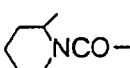
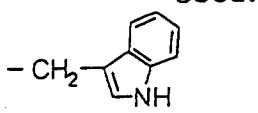
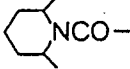
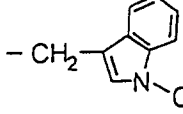
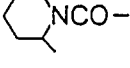
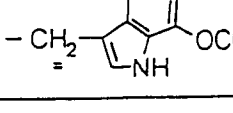
Exp. No.	Compd No.	A	R ⁵	Y
69	78			COOH
69	79			COOH
70	80			COOH
70	81			COOH
70	82			COOH
70	83			COOH
74	87			CH ₂ OH
74	88			CH ₂ OH
87	102			COOH
88	103			COOH
89	104			COOH

Table 6 (continued)

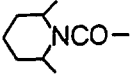
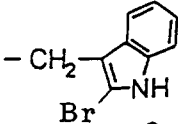
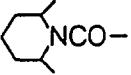
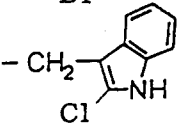
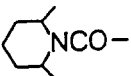
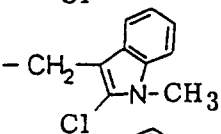
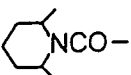
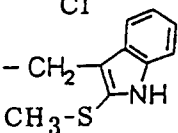
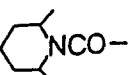
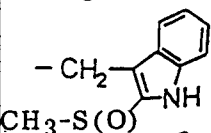
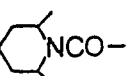
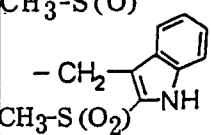
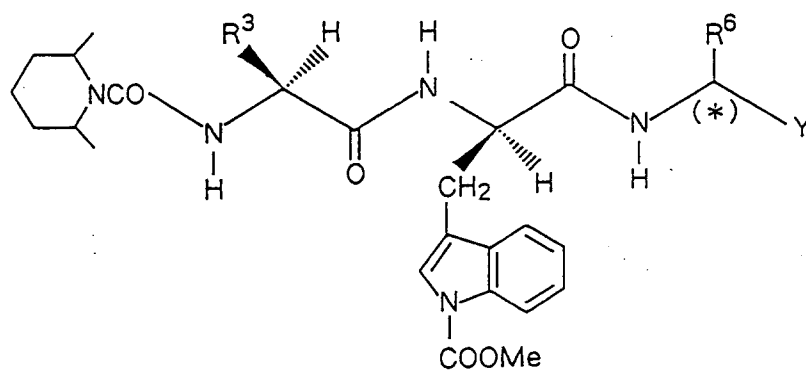
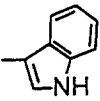
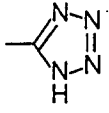
Exp. No.	Compd No.	A	R ⁵	Y
95	110			COOH
96	111			COOH
99	114			COOH
100	115			COOH
101	116			COOH
102	117			COOH

Table 7



Exp. No.	Compd. No.	R ³	R ⁶	*	Y
71	84	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	COOMe
72	85	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	COOEt
73	86	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	COO ⁱ Pr
75	89		-(CH ₂) ₃ CH ₃	(R)	COO ^t Bu
75	90		-(CH ₂) ₃ CH ₃	(R)	COOH
76	91		-CH ₃	(R)	CH ₂ COOH
77	92	-CH ₂ C(CH ₃) ₃	-H	-	COOH
78	93	-CH ₂ C(CH ₃) ₃	=H	-	CH ₂ COOH
81	96	-CH ₂ C(CH ₃) ₃	-H	-	SO ₃ H

Table 7 (continued)

Exp. No.	Compd No.	R ³	R ⁶	*	Y
79	94	-CH ₂ C(CH ₃) ₃	-CH ₂ - 	(R)	COOH
80	95	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	CH ₂ COOH
82	97	-CH ₂ C(CH ₃) ₃	-H	-	CH ₂ SO ₃ H
83	98	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	CH ₂ SO ₃ H
84	99	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(RS)	SO ₃ H
85	100	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(S)	PO ₃ H ₂
86	101	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	CONHCH ₂ CH ₂ OH
90	105	-CH ₂ C(CH ₃) ₃	-(CH ₂) ₃ CH ₃	(R)	
91	106	-CH(CH ₃) ₂	-(CH ₂) ₃ CH ₃	(RS)	SO ₃ H
92	107	-CH(CH ₃) ₂	-(CH ₂) ₃ CH ₃	(R)	CH ₂ SO ₃ H
93	108	-CH ₂ CH(CH ₃) ₂	-(CH ₂) ₃ CH ₃	(R)	CH ₂ SO ₃ H
94	109	-CH ₂ CH(CH ₃) ₂	-(CH ₂) ₃ CH ₃	(RS)	SO ₃ H

Now, the endothelin antagonistic properties of the peptide derivatives of the present invention will be described.

Endothelin binding inhibition test

The cerebellum of porcine was homogenized in a buffer solution of 10 mM MOPS, pH 7.4, at 4 °C by a polytron. To the homogenate, sucrose was added to a concentration of 20%, and the mixture was

centrifuged at 1,000 x g for 15 minutes, and the supernatant was further centrifuged at 10,000 x g for 15 minutes. The supernatant thereof was further centrifuged at 90,000 x g for 40 minutes. The membrane precipitate thereby obtained was suspended in a buffer solution of 5 mM HEPES/Tris, pH 7.4, at a concentration of 3.3 mg/ml.

5 Then, 16 μ l of this membrane suspension was added to 340 μ l of 50 mM tris/HCl buffer, pH 7.4, containing 10 μ l calcium chloride, 10 μ M magnesium chloride, 0.1 mM PMSF, 1 μ M pepstatin A, 2 μ M leupeptin, 1 mM 1,10-phenanthroline and 0.1% bovine serum albumin. To this suspension, 4 μ l of (A) endothelin-1 (for nonspecific binding; 0.2 μ M as the final concentration), (B) buffer solution A (for total control binding), or (C) a test compound (1.1 μ M as the final concentration), was added. Further, to each
 10 suspension, 40 μ l of 125 I-endothelin-1 (12000-18000 cpm) was added. These mixtures were incubated at 25°C for 4 hours, then subjected to filtration on a glass filter GF/C and then washed with 5 mM HEPES/Tris, pH 7.4, containing 0.3% bovine serum albumin. Then, the radioactivity trapped by the glass filter was measured, and the 125 I-endothelin-1 binding inhibition D (%) at 1.1 μ M of the test compound was determined by the following equation.

$$D(\%) = 100 - \frac{(C) - (A)}{(B) - (A)} \times 100$$

Each test was performed in triplicate.

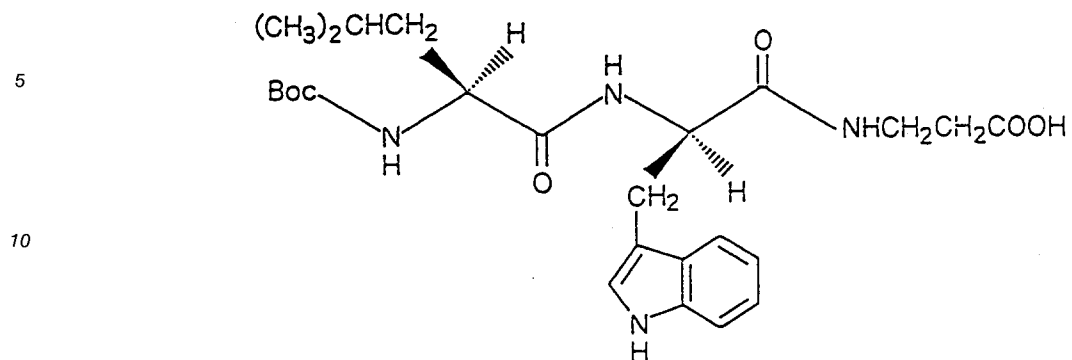
Table 8

	¹²⁵ I-endothelin-1 binding inhibition by 1.1 μM of the test compound							
5	Compd No.	Inhibition(%)	Compd No.	Inhibition (%)	Compd No.	Inhibition(%)	Compd No.	Inhibition(%)
10	1	90	28	99	55	100	83	57
	2	84	29	89	56	70	84	98
	3	75	30	73	57	89	85	98
	4	99	31	96	58	98	86	94
	5	97	32	72	59	95	87	99
	6	64	33	99	60	95	88	96
	7	78	34	99	61	64	89	96
15	8	81	35	97	62	75	90	100
	9	54	36	95	63	82	91	94
	10	81	37	90	64	77	92	59
	11	41	38	84	66	91	93	80
20	12	77	39	98	67	100	94	86
	13	26	40	99	68	100	95	100
	14	65	41	67	69	100	96	51
	15	89	42	97	70	100	97	58
	16	18	43	99	71	60	98	98
	17	18	44	97	72	86	99	99
	18	94	45	71	73	35	100	96
25	19	61	46	91	74	44	101	98
	20	75	47	35	75	42	102	92
	21	94	48	92	76	33	103	94
	22	93	49	99	77	69	104	74
30	23	100	50	99	78	84	105	100
	24	97	51	99	79	96	106	99
	25	88	52	100	80	94	107	100
	26	85	53	100	81	18	108	99
35							109	99
	27	83	54	97	82	50	110	98
							111	99
							112	97
							113	93
40							114	98
							115	88

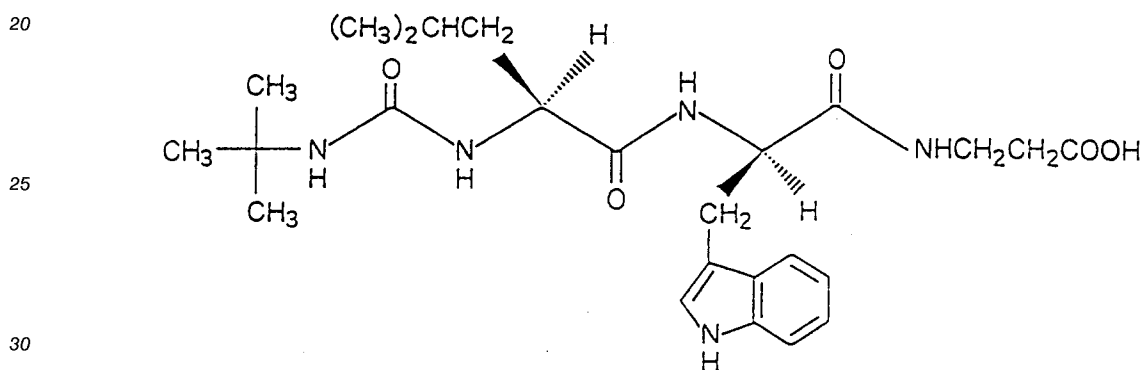
As shown in Table 8, the compounds of the present invention were found to be very potent inhibitor of endothelin binding to ET_B receptor. The test compounds are indicated by compound Nos.

On the other hand, each of the representative compounds (Reference Compounds 1, 2 and 3) of the endothelin antagonistic peptides described in EP 0460679A2 showed, at a concentration of 1.1 μ M, 69, 77 or 86% of the ¹²⁵I-ET-1 binding inhibition to porcine aorta membrane suspension containing a large amount of ET_A receptors, and, on the contrary, only 6.4, 3.8 or 10.1% of the ¹²⁵I-ET-1 binding inhibition to ET_B receptors of porcine cerebellum membrane suspension used in the present experiments.

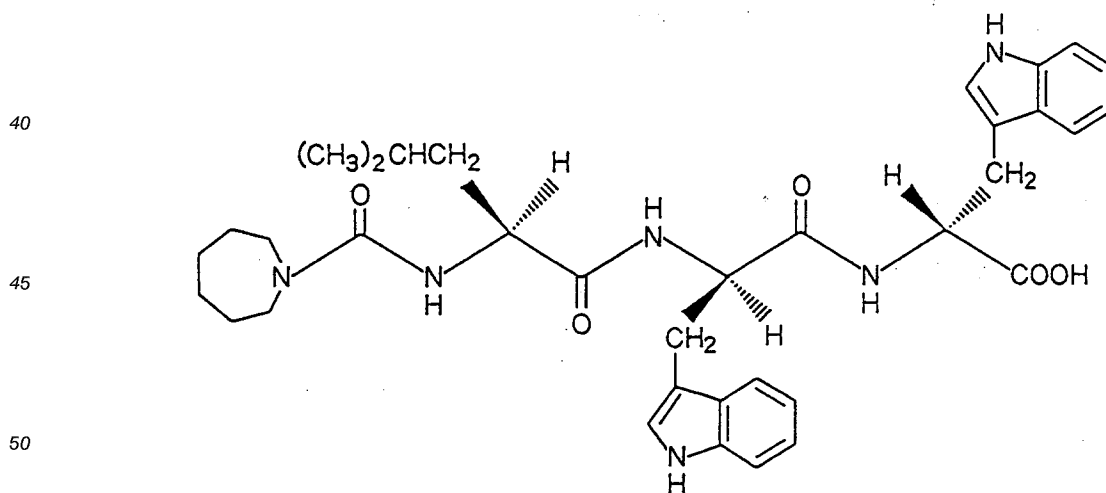
Reference Compound 1



Reference Compound 2



Reference Compound 3



55 Activities against endothelin-induced contraction of isolated rabbit pulmonary arteries

The pulmonary artery of rabbit was isolated, and a spiral preparation having a width of 1 mm and a length of 10 mm was prepared therefrom. The preparation having the endothelial cells denuded, was placed

in a 5 ml organ bath filled with a Krebs•Henseleit solution saturated with a gas mixture of 95% O₂ and 5% CO₂, and a change in the tension was isometrically measured and recorded.

Endothelin-3 was added into the organ bath in a cumulatively increasing manner, whereby the influence of a compound of the present invention to the concentration-response curve for endothelin-3 was examined.

The compound was added into the organ bath 20 minutes prior to the addition of endothelin-3.

As shown in Figures 1 to 3, Compound 4, compound 39 and Compound 40 (0.1 to 10 μ M) remarkably shifted the concentration-response curves of endothelin-3 to the right. Table 9 shows pA₂ values (-log of the concentration of the compound necessary for shifting the concentration-response curve for endothelin-3 twice to the right. Further, the compounds showed no effects to the isolated pulmonary artery when applied alone. As is evident from the above, the compounds showed remarkable antagonistic activities against endothelin-induced concentration of isolated rabbit pulmonary artery.

Table 9

Antagonistic activities of the compound of the present invention against endothelin-3-induced contraction of isolated rabbit pulmonary artery	
Compound No.	PA ₂ value
4	6.7
39	6.7
40	7.1

Activities against endothelin-induced contraction of isolated guinea pig bronchus

The bronchus of a guinea pig was isolated, and the bronchus was cut into rings having an outer diameter of 2 mm and a width of 4 mm to afford a preparation. The preparation was placed in a 5 ml organ bath filled with a Krebs•Henseleit solution saturated with a gas mixture of 95% O₂ and 5% CO₂, and a change in the tension was isometrically measured and recorded.

Endothelin-3 was added into the organ bath in a cumulatively increasing manner, and the influence of a compound of the present invention to the concentration-response curve for endothelin was examined. The compound was added into the organ bath 20 minutes prior to the addition of endothelin-3.

As shown in Figures 1 to 3, Compound 4, Compound 39 and Compound 40 (10 μ M) remarkably shifted to concentration-response curves for endothelin-3 to the right in isolated bronchus. Further, the compounds showed no effects to the isolated bronchus when applied alone. As is evident from the foregoing, the compounds showed remarkable antagonistic activities against endothelin-induced contraction of isolated guinea pig bronchus.

Thus, the compounds of the present invention have excellent endothelin antagonistic activities and are useful as vasodilators for bronchodilators in the field of medicines, and they can be drugs for treating hypertension, pulmonary hypertension, Raynaud's disease, acute renal failure, myocardial infarction, angina pectoris, cerebral infarction, cerebral vasospasm, arteriosclerosis, asthma, gastric ulcer, diabetes, endotoxin shock, endotoxin-induced multiple organ failure or disseminated intravascular coagulation, and/or cyclosporin-induced renal failure or hypertension. When used as drugs for treating such diseases, the compounds of the present invention can be used alone or in combination with other drugs for treatment.

The compounds of the present invention may be used in the form of drug formulations suitable for parenteral administration, oral administration or external administration by mixing them with solid or liquid excipient carriers known in this field. The drug formulations include a liquid formulation such as an injection formulation, an inhalant formulation, a syrup formulation or an emulsion, a solid formulation such as tablets, capsules or granules, and an external drug such as an ointment or a suppository. Further, these drug formulations may contain additives which are commonly employed, such as an adjuvant, a stabilizer, a wetting agent, an emulsifier, an absorption-promoting agent or a surfactant, as the case requires. As the additives, distilled water for injection, physiological saline, Ringer's solution, glucose, sugar syrup, gelatin, vegetable oil, cacao butter, ethylene glycol, hydroxypropyl cellulose, lactose, sucrose, corn starch, magnesium stearate and talc may be mentioned.

The dose of a compound of the present invention as an endothelin antagonist varies depending upon the manner of administration, the age and body weight of the patient and the condition of the patient to be treated. However, a typical administration method for an adult is oral administration or parenteral administra-

tion. The daily dose in the case of oral administration to an adult patient is from 0.1 to 100 mg/kg body weight, and the daily dose in the case of parenteral administration is from 0.01 to 10 mg/kg body weight.

The following Examples and Referential Examples illustrate the present invention more specifically. It should be understood that the present invention is not limited to these examples alone.

Example 1

Synthesis of Compound 1

(1) Preparation of Boc-Leu-DTrp(COOMe)-OH

To a mixture of Boc-Leu-OH•H₂O (2.5 g), H-DTrp-OBzl (2.9 g) and HOBT•H₂O (1.6 g) in dichloromethane (50 ml) was added EDCI•HCl (2.0 g) under ice cooling. After being stirred at room temperature for 16 h, the mixture was washed with water, 10% aq. citric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was triturated with hexane to give Boc-Leu-DTrp-OBzl (4.79 g). To a solution of the dipeptide (753 mg) in dichloromethane (7 ml) were added methyl chloroformate (1 ml), pulverized NaOH (89 mg) and TBAHS (20 mg) under ice cooling. After being stirred of the mixture at room temperature for 30 min, pulverized NaOH (89 mg) was added. This procedure was repeated twice more. The reaction mixture was diluted with dichloromethane, washed with sat. aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo.

The residue was recrystallized from dichloromethane and petroleum ether to give Boc-Leu-DTrp(COOMe)-OBzl (721 mg) as colorless crystals. The dipeptide (708 mg) was dissolved in methanol (7 ml) - THF (5 ml) and 10% Pd/C (70 mg) was added. The mixture was vigorously stirred at room temperature under an atmospheric pressure of hydrogen for 1 h. The reaction mixture was filtered and the filtrate was evaporated in vacuo to give Boc-Leu-DTrp(COOMe)-OH (580 mg).

FAB-MS(m/e, (C₂₄H₃₃N₃O₇ + H)⁺): 476

(2) Preparation of Boc-Leu-DTrp(COOMe)-DNle-OBzl

To a mixture of Boc-Leu-DTrp(COOMe)-OH (50 mg), H-DNle-OBzl•TsOH (46 mg), NMM (13 μl) and HOBT•H₂O (16 mg) in dichloromethane (1 ml) was added EDCI•HCl (22 mg) under ice cooling. After being stirred at room temperature for 2 h, the mixture was diluted with dichloromethane, washed with water, 10% aq. citric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 100/10/1 for development to give Boc-Leu-DTrp(COOMe)-DNle-OBzl (67 mg).

FAB-MS(m/e, (C₃₇H₅₀N₄O₈ + H)⁺): 679

(3) Preparation of Compound 1

A solution of Boc-Leu-DTrp(COOMe)-DNle-OBzl (25 mg) in methanol (1 ml) was hydrogenated over 10% Pd/C at an atmospheric pressure of hydrogen for 1 h. After removal of the catalyst by filtration, the filtrate was concentrated in vacuo. The residue was purified by preparative TLC (3M, Empore Silica Sheet) with chloroform/methanol/acetic acid = 100/10/1 for development and the product was reprecipitated from methanol and water to give compound 1 (11 mg).

m.p.: 154-163 °C

IR(KBr, cm⁻¹):

3328, 2962, 2872, 1728, 1695, 1650, 1536, 1461, 1386, 1371, 1341, 1311, 1260, 1167, 1092, 762, 747

High Resolution FAB-MS(m/e, (C₃₀H₄₄N₄O₈ + H)⁺):

Calcd: 589,3237

Found: 589,3260

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(6H,d,J = 6.2Hz), 0.85(3H,t,J = 7.0Hz), 0.93-1.10 (2H,m), 1.10-1.55(5H,m), 1.31(9H,s), 1.55-1.82(2H,m), 2.86(1H,dd,J = 10.6Hz, 15.2Hz), 3.00-3.30(1H,m), 3.80-3.90(1H,m), 3.95(3H,s), 4.05-4.20(1H,m), 4.59-4.70 (1H,m)-, 6.74(1H,d,J = 7.1Hz), 7.23(1H,t,J = 7.5Hz), 7.31 (1H,t,J = 7.5Hz), 7.49(1H,s), 7.70(1H,d,J = 7.5Hz), 8.04 (2H,d,J = 7.5Hz), 8.20(1H,d,J = 8.3Hz)

Example 2

Synthesis of Compound 2

5 Compound 2 was prepared using H-DNle-OBzl•TsOH as a starting material in the same manner described in Example 1-(2) and (3).

m.p.: 179-185 °C

IR(KBr, cm⁻¹):

3322, 2968, 1720, 1656, 1530, 1461, 1389, 1341, 1311, 1260, 1167, 1092, 765, 747

10 High Resolution FAB-MS(m/e, (C₂₉H₄₂N₄O₈ + H)⁺):

Calcd: 575.3080

Found: 575.3101

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 7.2Hz), 0.66(3H, d, J = 7.2Hz), 0.85(3H, t, J = 7.2Hz), 0.95-1.75(7H, m), 1.31(9H, s), 2.88(1H, dd, J = 10.6Hz, 14.9Hz), 3.15(1H, dd, J = 4.3Hz, 14.9Hz), 3.85(1H, m), 3.95(3H, s), 4.06(1H, dt, J = 5.6Hz, 7.6Hz), 4.61(1H, ddd, J = 4.3Hz, 7.6Hz, 10.6Hz), 6.72(1H, d, J = 7.6Hz), 7.23(1H, t, J = 7.6Hz), 7.31(1H, t, J = 7.6Hz), 7.48(1H, s), 7.69(1H, d, J = 7.6Hz), 7.93(1H, d, J = 7.6Hz), 8.03(1H, d, J = 7.6Hz), 8.19(1H, d, J = 8.4Hz)

Example 3

20

Synthesis of Compound 3

(1) Preparation of Boc-Leu-DTrp(COOEt)-DNle-OBzl

25 The title compound was prepared using ethyl chloroformate instead of methyl chloroformate in the same manner described in Example 1-(1) and (2).

FAB-MS(m/e, (C₃₈H₅₂N₄O₈ + H)⁺): 693

(2) Preparation of Compound 3

30

Compound 3 was prepared using Boc-Leu-DTrp(COOEt)-DNle-OBzl as a starting material in the same manner described in Example 1-(3).

m.p.: 104-105 °C

IR(KBr, cm⁻¹):

35 3322, 2968, 2872, 1720, 1662, 1536, 1461, 1386, 1254, 1170, 747

High Resolution FAB-MS(m/e, (C₃₁H₄₆N₄O₈ + H)⁺):

Calcd: 603.3394

Found: 603.3406

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

40 0.65(6H, d, J = 6.4Hz), 0.77-1.90(12H, m), 1.31(9H, s), 1.37(3H, t, J = 7.0Hz), 2.88(1H, dd, J = 10.7Hz, 15.4Hz), 3.15(1H, dd, J = 4.2Hz, 15.4Hz), 3.80-3.91(1H, m), 4.05-4.25(1H, m), 4.40(2H, q, J = 7.0Hz), 4.60-4.71(1H, m), 6.73(1H, d, J = 7.6Hz), 7.23(1H, t, J = 7.5Hz), 7.31(1H, t, J = 7.5Hz), 7.51(1H, s), 7.70(1H, d, J = 7.5Hz), 8.03(1H, d, J = 7.5Hz), 8.06(1H, brs), 8.19(1H, d, J = 8.8Hz)

45 Example 4

Synthesis of Compound 4

A mixture of Boc-Leu-DTrp(COOMe)-DNle-OBzl (20 mg, prepared in Example 1-(2)) in TFA (0.5 ml) was stirred under ice cooling for 15 min and evaporated in vacuo. To a solution of the residue in chloroform (0.5 ml) were added TEA (6 μl) and 2-chlorophenylisocyanate (4 μl) under ice cooling. After being stirred at the same temperature for 1 h, the mixture was diluted with chloroform, washed with 1N-hydrochloric acid, 5% aq. NaHCO₃, and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 50/1 for development, and the product was hydrogenated over 10% Pd/C at an atmospheric pressure of hydrogen for 1 h. The reaction mixture was filtered and the filtrate was evaporated in vacuo. The residue was purified by reverse phase MPLC (Merck, LiChroprep RP-18) with methanol/water = 5/1 for elution to give Compound 4 (5 mg).
m.p.: 193-201 °C

IR(KBr, cm⁻¹):

3328, 2962, 2872, 1740, 1647, 1593, 1548, 1461, 1446, 1386, 1341, 1311, 1263, 1230, 1092, 747

High Resolution FAB-MS(m/e, (C₃₂H₄₀ClN₅O₇ + H)⁺):

Calcd: 642.2695

Found: 642.2661

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 5.8Hz), 0.67(3H, d, J = 5.8Hz), 0.82(3H, t, J = 7.1Hz), 1.00-1.40(7H, m), 1.57-1.80(2H, m), 2.89(1H, dd, J = 3.2Hz, 11.5Hz), 3.10-3.20(1H, m), 3.95(3H, s), 4.08-4.22(2H, m), 4.62-4.72(1H, m), 6.91(1H, dt, J = 1.4Hz, 7.8Hz), 7.14-7.32(4H, m), 7.35(1H, dd, J = 1.4Hz, 7.8Hz), 7.54(1H, s), 7.76(1H, d, J = 7.8Hz), 8.05(1H, d, J = 7.8Hz), 8.08(1H, dd, J = 1.4Hz, 7.8Hz), 8.11(1H, s), 8.13-8.20(1H, m), 8.50(1H, d, J = 8.8Hz)

Example 5

Synthesis of Compound 5

Compound 5 was prepared using Boc-Leu-DTrp(COOEt)-DNle-OBzl as a starting material in the same manner described in Example 4.

m.p.: 161-168 °C

IR(KBr, cm⁻¹):

3328, 2962, 2872, 1740, 1644, 1593, 1551, 1461, 1446, 1206, 1092, 747

High Resolution FAB-MS(m/e, (C₃₃H₄₂ClN₅O₇ + H)⁺):

Calcd: 656.2852

Found: 656.2851

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 6.4Hz), 0.66(3H, d, J = 6.4Hz), 0.82(3H, t, J = 7.2Hz), 0.88-1.80(9H, m), 1.38(3H, t, J = 7.0Hz), 2.90(1H, dd, J = 11.8Hz, 14.4Hz), 3.15(1H, dd, J = 2.7Hz, 14.4Hz), 4.10-4.25(2H, m), 4.41(2H, q, J = 7.0Hz), 4.64-4.75(1H, m), 6.86(1H, s), 6.91(1H, dt, J = 1.6Hz, 7.6Hz), 7.12-7.40(5H, m), 7.56(1H, s), 7.75(1H, d, J = 7.1Hz), 8.05(1H, d, J = 7.1Hz), 8.09(1H, d, J = 10.1Hz), 8.17(1H, d, J = 7.9Hz), 8.47(1H, d, J = 8.8Hz)

Example 6

Synthesis of Compound 6

To a solution of Boc-Leu-DTrp(COOMe)-OH (104 mg, prepared in Example 1-(1)) in DMF (3.5 ml) were added HOSu (51 mg) and EDCI·HCl (64 mg) under ice cooling. The mixture was stirred at the same temperature for 1.5 h and at room temperature for 5 h. To the mixture was added an aqueous solution of H-DMet-ONa (prepared from DMet (49 mg) and sodium bicarbonate (28 mg) in water (2 ml)) under ice cooling. After being stirred at room temperature for 2 h, the reaction mixture was diluted with ethyl acetate, washed with 10% aq. citric acid and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by reverse phase chromatography (Nacalai Tesque, Cosmosil 75C18-OPN) with methanol/water = 7/3 - 8/2 for elution to give Compound 6 (65 mg).

m.p.: 140-143 °C

IR(KBr, cm⁻¹):

3322, 2962, 1740, 1653, 1533, 1461, 1389, 1341, 1260, 1167, 1092, 762, 747

High Resolution FAB-MS(m/e, (C₂₉H₄₂N₄O₈S + H)⁺):

Calcd: 607.2801

Found: 607.2761

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(6H, d, J = 5.9Hz), 1.00-1.25(3H, m), 1.32(9H, s), 1.85-2.05(2H, m), 2.04(3H, s), 2.52-2.75(2H, m), 2.87(1H, dd, J = 10.7Hz, 15.4Hz), 3.16(1H, dd, J = 6.2Hz, 15.4Hz), 3.78-3.85(1H, m), 3.95(3H, s), 4.28-4.37(1H, m), 4.59-4.70(1H, m), 6.79(1H, d, J = 7.6Hz), 7.25(1H, t, J = 7.6Hz), 7.32(1H, t, J = 7.6Hz), 7.50(1H, s), 7.69(1H, d, J = 7.6Hz), 8.04(1H, d, J = 7.6Hz), 8.11(1H, d, J = 8.4Hz), 8.22(1H, d, J = 8.3Hz)

Each Compound 7 - 12 described in the following Examples 7 - 12 was prepared using each corresponding amino acid sodium salt as a starting material in the same manner described in Example 6.

Example 7

Compound 7

5 m.p.: 147-150 °C

IR(KBr, cm⁻¹):

3334, 2962, 1740, 1650, 1553, 1461, 1386, 1341, 1311, 1260, 1167, 1092, 767, 747

High Resolution FAB-MS(m/e, (C₂₈H₄₀N₄O₈S + H)⁺):

Calcd: 593.2645

10 Found: 593.2628

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 6.3Hz), 0.66(3H, d, J = 5.4Hz), 0.70-1.15 (3H, m), 1.31(9H, s), 2.09(3H, s), 2.75-2.92(3H, m), 3.15
 (1H, dd, J = 3.5Hz, 14.9Hz), 3.85-3.94(1H, m), 3.95(3H, s), 4.42(1H, dt, J = 5.0Hz, 7.6Hz), 4.64-4.73(1H, m), 6.72(1H,
 d, J = 7.6Hz), 7.24(1H, t, J = 7.4Hz), 7.31(1H, t, J = 7.4Hz), 7.50(1H, s), 7.72(1H, d, J = 7.4Hz), 8.04(1H, d, J = 7.4Hz),
 15 8.20(1H, d, J = 8.5Hz), 8.36(1H, d, J = 7.8Hz)

Example 8

Compound 8

20 m.p.: 170-173 °C

IR(KBr, cm⁻¹):

3346, 2968, 1743, 1650, 1611, 1530, 1461, 1389, 1341, 1260, 1170, 1092, 762, 747

High Resolution FAB-MS(m/e, (C₂₉H₄₂N₄O₈S + H)⁺):

25 Calcd: 607.2802

Found: 607.2792

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 6.4Hz), 0.67(3H, d, J = 3.9Hz), 0.70-1.20 (3H, m), 1.16(3H, t, J = 7.3Hz), 1.31(9H, s), 2.53(2H, q,
 J = 7.3Hz), 2.78-2.97(3H, m), 3.14(1H, dd, J = 3.6Hz, 14.7Hz), 3.82-3.96(1H, m), 3.95(3H, s), 4.36(1H, dt,
 30 J = 5.4Hz, 7.8Hz), 4.62-4.75(1H, m), 6.69(1H, d, J = 7.8Hz), 7.24(1H, t, J = 7.3Hz), 7.31(1H, t, J = 7.3Hz), 7.50(1H, s),
 7.72(1H, d, J = 7.3Hz), 8.03(1H, d, J = 7.3Hz), 8.18(1H, d, J = 8.9Hz), 8.31(1H, d, J = 7.9Hz)

Example 9

35 Compound 9

m.p.: 106-108 °C

IR(KBr, cm⁻¹):

3322, 2968, 1737, 1695, 1653, 1530, 1461, 1386, 1341, 1311, 1260, 1167, 1092, 765, 747

40 High Resolution FAB-MS(m/e, (C₃₀H₄₄N₄O₈S + H)⁺):

Calcd: 621.2958

Found: 621.2938

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.65(3H, d, J = 6.4Hz), 0.66(3H, d, J = 6.4Hz), 0.91(3H, t, J = 7.3Hz), 0.98-1.28(3H, m), 1.31(9H, s), 1.52(2H, sext,
 45 J = 7.3Hz), 2.48-2.53(2H, m), 2.75-2.95(3H, m), 3.15(1H, dd, J = 4.2Hz, 15.4Hz), 3.85-3.94(1H, m), 3.95(3H, s), 4.32-
 4.40(1H, m), 4.65-4.72(1H, m), 6.70(1H, d, J = 7.8Hz), 7.24 (1H, t, J = 7.4Hz), 7.31(1H, t, J = 7.4Hz), 7.50(1H, s), 7.72
 (1H, d, J = 7.4Hz), 8.04(1H, d, J = 7.4Hz), 8.18(1H, d, J = 8.3Hz), 8.33(1H, d, J = 7.0Hz)

Example 10

50 Compound 10

m.p.: 188-190 °C

IR(KBr, cm⁻¹):

55 3304, 2962, 1740, 1653, 1533, 1461, 1386, 1341, 1260, 1167, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₂₉H₄₀N₄O₈ + H)⁺):

Calcd: 573.2924

Found: 573.2919

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.65(6H,d,J = 6.1Hz),0.94-1.30(3H,m),1.32(9H,s),2.37-3.52(2H,m),2.87(1H,dd,J = 10.7Hz,15.4Hz),3.16-
(1H,dd, J = 4.2Hz,15.4Hz),3.82-3.90(1H,m),3.95(3H,s),4.24-4.31(1H,m),4.60-4.72(1H,m),5.05(1H,dd,J = 1.7Hz,
9.8Hz),5.12(1H,dd,J = 1.7Hz,17.7Hz),5.72-5.85(1H,m), 6.75(1H,d,J = 7.6Hz),7.24(1H,t,J = 7.3Hz),7.32(1H,t,
5 J = 7.3Hz),7.50(1H,s),7.70(1H,d,J = 7.3Hz),8.04(1H,d, J = 7.3Hz),8.16(1H,d,J = 8.7Hz),8.20(1H,d,J = 8.5Hz)

Example 11

Compound 11

10

m.p.: 99-126 °C(dec.)

IR(KBr,cm⁻¹):

3304,2962,1740,1656,1553,1461,1389,1260,1167,1092, 1047,765,747

High Resolution FAB-MS(m/e,(C₃₀H₄₄N₄O₈ + H)⁺):

15

Calcd: 589.3273

Found: 589.3243

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.66(3H,d,J = 6.4Hz),0.67(3H,d,J = 6.4Hz),0.70-1.73 (12H,m),1.31(9H,s),2.88(1H,dd,J = 10.7Hz,15.4Hz)-
,3.15 (1H,dd,J = 4.2Hz,15.4Hz),3.82-4.00(1H,m),3.95(3H,s), 4.00-4.12(1H,m),4.51-4.68(1H,m),6.69-
20 (1H,d,J = 7.6Hz), 7.23(1H,t,J = 7.9Hz),7.29(1H,t,J = 7.9Hz),7.48(1H,s), 7.69(1H,d,J = 7.9Hz),7.80-
(1H,d,J = 8.8Hz),8.03(1H,d, J = 7.9Hz),8.16(1H,d,J = 7.8Hz)

Example 12

25 Compound 12

m.p.: 185-189 °C

IR(KBr,cm⁻¹):

3310,2968,1740,1656,1539,1461,1386,1260,1170,1092, 1047,765,747

30 High Resolution FAB-MS(m/e,(C₃₀H₄₄N₄O₈ + H)⁺):

Calcd: 589.3237

Found: 589.3227

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.65(3H,d,J = 6.5Hz),0.68(3H,d,J = 6.5Hz),0.79(3H,d, J = 6.8Hz),0.84(3H,t,J = 7.3Hz),0.93-1.60(6H,m),1.31
35 (9H,s),2.92(1H,dd,J = 10.7Hz,15.4Hz),3.12(1H,dd, J = 4.2Hz,15.4Hz),3.85-4.02(1H,m),3.95(3H,s),4.06-4.17-
(1H,m),4.56-4.69(1H,m),6.66(1H,d,J = 8.6Hz),7.23 (1H,t,J = 7.3Hz),7.30(1H,t,J = 7.3Hz),7.50(1H,s),7.53
(1H,d,J = 8.5Hz),7.68(1H,d,J = 7.3Hz),8.03(1H,d, J = 7.3Hz),8.16(1H,d,J = 7.8Hz)

Example 13

40

Synthesis of Compound 13

To a solution of Boc-Leu-DTrp(COOMe)-OH (25 mg, prepared in Example 1-(1)) in dichloromethane (0.5 ml) were added D-norleucinol hydrochloride (12 mg), NMM (8.5 μl), HOBT·H₂O (8 mg) and EDCI·HCl (15
45 mg) successively under ice cooling. The mixture was stirred at 0-5 °C for 10 min and at room temperature for 1 h. The reaction mixture was diluted with ethyl acetate, washed with 4% aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo.

The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 10/1 for development to give Compound 13 (26 mg).

50 m.p.: 141-148 °C

IR(KBr,cm⁻¹):

3304,2962,1743,1647,1536,1461,1386,1371,1341,1311, 1260,1170,1092,762,747

High Resolution FAB-MS(m/e,(C₃₀H₄₆N₄O₇ + H)⁺):

Calcd: 575.3444

55

Found: 575.3481

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.69-0.95(9H,m),0.96-1.70(17H,m),2.88(1H,dd, J = 10.5Hz,14.2Hz),3.08-3.25(2H,m),3.60-3.76(1H,m), 3.76-
3.88(1H,m),3.95(3H,s),4.42-4.66(3H,m),6.80 (1H,d,J = 6.9Hz),7.24(1H,t,J = 7.4Hz),7.32(1H,t, J = 7.4Hz),7.48-

(1H,s),7.49(1H,d,J = 7.4Hz),7.67(1H,d, J = 7.4Hz),8.04(1H,d,J = 9.1Hz),8.18(1H,d,J = 8.4Hz)

Example 14

5 Synthesis of Compound 14

(1) Preparation of Boc-DTrp-DNle-OBzl

To a solution of H-DNle-OBzl•TsOH (1.27 g) in dichloromethane (20 ml) were added NMM (0.53 ml),
 10 Boc-DTrp-OH (1.22 g), HOBT•H₂O (734 mg) and EDCI•HCl (926 mg) successively under ice cooling. The mixture was stirred at the same temperature for 1 h and at room temperature for 2 h. The mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃, 10% aq. citric acid, and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was recrystallized from ethyl acetate and hexane to give Boc-DTrp-DNle-OBzl (1.56 g).
 15 FAB-MS(m/e, (C₂₉H₃₇N₃O₅ + H)⁺):508

(2) Preparation of Boc-DTrp(COOMe)-DNle-OBzl

To a solution of the dipeptide obtained in (1) (1.00 g) in dichloromethane (20 ml) were added pulverized
 20 NaOH (118 mg), TBAHS (134 mg) and methyl chloroformate (228 μ l) at room temperature under argon atmosphere. After being stirred at room temperature for 8 h, the reaction mixture was diluted with dichloromethane, washed with 10% aq. citric acid, sat. aq. NaHCO₃, water and brine successively, dried over MgSO₄ and evaporated in vacuo to give Boc-DTrp(COOMe)-DNle-OBzl (1.11 g).
 FAB-MS(m/e,(C₃₁H₃₉N₃O₇ + H)⁺):566

(3) Preparation of Compound 14

A mixture of Boc-DTrp(COOMe)-DNle-OBzl (83 mg) in TFA (2 ml) was stirred at room temperature for 1 h and evaporated in vacuo. To a solution of the residue in DMF (0.6 ml) were added NMM (18 μ l), Boc-Nle-OH (51 mg) and EDCI•HCl (43 mg) successively under ice cooling. The mixture was stirred at the same
 30 temperature for 1 h and at room temperature for 12 h. The mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by MPLC (Merck, LiChroprep Si 60) with chloroform/methanol = 500/6 for elution to give Boc-Nle-DTrp(COOMe)-DNle-OBzl (88 mg). A solution of the tripeptide derivative (22.4
 35 mg) in methanol was hydrogenated over 10% Pd/C (10 mg) at an atmospheric pressure of hydrogen for 2 h. After removal of the catalyst by filtration, the filtrate was concentrated in vacuo. The residue was purified by preparative TLC (3M, Empore Silica Sheet) with chloroform/methanol = 20/1 for development to give Compound 14 (14 mg).

m.p.: 152-159 °C

40 IR(KBr,cm⁻¹):

3322,2962,2872,1725,1656,1539,1461,1386,1371,1341, 1311,1260,1167,1092,762,747

High Resolution FAB-MS(m/e,(C₃₀H₄₄N₄O₈ + H)⁺):

Calcd: 589.3237

Found: 589.3248

45 ¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.64(3H,d,J = 7.2Hz),0.78-1.82(12H,m),0.85(3H,t, J = 6.9Hz),1.32(9H,s),2.87(1H,dd,J = 10.4Hz,14.8Hz),
 3.10-3.30(1H,m),3.76-3.90(1H,m),3.95(3H,s),4.45-4.60(1H,m),4.60-4.72(1H,m),6.68(1H,d,J = 6.8Hz),7.24
 (1H,t,J = 7.4Hz),7.31(1H,t,J = 7.4Hz),7.51(1H,s),7.71 (1H,d,J = 7.4Hz),8.02(1H,d,J = 7.4Hz),8.03(1H,d,
 J = 8.1Hz),8.18(1H,d,J = 7.8Hz)

Example 15

Synthesis of Compound 15

55 Compound 15 was prepared using Boc- γ MeLeu instead of Boc-Nle in the same manner described in Example 14-(3).

m.p.: 118-125 °C

IR(KBr,cm⁻¹):

3328,2962,2872,1720,1698,1660,1539,1494,1461,1395, 1371,1341,1257,1197,1170,1092,1053,765,750
High Resolution FAB-MS(m/e,(C₃₁H₄₆N₄O₈ + H)⁺):

Calcd: 603.3394

Found: 603.3412

5 ¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.71(3H,t,J = 7.7Hz),0.88(9H,s),1.00-1.98(8H,m),1.31 (9H,s),2.90(1H,dd,J = 10.5Hz,14.8Hz),3.11(1H,dd,
J = 3.9Hz,14.8Hz),3.60-3.85(1H,m),3.94(3H,s),4.35-4.68(2H,m),6.73(1H,d,J = 8.3Hz),7.24(1H,t,J = 7.5Hz), 7.31-
(1H,t,J = 7.5Hz),7.52(1H,s),7.72(1H,d,J = 7.5Hz), 8.03(1H,d,J = 7.5Hz),8.12(1H,d,J = 7.4Hz),8.23(1H,d,
J = 7.6Hz)

10

Example 16

Synthesis of Compound 16

15 (1) Preparation of Boc-DNal-DNle-OBzl

Boc-DNal-DNle-OBzl was prepared using Boc-DNal-OH instead of Boc-DTrp-OH in the same manner described in Example 14-(1).

FAB-MS(m/e,(C₃₁H₃₈N₂O₅ + H)⁺):519

20

(2) Preparation of H-DNal-DNle-OBzl

To a solution of Boc-DNal-DNle-OBzl (233 mg) in dichloromethane (2 ml) was added TFA (5 ml) and the mixture was stirred at room temperature for 13 h. The mixture was concentrated in vacuo. The residue
25 was dissolved in dichloromethane, washed with sat. aq. NaHCO₃, water and brine successively, dried over MgSO₄ and evaporated in vacuo to give the product (185 mg).

FAB-MS(m/e,(C₂₆H₃₀N₂O₃ + H)⁺):419

(3) Preparation of Compound 16

30

To a solution of H-DNal-DNle-OBzl (180 mg, prepared in (2)) in dichloromethane (5 ml) were added Boc-Leu-OH·H₂O (180 mg), HOBT·H₂O (70 mg) and EDCI·HCl (99 mg) successively under ice cooling. After being stirred at room temperature for 14 h, the mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo.
35 The residue was purified by dry column flash chromatography (Merck, Kieselgel 60) with ethyl acetate/hexane = 1/2 for elution to give Boc-Leu-DNal-DNle-OBzl (239 mg). A solution of the tripeptide derivative (87 mg) in 95% ethanol (10 ml) was hydrogenated over 10% Pd/C (21 mg) at an atmospheric pressure of hydrogen for 1 h. After removal of the catalyst by filtration, the filtrate was concentrated in vacuo to give Compound 16 (74 mg).

40

m.p.: 88-90 °C

IR(KBr,cm⁻¹):

3304,3070,2962,2872,1725,1698,1656,1536,1461,1398, 1371,1251,1167,777

High Resolution FAB-MS(m/e,(C₃₀H₄₃N₃O₆ + H)⁺):

Calcd: 542.3230

45

Found: 542.3223

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.77-0.97(9H,m),1.12-1.89(9H,m),1.39 + 1.45(9H,s×2),3.48-3.66(2H,m),3.92-3.99(1H,m),4.33-4.40(1H,m),
4.82-4.90(1H,m),5.07(1H,brs),6.81(1H,brd,J = 5.4Hz), 6.85(1H,brs),7.32-7.38(2H,m),7.46(1H,dd,J = 8.0Hz,
7.0Hz),7.53(1H,dd,J = 8.0Hz,7.0Hz),7.69-7.75(1H,m), 7.82(1H,d,J = 8.0Hz),8.16(1H,d,J = 8.0Hz)

50

Example 17

Synthesis of Compound 17

55 Compound 17 was prepared using DLTrp(Me) instead of DNal in the same manner described in Example 16.

m.p.: 91-98 °C

IR(KBr,cm⁻¹):

3322,2962,1653,1530,1476,1371,1251,1167,738

High Resolution FAB-MS(m/e, (C₂₉H₄₄N₄O₆ + H)⁺):

Calcd: 545.3339

Found: 545.3311

5 ¹H-NMR(300MHz,DMSO-d₆,70 ° C, δ ppm):

0.75(3H,d,J = 6.3Hz),0.78-0.89(6H,m),1.09-1.72(9H,m), 1.34 + 1.36(9H,s×2),2.81-3.10(2H,m),3.69 + 3.70-(3H, s×2),3.88-3.94(1H,m),4.10-4.22(1H,m),4.55-4.61 (1H,m),6.45 + 6.60(1H,brs×2),6.98(1H,t,J = 8.1Hz),7.02 + 7.05(1H,s×2),7.10(1H,t,J = 8.1Hz),7.32(1H,d, J = 8.1Hz),7.53-7.78(2H,m),7.54 + 7.57(1H,d×2,J = 8.1Hz)

10 Example 18

Synthesis of Compound 18

(1) Preparation of H-DBal-DNle-O^tBu

15

To a solution of Boc-DBal-OH (80 mg) in dichloromethane (10 ml) were added H-DNle-O^tBu (67 mg), HOBT•H₂O (41 mg) and EDCI•HCl (58 mg) successively at room temperature. After being stirred at room temperature for 6 h, the mixture was diluted with dichloromethane, washed with sat. aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was

20

purified by silica gel flash chromatography (Merck, Kieselgel 60) with chloroform for elution to give an oily product (123 mg), which was dissolved in formic acid. The mixture was stirred at room temperature for 2 h. The mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃ and brine, dried over MgSO₄ and evaporated in vacuo to give H-DBal-DNle-O^tBu (97 mg).

25

(2) Preparation of (2-chlorophenyl)carbamoyl-Leu-OH

(2-Chlorophenyl)carbamoyl-Leu-OH was prepared using H-Leu-OBzl•TsOH as a starting material in the same manner described in Example 4.

30 FAB-MS(m/e, (C₁₃H₁₇ClN₂O₃ + H)⁺):285

(3) Preparation of Compound 18

To a solution of H-DBal-DNle-O^tBu (98 mg, prepared in (1)) in DMF (10 ml) were added (2-chlorophenyl)carbamoyl-Leu-OH (85 mg, prepared in (2)), HOBT•H₂O (46 mg) and EDCI•HCl (62 mg) successively under ice cooling. After being stirred at room temperature for 11 h, the reaction mixture was concentrated under reduced pressure. The residue was dissolved in chloroform and the solution was washed with sat. aq. NaHCO₃, 10% aq. citric acid and brine successively, dried over MgSO₄ and evaporated to give a corresponding tripeptide derivative (148 mg). The tripeptide (102 mg) was dissolved in TFA (2 ml).

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The solution was stirred at room temperature for 2 h and evaporated in vacuo. The residue was purified by reprecipitation from methanol and water to give Compound 18 (81 mg).

m.p.: 182-183 ° C

IR(KBr,cm⁻¹):

3304,3076,2962,1716,1644,1590,1551,1443,750,729

45 High Resolution FAB-MS(m/e, (C₃₀H₃₇ClN₄O₅S + H)⁺):

Calcd: 601.2252

Found: 601.2255

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.77-0.82(9H,m),0.88-1.65(9H,m),3.32(1H,dd,J = 7.8Hz, 14.5Hz),3.55(1H,dd,J = 5.5Hz,14.5Hz),4.06-4.15-(2H,m), 4.72-4.80(1H,m),6.80(1H,brs),6.88-6.97(2H,m),7.10-7.42(6H,m),7.56(1H,brs),7.78-7.88(2H,m),7.94-8.00 (1H,m)

50

Example 19

Synthesis of Compound 19

Compound 19 was prepared using Boc-DLPhe(3-COOEt)-OH as a starting material in the same manner described in Example 18.

IR(KBr,cm⁻¹):

3322,2962,1725,1647,1542,1287,1203,750

High Resolution FAB-MS(m/e,(C₃₁H₄₁ClN₄O₇ + H)⁺):

Calcd: 617.2742

Found: 617.2720

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.69(3H,t,J = 6.6Hz),0.79-0.88(6H,m),1.00-1.80(9H,m), 1.30 + 1.33(3H,t×2,J = 7.0Hz,J = 7.0Hz),2.78 + 2.89-
(1H, dd×2,J = 11.7Hz,13.5Hz,J = 9.0Hz,13.5Hz),3.07 + 3.18 (1H,dd×2,J = 5.5Hz,13.5Hz,J = 3.2Hz,13.5Hz),4.11-
4.22 (2H,m),4.28 + 4.29(2H,q×2,J = 7.0Hz,J = 7.0Hz),4.56-4.68 (1H,m),6.90-6.97(1H,m),7.16-7.24(2H,m),7.31-
7.41 (2H,m),7.51 + 7.58(1H,d×2,J = 7.8Hz,J = 7.8Hz),7.75 + 7.79 (1H,d×2,J = 7.8Hz,J = 7.8Hz),7.86 + 7.94-
(1H,s×2),8.06-8.24(3H,m),8.19 + 8.52(1H,d×2,J = 9.0Hz,J = 9.0Hz)

Example 20

15 Synthesis of Compound 20

Compound 20 was prepared using Boc-DLPhe(3-COOMe)-OH as a starting material in the same manner described in Example 18.

IR(KBr,cm⁻¹):

3322,2962,1728,1647,1545,1446,1290,1209,750

High Resolution FAB-MS(m/e,(C₃₀H₃₉ClN₄O₇ + H)⁺):

Calcd: 603.2585

Found: 603.2598

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.69(3H,t,J = 6.9Hz),0.80-0.93(6H,m),1.00-1.80(9H,m), 2.78 + 2.89-
(1H,dd×2,J = 11.7Hz,13.5Hz,J = 9.0Hz,13.5Hz), 3.07 + 3.18(1H,dd×2,J = 5.5Hz,13.5Hz,J = 3.2Hz,13.5Hz),
3.81 + 3.84(3H,s×2),4.10-4.22(2H,m),4.57-4.68 (1H,m),6.90-6.97(1H,m),7.16-7.24(2H,m),7.31-7.41 (2H,m)-
,7.51 + 7.58(1H,d×2,J = 7.8Hz,J = 7.8Hz),7.75 + 7.79 (1H,d×2,J = 7.8Hz,J = 7.8Hz),7.87 + 7.95(1H,s×2),8.08-
8.26(3H,m),8.22 + 8.52(1H,d×2,J = 8.8Hz,J = 8.8Hz)

Example 21

Synthesis of Compound 21

35 (1) Preparation of H-DTrp(COOMe)-DNle-O^tBu•HCl

To a suspension of H-DNle-O^tBu•HCl (603 mg) in dichloromethane (10 ml) were added NMM (297 μl), Boc-DTrp-OH (903 mg), HOBT•H₂O (413 mg) and EDCI•HCl (569 mg) successively under ice cooling. The mixture was stirred at the same temperature for 2 h and at room temperature for 1 h. The reaction mixture was diluted with ethyl acetate, washed with 10% aq. citric acid, sat. aq. NaHCO₃, water and brine successively, dried over MgSO₄ and evaporated in vacuo to give Boc-DTrp-DNle-O^tBu (1.22 g). To a solution of the dipeptide derivative in dichloromethane (12 ml) were added methyl chloroformate (1 ml), pulverized NaOH (103 mg) and TBAHS (20 mg). After being stirred of the mixture at room temperature for 30 min, pulverized NaOH (103 mg) was added. This procedure was repeated twice more. The reaction mixture was diluted with dichloromethane, washed with sat. aq. NaHCO₃, 10% aq.citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was recrystallized from ethyl acetate and hexane to give Boc-DTrp(COOMe)-DNle-O^tBu (1.15 g). The dipeptide derivative (1.03 g) was dissolved in TFA (10 ml) and the solution was stirred under ice cooling for 10 min. After evaporation of the reaction mixture, the residue was dissolved in ethyl acetate. The solution was washed with sat. aq NaHCO₃, water and brine successively, dried over MgSO₄ and concentrated under reduced pressure to give H-DTrp-(COOMe)-DNle-O^tBu as a free base. To a solution of the free base (750 mg) in methanol was added 3.78N - hydrogen chloride / 1,4-dioxane (0.47 ml) under ice cooling. The solution was evaporated in vacuo to give H-DTrp(COOMe)-DNle-O^tBu•HCl (810 mg).

FAB-MS(m/e, (C₂₃H₃₃N₃O₅ + H)⁺):432

(2) Preparation of H-Leu-DTrp(COOMe)-DNle-O^tBu

To a solution of H-DTrp(COOMe)-DNle-O^tBu·HCl (60 mg, prepared in (1)) in DMF (2 ml) were added Z-Leu-OH (45 mg), HOBT·H₂O (33 mg) and EDCI·HCl (41 mg) under ice cooling. After being stirred at the same temperature for 1 h, the mixture was diluted with ethyl acetate, washed with 10% aq. citric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel flash chromatography (Merck, Kieselgel 60) with ethyl acetate/hexane = 1/2 for elution to give an oily product (72 mg), which was dissolved in methanol (3 ml). The solution was hydrogenated over 10% Pd/C at an atmospheric pressure of hydrogen for 1 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo to give H-Leu-DTrp(COOMe)-DNle-O^tBu (48 mg).

(3) Preparation of Compound 21

To a solution of H-Leu-DTrp(COOMe)-DNle-O^tBu (15 mg, prepared in (2)) in chloroform (1 ml) was added 2,6-dichlorophenylisocyanate (5.6 mg) at 0-5 °C under argon atmosphere. The mixture was stirred at the same temperature for 1.5 h, and diluted with chloroform. The solution was washed with 1N-hydrochloric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 30/1 for development to give an oily product (6.7 mg), which was dissolved in TFA (1 ml). After being stirred at room temperature for 1 h, the mixture was concentrated under reduced pressure. The residue was triturated with water to give Compound 21 (4.3 mg).

m.p.: 211-212 °C

IR(KBr,cm⁻¹):

3310,2962,1740,1647,1545,1461,1386,1260,1092,765

High Resolution FAB-MS(m/e,(C₃₂H₃₉Cl₂N₅O₇ + H)⁺):

Calcd: 676.2305

Found: 676.2297

¹H-NMR(300MHz,Acetone-d₆, δ ppm):

0.78(3H,d,J = 6.1Hz),0.79(3H,d,J = 6.1Hz),0.80(3H,t, J = 8.4Hz),1.10-1.76(9H,m),3.08(1H,dd,J = 9.6Hz, 15.1Hz),3.37(1H,dd,J = 4.7Hz,15.1Hz),4.00(3H,s),4.14-4.31(2H,m),4.77-4.91(1H,m),6.32(1H,d,J = 7.1Hz),7.23 (1H,t,J = 7.8Hz),7.28(1H,t,J = 7.7Hz),7.33(1H,t, J = 7.7Hz),7.40(2H,d,J = 7.8Hz),7.58(1H,s),7.60(1H,s), 7.66- (1H,d,J = 7.6Hz),7.70(1H,d,J = 7.3Hz),7.70(1H,d, J = 7.7Hz),8.14(1H,d,J = 7.7Hz)

Each Compound 22-24 in the following Examples 22-24 was prepared using corresponding isocyanate in the same manner described in Example 21-(3).

Example 22

Compound 22

m.p.: 208-209 °C

IR(KBr,cm⁻¹):

3352,2962,2872,1737,1647,1551,1461,1389,1260,1095, 750

High Resolution FAB-MS(m/e,(C₃₂H₄₀FN₅O₇ + H)⁺):

Calcd: 626.2990

Found: 626.2984

¹H-NMR(300MHz,Acetone-d₆, δ ppm):

0.78(3H,d,J = 6.3Hz),0.80(3H,d,J = 6.3Hz),0.86(3H,t, J = 6.9Hz),1.20-1.50(7H,m),1.65-1.90(2H,m),3.12(1H, dd,J = 9.6Hz,15.0Hz),3.38(1H,dd,J = 4.7Hz,15.0Hz),4.01 (3H,s),4.19-4.29(1H,m),4.34-4.46(1H,m),4.79-4.90 (1H,m),6.46(1H,d,J = 7.1Hz),6.88-6.99(1H,m),7.00-7.14 (2H,m),7.25(1H,t,J = 7.8Hz),7.32(1H,t,J = 7.8Hz),7.61 (1H,s) 7.67(1H,d,J = 8.0Hz),7.70(1H,d,J = 7.8Hz),7.81 (1H,d,J = 8.2Hz),7.91(1H,d,J = 2.1Hz),8.14(1H,d, J = 7.8Hz),8.17-8.27(1H,m)

Example 23

Compound 23

m.p.: 212-224 °C

IR(KBr,cm⁻¹):

3322,3088,2962,2872,1740,1647,1554,1461,1389,1326, 1260,1122,762

High Resolution FAB-MS(m/e,(C₃₃H₄₀F₃N₅O₇ + H)⁺):

Calcd: 676.2958

Found: 676.2957

5 ¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.64(3H,d,J = 6.1Hz),0.65(3H,d,J = 6.1Hz),0.81(3H,t, J = 7.1Hz),0.78-1.82(9H,m),2.88(1H,dd,J = 11.2Hz, 14.7Hz),3.10-3.25(1H,m),3.95(3H,s),4.10-4.28(2H,m), 4.62-4.76(1H,m),7.10-7.24(2H,m),7.26(1H,t,J = 7.6Hz), 7.33(1H,t,J = 7.6Hz),7.54(1H,s),7.43-7.63(2H,m),7.76 (1H,d,J = 7.6Hz),7.88(1H,d,J = 7.9Hz),7.88(1H,s),8.05 (1H,d,J = 7.6Hz),8.22(1H,d,J = 7.9Hz),8.49(1H,d, J = 8.4Hz)

10

Example 24

Compound 24

15 m.p.: 186-188 °C

IR(KBr,cm⁻¹):

3346,2962,1740,1653,1506,1458,1386,1344,1260,1092, 741

High Resolution FAB-MS(m/e,(C₃₂H₄₀N₆O₉ + H)⁺):

Calcd: 653.2935

20 Found: 653.2939

¹H-NMR(300MHz,Acetone-d₆, δ ppm):

0.79(3H,d,J = 6.4Hz),0.81(3H,d,J = 6.4Hz),0.83(3H,t, J = 7.2Hz),1.14-1.50(7H,m),1.65-1.90(2H,m),3.11 (1H,dd,J = 9.6Hz,14.8Hz),3.39(1H,dd,J = 4.2Hz,14.8Hz), 4.01(3H,s),4.17-4.29(1H,m),4.37-4.49(1H,m),4.79- 4.90(1H,m),7.12(1H,dt,J = 1.4Hz,7.2Hz),7.25(1H,t, J = 7.6Hz),7.31(1H,t,J = 7.6Hz),7.20-7.37(1H,m),7.55-7.67- (2H,m),7.60(1H,s),7.69(1H,d,J = 7.6Hz),7.78(1H,d, J = 9.1Hz),8.12(1H,d,J = 7.6Hz),8.13(1H,d,J = 8.2Hz),8.54 (1H,dd,J = 1.4Hz,7.2Hz),9.56(1H,s)

25

Example 25

30 Synthesis of Compound 25

A solution of Compound 24 (30 mg, obtained in Example 24) in 95% ethanol (3 ml) was hydrogenated over 10% Pd/C (20 mg) at an atmospheric pressure of hydrogen for 1.5 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo to give Compound 25 (26 mg).

35 m.p.: 188-190 °C

IR(KBr,cm⁻¹):

3370,2962,1740,1650,1551,1461,1386,1260,747

High Resolution FAB-MS(m/e,(C₃₂H₄₂N₆O₇ + H)⁺):

Calcd: 623.3193

40 Found: 623.3157

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.64(3H,d,J = 6.5Hz),0.66(3H,d,J = 6.5Hz),0.83(3H,t, J = 7.0Hz),0.78-1.40(7H,m),1.56-1.78(2H,m),2.88 (1H,dd,J = 11.8Hz,14.9Hz),3.15(1H,dd,J = 2.8Hz,14.9Hz), 3.95(3H,s),4.09-4.21(2H,m),4.60-4.73(1H,m),6.28 (1H,d,J = 8.1Hz),6.47(1H,dt,J = 1.7Hz,7.7Hz),6.64 (1H,dd,J = 1.7Hz,7.7Hz),6.74(1H,dt,J = 1.7Hz,7.7Hz), 7.26- (1H,t,J = 7.6Hz),7.27(1H,dd,J = 1.7Hz,7.7Hz),7.33 (1H,t,J = 7.6Hz),7.54(1H,s),7.59(1H,s),7.76(1H,d, J = 7.6Hz)- ,8.05(1H,d,J = 7.6Hz),8.23(1H,d,J = 7.3Hz),8.46 (1H,d,J = 8.6Hz)

45

Example 26

50 Synthesis of Compound 26

To a solution of Compound 25 (15 mg, obtained in Example 25) were added TEA (21 μl) and formic pivalic anhydride (15 μl) at 0-5 °C under argon atmosphere. The mixture was stirred at the same temperature for 5 h.

55 Water was added to the reaction mixture. The mixture was acidified with 1N-hydrochloric acid and extracted with ethyl acetate. The ethyl acetate extract was dried over anhydrous sodium sulfate and evaporated in vacuo. The residue was purified by preparative TLC (3M, Empore Silica Sheet) with chloroform/methanol/acetic acid = 20/1/1 for development and the product was triturated with methanol and

water to give Compound 26 (11 mg).

m.p.: 174-178 °C

IR(KBr, cm⁻¹):

3310, 2962, 1743, 1665, 1554, 1461, 1389, 1260, 1092, 747

5 High Resolution FAB-MS(m/e, (C₃₃H₄₂N₆O₈ + H)⁺):

Calcd: 651.3143

Found: 651.3134

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

10 0.67(3H, d, J = 6.1 Hz), 0.74(3H, d, J = 6.1 Hz), 0.70(3H, t, J = 7.2 Hz), 1.00-2.00(9H, m), 2.81-2.99(1H, m), 3.07-3.20
(1H, m), 3.87(3H, s), 4.20-4.48(2H, m), 4.60-4.90(1H, m), 6.87(1H, dt, J = 1.4 Hz, 7.1 Hz), 7.00(1H, dt, J = 1.4 Hz, 7.1 Hz)-
, 7.05-7.34(2H, m), 7.16(1H, t, J = 7.7 Hz), 7.28 (1H, t, J = 7.7 Hz), 7.48(1H, s), 7.45-7.59(2H, m), 7.61 (1H, d, J = 7.7 Hz)-
, 7.78(1H, brs), 8.03(1H, d, J = 7.7 Hz), 8.47 (1H, brs), 8.78(1H, d, J = 8.3 Hz), 11.77(1H, brs)

Example 27

15

Synthesis of Compound 27

(1) Preparation of PhOCO-Leu-DTrp(COOMe)-DNle-O^tBu

20 To a solution of H-Leu-DTrp(COOMe)-DNle-O^tBu (320 mg, prepared in Example 21-(2)) in pyridine (2.3 ml) was added phenyl chloroformate at 0-5 °C under argon atmosphere. The mixture was stirred at the same temperature for 1.5 h and concentrated in vacuo. The residue was dissolved in ethyl acetate and the solution was washed with 10% aq. citric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel flash chromatography (Merck, Kieselgel 60)
25 with ethyl acetate/hexane = 2/1 for elution to give the product (337 mg).

FAB-MS(m/e, (C₃₆H₄₈N₄O₈ + H)⁺): 665

(2) Preparation of Compound 27

30 A solution of PhOCO-Leu-DTrp(COOMe)-DNle-O^tBu (10 mg, prepared in (1)) in TFA (1 ml) was stirred at room temperature for 1 h. The mixture was evaporated in vacuo and the residue was triturated with water to give Compound 27 (5.8 mg).

m.p.: 198-199 °C

IR(KBr, cm⁻¹):

35 3316, 2962, 2872, 1740, 1653, 1536, 1461, 1386, 1260, 1209, 1092, 762

High Resolution FAB-MS(m/e, (C₃₂H₄₀N₄O₈ + H)⁺):

Calcd: 609.2924

Found: 609.2899

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

40 0.68(3H, d, J = 6.3 Hz), 0.69(3H, d, J = 6.3 Hz), 0.81(3H, t, J = 7.1 Hz), 1.00-1.75(9H, m), 2.88(1H, dd, J = 11.0 Hz, 14.4 Hz), 3.14(1H, dd, J = 5.6 Hz, 14.4 Hz), 3.94(3H, s), 3.96-4.20(2H, m), 4.63-4.75(1H, m), 7.01(2H, dd, J = 1.3 Hz, 8.3 Hz), 7.16(1H, dt, J = 1.3 Hz, 8.3 Hz), 7.25(1H, t, J = 7.7 Hz), 7.27-7.37(3H, m), 7.52(1H, s), 7.75(1H, d, J = 7.7 Hz), 7.83-
(1H, d, J = 7.8 Hz), 8.05(1H, d, J = 7.8 Hz), 8.14(1H, d, J = 7.7 Hz), 8.41(1H, d, J = 8.8 Hz)

45 Example 28

Synthesis of Compound 28

50 To a solution of PhOCO-Leu-DTrp(COOMe)-DNle-O^tBu (33 mg, prepared in Example 27-(1)) in chloroform were added hexamethyleneimine (56 μl) and TEA (100 μl) at room temperature under argon atmosphere. The mixture was stirred at 55 °C for 2 h and cooled to room temperature. The mixture was diluted with ethyl acetate, washed with 1N-hydrochloric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with ethyl acetate/hexane = 1/1 for development to give an oily product (28 mg), which was dissolved
55 in TFA (1 ml). After being stirred at room temperature for 1 h, the mixture was evaporated in vacuo. The residue was triturated with water to give Compound 28 (22 mg).

m.p.: 94-97 °C

IR(KBr, cm⁻¹):

3352,2938,1743,1635,1536,1461,1386,1260,1221,1092, 765,747

High Resolution FAB-MS(m/e , $(C_{32}H_{47}N_5O_7 + H)^+$):

Calcd: 614.3554

Found: 614.3532

5 1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.79(3H,d,J = 6.1Hz), 0.80(3H,d,J = 6.1Hz), 0.83(3H,t, J = 6.8Hz), 1.10-1.93(17H,m), 3.10-3.45(6H,m), 3.88-3.98 (1H,m), 4.02(3H,s), 4.35-4.45(1H,m), 4.77-4.85(1H,m), 5.21(1H,brs), 6.74(1H,d,J = 8.0Hz), 7.25-(1H,t,J = 7.6Hz), 7.34(1H,t,J = 7.6Hz), 7.49(1H,s), 7.58(1H,d,J = 7.6Hz), 7.66(1H,d,J = 8.4Hz), 8.16-(1H,d,J = 7.6Hz)

10 Each Compound 29-32 in the following Examples 29-32 was prepared using each corresponding amine as a starting material in the same manner described in Example 28.

Example 29

15 Compound 29

m.p.: 106-110 °C

IR(KBr, cm^{-1}):

3328,2962,1743,1635,1539,1461,1386,1257,1092,765, 747

20 High Resolution FAB-MS(m/e , $(C_{30}H_{43}N_5O_7S + H)^+$):

Calcd: 618.2961

Found: 618.2987

1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.80(3H,d,J = 5.5Hz), 0.81(3H,d,J = 5.5Hz), 0.83(3H,t, J = 7.1Hz), 1.08-1.90(9H,m), 2.49-2.64(4H,m), 3.24(1H, dd,J = 7.3Hz, 15.1Hz), 3.31(1H,dd,J = 6.1Hz, 15.1Hz), 3.55-3.73(4H,m), 3.88-3.98(1H,m), 4.03(3H,s), 4.27-4.37 (1H,m), 4.70-4.80(1H,m), 5.73(1H,brs), 6.68(1H,d, J = 8.1Hz), 7.26(1H,t,J = 7.8Hz), 7.35(1H,t,J = 7.8Hz), 7.49-(1H,s), 7.50(1H,d,J = 6.6Hz), 7.57(1H,d,J = 7.8Hz), 8.16(1H,d,J = 7.8Hz)

Example 30

30

Compound 30

m.p.: 104-109 °C

IR(KBr, cm^{-1}):

3322,2962,1743,1644,1536,1461,1386,1260,1092,765, 747

35 High Resolution FAB-MS(m/e , $(C_{30}H_{43}N_5O_7 + H)^+$):

Calcd: 586.3241

Found: 586.3264

1H -NMR(300MHz, $CDCl_3$, δ ppm):

40 0.79(3H,d,J = 6.0Hz), 0.81(3H,d,J = 6.0Hz), 0.83(3H,t, J = 7.1Hz), 1.10-1.98(13H,m), 3.13-3.40(6H,m), 3.87-3.98 (1H,m), 4.02(3H,s), 4.39-4.48(1H,m), 4.78-4.85(1H,m), 5.14(1H,d,J = 5.9Hz), 6.69(1H,d,J = 8.0Hz), 7.25(1H,t, J = 7.4Hz), 7.34(1H,t,J = 7.4Hz), 7.49(1H,s), 7.59(1H,d, J = 7.4Hz), 7.74(1H,d,J = 7.9Hz), 8.16(1H,d,J = 7.4Hz)

Example 31

45

Compound 31

m.p.: 96-101 °C

IR(KBr, cm^{-1}):

50 3322,2956,2866,1743,1638,1539,1461,1386,1260,1092, 765,747

High Resolution FAB-MS(m/e , $(C_{31}H_{45}N_5O_7 + H)^+$):

Calcd: 600.3397

Found: 600.3385

1H -NMR(300MHz, $CDCl_3$, δ ppm):

55 0.77(3H,d,J = 6.3Hz), 0.79(3H,d,J = 6.3Hz), 0.83(3H,t, J = 7.1Hz), 1.10-1.92(15H,m), 3.15-3.37(6H,m), 3.86-3.98 (1H,m), 4.02(3H,s), 4.31-4.40(1H,m), 4.73-4.80(1H,m), 5.52(1H,brs), 6.80(1H,d,J = 6.4Hz), 7.25-(1H,t,J = 7.6Hz), 7.34(1H,t,J = 7.6Hz), 7.50(1H,s), 7.58(1H,d,J = 7.6Hz), 7.62(1H,d,J = 7.2Hz), 8.16(1H,d,J = 7.6Hz)

Example 32

Compound 32

5 m.p.: 104-108 °C
 IR(KBr, cm⁻¹):
 3328, 2962, 1743, 1647, 1539, 1461, 1386, 1260, 1092, 765, 747
 High Resolution FAB-MS(m/e, (C₂₉H₄₁N₅O₇S + H)⁺):
 Calcd: 604.2805
 10 Found: 604.2799
¹H-NMR(300MHz, CDCl₃, δ ppm):
 0.75-0.86(9H, m), 1.05-1.92(9H, m), 2.90-3.00(2H, m), 3.28(2H, d, J = 7.4 Hz), 3.39-3.48(1H, m), 3.62-3.70(1H, m),
 3.85-3.95(1H, m), 4.03(3H, s), 4.17(1H, d, J = 9.1 Hz), 4.43 (1H, d, J = 9.1 Hz), 4.48-4.56(1H, m), 4.73-4.82(1H, m), 5.93
 (1H, brs), 6.57(1H, d, J = 8.5 Hz), 7.26(1H, t, J = 7.8 Hz), 7.35 (1H, t, J = 7.8 Hz), 7.49(1H, s), 7.58(1H, d, J = 7.8 Hz), 7.59
 15 (1H, d, J = 11.9 Hz), 8.16(1H, d, J = 7.8 Hz)

Example 33

Synthesis of Compound 33

20

(1) Preparation of PhOCO-Leu-OBzl

To a solution of H-Leu-OBzl•TsOH (1.01 g) in pyridine (9 ml) was added phenyl chloroformate (386 μl) at 0-5 °C under argon atmosphere. The mixture was stirred at the same temperature for 2 h and concentrated in vacuo. The residue was dissolved in ethyl acetate and the solution was washed with 1N-hydrochloric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with ethyl acetate/hexane = 1/10 for elution to give the product (828 mg).
 FAB-MS(m/e, (C₂₀H₂₃NO₄ + H)⁺): 342

30

(2) Preparation of ⁱPr₂NCO-Leu-OBzl

To a solution of PhOCO-Leu-OBzl (70 mg, prepared in (1)) in chloroform (2 ml) were added diisopropylamine (575 μl) and TEA (286 μl) at room temperature under argon atmosphere. The mixture was stirred at 55 °C for 17 h and cooled to room temperature. The mixture was diluted with ethyl acetate, washed with 1N-hydrochloric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with ethyl acetate/hexane = 1/4 for development to give the product (32 mg).
 FAB-MS(m/e, (C₂₀H₃₂N₂O₃ + H)⁺): 349

40

(3) Preparation of Compound 33

A mixture of ⁱPr₂NCO-Leu-OBzl (32 mg, obtained in (2)) and 10% Pd/C (10 mg) in methanol (1 ml) was stirred at room temperature under an atmospheric pressure of hydrogen for 16 h. The catalyst was filtered off and the filtrate was concentrated under reduced pressure. The residue was dissolved in dichloromethane (1.5 ml). To the solution were added H-DTrp(COOMe)-DNle-O^tBu•HCl (40 mg, prepared in Example 21-(1)), NMM (10 μl), HOBT•H₂O (13 mg) and EDCI•HCl (18 mg) under ice cooling.

After being stirred at room temperature for 1.5 h, the mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃, 10% aq. citric acid, water and brine successively, dried over MgSO₄ and evaporated in vacuo.

The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with ethyl acetate/hexane = 1/1 for development to give the product (52 mg), which was dissolved in TFA (1 ml). After being stirred at room temperature for 1 h, the mixture was evaporated in vacuo. The residue was triturated with water to give Compound 33 (18 mg).

55 m.p.: 94-96 °C

IR(KBr, cm⁻¹):

3322, 2962, 2872, 1743, 1650, 1524, 1461, 1386, 1341, 1311, 1260, 1221, 1149, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₃₂H₄₉O₇N₅ + H)⁺):

Calcd: 616,3718

Found: 616,3699

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.66(3H,d,J = 6.0Hz),0.67(3H,d,J = 6.0Hz),0.85(3H,t, J = 7.0Hz),1.08 + 1.10(12H,d×2,J = 6.8Hz),1.10-1.40
 5 (7H,m),1.15-1.30(2H,m),2.86(1H,dd,J = 11.0Hz,14.8Hz), 3.18(1H,dd,J = 3.5Hz,14.8Hz),3.70(2H,sept,J = 6.8Hz),
 3.95(3H,s),3.95-4.07(1H,m),4.10-4.20(1H,m),4.60-4.70(1H,m),5.56(1H,d,J = 7.3Hz),7.25(1H,t,J = 7.6Hz), 7.32-
 (1H,t,J = 7.6Hz),7.49(1H,s),7.70(1H,d,J = 7.6Hz), 8.04(1H,d,J = 7.6Hz),8.15(1H,d,J = 8.1Hz),8.18(1H,d,
 J = 9.3Hz)

Each Compound 34 and 35 in the following Examples 34 and 35 was prepared using corresponding
 10 amine in the same manner described in Example 33-(2) and (3).

Example 34

Compound 34

15 m.p.: 85-94 °C

IR(KBr,cm⁻¹):

3328,2962,2872,1743,1638,1533,1461,1383,1344,1311, 1260,1224,1200,1092,765,747

High Resolution FAB-MS(m/e,(C₃₂H₄₇N₅O₇ + H)⁺):

20 Calcd: 614.3554

Found: 614.3569

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.66(3H,d,J = 6.0Hz),0.68(3H,d,J = 6.0Hz),0.86(3H,t, J = 7.0Hz),1.06(3H,d,J = 6.1Hz),1.09(3H,d,J = 6.1Hz),
 1.10-1.40(7H,m),1.40-1.60(2H,m),1.65-1.78(2H,m), 1.80-1.94(2H,m),2.86(1H,dd,J = 4.0Hz,10.7Hz),3.12-3.30-
 25 (1H,m),3.78(2H,sext,6.1Hz),3.96(3H,s),4.08-4.20(2H,m),4.50-4.68(1H,m),5.65(1H,d,J = 6.8Hz), 7.27-
 (1H,t,J = 7.4Hz),7.32(1H,t,J = 7.4Hz),7.49(1H,s), 7.68(1H,d,J = 7.4Hz),8.05(1H,d,J = 7.4Hz),8.20-8.31 (2H,m)

Example 35

Compound 35

m.p.: 100-110 °C

IR(KBr,cm⁻¹):

3376,2962,2866,1743,1632,1539,1461,1386,1341,1311, 1260,1188,1152,747

35 High Resolution FAB-MS(m/e,(C₃₂H₄₇N₅O₇ + H)⁺):

Calcd: 614.3554

Found: 614.3540

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.62-0.75(6H,m),0.86(3H,t,J = 7.0Hz),0.95 + 0.97 (3H,d×2,J = 6.8Hz),1.03-1.58(13H,m),1.65-1.78(2H,m),
 40 2.77-2.91(1H,m),3.15-3.50(2H,m),3.63-3.77(1H,m), 3.86-4.00(1H,m),4.01(3H,s),4.04-4.30(2H,m),4.53-4.64-
 (1H,m),6.15-6.25(1H,m),7.25(1H,t,J = 7.3Hz), 7.32(1H,t,J = 7.3Hz),7.47 + 7.48(1H,s×2),7.66 + 7.67
 (1H,d×2,J = 7.3Hz),8.05(1H,d,J = 7.3Hz),8.15-8.28 (2H,m)

Example 36

45

Synthesis of Compound 36

(1) Preparation of (c-Pent)_nPrNCO-Leu-OBzl

50 To a mixture of H-Leu-OBzl•TsOH (1.0 g) and CDI (469 mg) in chloroform (10 ml) was added TEA
 (0.43 ml) at room temperature and the mixture was stirred at the same temperature for 1 h. Cyclopentyl-
 propylamine (0.47 ml) was added and the reaction mixture was stirred at room temperature for 6 h. The
 mixture was diluted with ethyl acetate, washed with 1N-hydrochloric acid and brine, dried over MgSO₄ and
 evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with ethyl
 55 acetate/hexane = 3/7 for elution to give the product (850 mg).

FAB-MS(m/e, (C₂₂H₃₄NO₃ + H)⁺):375

(2) Preparation of Compound 36

Compound 36 was prepared using (c-Pent)nPrNCO-Leu-OBzl (prepared in (1)) as a starting material in the same manner described in Example 33-(3).

5 m.p.: 68-73 °C

IR(KBr, cm⁻¹):

3382, 2962, 2872, 1743, 1632, 1533, 1461, 1386, 1344, 1311, 1260, 1227, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₃₄H₅₁N₅O₇ + H)⁺): 642

¹H-NMR(300MHz, CDCl₃, δ ppm):

10 0.81(3H,d,J = 5.8Hz), 0.82(3H,d,J = 6.7Hz), 0.83(3H,t, J = 6.7Hz), 0.87(3H,t,J = 7.3Hz), 1.12-1.95(19H,m),
2.91(1H,dt,J = 7.9Hz, 15.8Hz), 3.00(1H,dt,J = 7.9Hz, 15.8Hz), 3.27(1H,dd,J = 4.5Hz, 15.0Hz), 3.33(1H,dd,
J = 6.0Hz, 15.0Hz), 3.84-3.98(1H,m), 4.02(3H,s), 4.14(1H,quint,J = 7.9Hz), 4.34(1H,ddd,J = 4.6Hz, 6.4Hz, 8.2Hz)-
4.75(1H,d,J = 6.4Hz), 4.78-4.86(1H,m), 6.60 (1H,d,J = 7.9Hz), 7.26(1H,t,J = 7.6Hz), 7.35(1H,t, J = 7.6Hz), 7.48-
(1H,s), 7.59(1H,d,J = 7.6Hz), 7.61 (1H,d,J = 8.2Hz), 8.16(1H,d,J = 7.6Hz)

15 Each Compound 37-39 in the following Examples 37-39 was prepared using each corresponding amine as a starting material in the same manner described in Example 36.

Example 37

20 Compound 37

m.p.: 98-107 °C

IR(KBr, cm⁻¹):

3388, 2914, 2860, 1743, 1653, 1551, 1461, 1386, 1260, 1092, 765, 747

25 High Resolution FAB-MS(m/e, (C₃₅H₄₅N₅O₇ + H)⁺):

Calcd: 648.3398

Found: 648.3346

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

30 0.65(3H,d,J = 5.6Hz), 0.69(3H,d,J = 5.6Hz), 0.86(3H,t, J = 7.0Hz), 1.07-1.45(7H,m), 1.65-1.83(2H,m), 2.60-2.80
(2H,m), 2.86(1H,dd,J = 11.0Hz, 14.4Hz), 3.15-3.30(1H,m), 3.30-3.60(2H,m), 3.94(3H,s), 3.94-4.05(1H,m), 4.06-4.20-
(1H,m), 4.39(1H,d,J = 16.4Hz), 4.50(1H,d,J = 16.4Hz), 4.52-4.66(1H,m), 6.44(1H,d,J = 7.1Hz), 7.00-7.19(4H,m),
7.23(1H,t,J = 7.5Hz), 7.31(1H,t,J = 7.5Hz), 7.49(1H,s), 7.68(1H,d,J = 7.5Hz), 8.04(1H,d,J = 7.5Hz), 8.19(1H,d,
J = 7.6Hz), 8.30(1H,d,J = 8.7Hz)

35 Example 38Compound 38

m.p.: 79-82 °C

40 IR(KBr, cm⁻¹):

3316, 2692, 2872, 1743, 1629, 1533, 1461, 1386, 1341, 1311, 1260, 1227, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₃₅H₅₃N₅O₇ + H)⁺):

Calcd: 656.4024

Found: 656.4029

45 ¹H-NMR(300MHz, CDCl₃, δ ppm):

0.81(3H,d,J = 6.3Hz), 0.82(3H,d,J = 6.3Hz), 0.83(3H,t, J = 6.3Hz), 0.91(3H,t,J = 7.4Hz), 1.10-1.95(21H,m), 2.93
(1H,dt,J = 7.5Hz, 15.0Hz), 3.00(1H,dt,J = 7.5Hz, 15.0Hz), 3.27(1H,dd,J = 6.6Hz, 15.0Hz), 3.31(1H,dd,J = 6.0Hz,
15.0Hz), 3.91(1H,q,J = 6.7Hz), 4.02(3H,s), 4.14(1H, quint,J = 8.2Hz), 4.34(1H,ddd,J = 5.4Hz, 6.4Hz, 8.7Hz), 4.72-
(1H,d,J = 6.4Hz), 4.82(1H,ddd,J = 6.0Hz, 6.6Hz, 7.6Hz), 6.56(1H,d,J = 7.6Hz), 7.26(1H,t,J = 7.6Hz), 7.35-
50 (1H,t,J = 7.6Hz), 7.48(1H,s), 7.59(1H,d,J = 6.7Hz), 7.64(1H,d,J = 7.6Hz), 8.16(1H,d,J = 7.6Hz)

Example 39

Compound 39

55 m.p.: 95-110 °C

IR(KBr, cm⁻¹):

3334, 2962, 2872, 1743, 1662, 1536, 1461, 1386, 1344, 1311, 1260, 1149, 1128, 1092, 765, 747

High Resolution FAB-MS(m/e , $(C_{33}H_{49}N_5O_7 + H)^+$):

Calcd: 628.3718

Found: 628.3699

1H -NMR(300MHz, DMSO- d_6 , δ ppm):

5 0.66(3H,d,J = 6.0Hz), 0.70(3H,d,J = 6.0Hz), 0.86(3H,t, J = 7.1Hz), 1.01(3H,d,J = 7.2Hz), 1.04(3H,d,J = 7.2Hz),
1.05-1.55(11H,m), 1.60-1.80(4H,m), 2.84(1H,dd, J = 11.5Hz, 15.0Hz), 3.15-3.25(1H,m), 3.95(3H,s), 4.01-4.18-
(4H,m), 4.54-4.65(1H,m), 6.04(1H,d,J = 6.9Hz), 7.25 (1H,t,J = 7.5Hz), 7.32(1H,t,J = 7.5Hz), 7.48(1H,s), 7.67
(1H,d,J = 7.5Hz), 8.05(1H,d,J = 7.5Hz), 8.20(1H,d, J = 7.6Hz), 8.24(1H,d,J = 8.8Hz)

10 Each Compound 40-43 in the following Examples 40-43 was prepared using each corresponding amino
acid benzyl ester tosylate as a starting material in the same manner described in Example 39.

Example 40

Compound 40

15 m.p.: 103-107 °C

IR(KBr, cm^{-1}):

3376, 2956, 2872, 1743, 1656, 1530, 1461, 1386, 1341, 1260, 1227, 1137, 1092, 765, 747

FAB-MS(m/e , $(C_{34}H_{51}N_5O_7 + H)^+$): 642

20 1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.82(3H,t,J = 7.3Hz), 0.85(9H,s), 1.14(6H,d,J = 7.1Hz), 1.20-1.90(14H,m), 3.26(1H,dd,J = 5.5Hz, 14.6Hz), 3.35
(1H,dd,J = 6.3Hz, 14.6Hz), 3.85-3.95(2H,m), 4.02(3H,s), 4.09-4.21(1H,m), 4.28(1H,ddd,J = 5.3Hz, 6.2Hz, 8.3Hz),
4.76(1H,ddd,J = 5.5Hz, 6.3Hz, 8.3Hz), 4.93(1H,d, J = 6.2Hz), 6.48(1H,d,J = 8.3Hz), 7.27(1H,t,J = 7.6Hz), 7.35-
(1H,t,J = 7.6Hz), 7.48(1H,s), 7.58(1H,d,J = 7.6Hz), 7.72(1H,d,J = 7.3Hz), 8.16(1H,d,J = 7.6Hz)

25 Example 41

Compound 41

30 m.p.: 101-104.5 °C

IR(KBr, cm^{-1}):

3286, 2944, 2872, 1743, 1653, 1560, 1530, 1461, 1386, 1341, 1311, 1260, 1143, 1092, 756

High Resolution FAB-MS(m/e , $(C_{36}H_{47}N_5O_7 + H)^+$):

Calcd: 662.3554

35 Found: 662.3530

1H -NMR(300MHz, $CDCl_3$, δ ppm):

40 0.70-1.92(12H,m), 0.77(3H,t,J = 7.4Hz), 1.01(3H,d, J = 7.1Hz), 1.04(3H,d,J = 7.1Hz), 2.90(1H,dd,J = 5.1Hz,
14.5Hz), 2.94(1H,dd, 6.5Hz, 12.6Hz), 3.06(1H,dd, J = 8.1Hz, 12.6Hz), 3.32(1H,dd,J = 5.1Hz, 14.5Hz), 3.74-3.93-
(1H,m), 3.93-4.12(2H,m), 4.03(3H,s), 4.27(1H,ddd, J = 4.6Hz, 6.5Hz, 8.1Hz), 4.69(1H,dt,J = 5.1Hz, 7.6Hz), 5.05-
(1H,brs), 6.32(1H,d,J = 6.7Hz), 7.06-7.45(9H,m), 7.60(1H,brs), 8.14(1H,d,J = 7.9Hz)

Example 42

Compound 42

45 m.p.: 102-110 °C

IR(KBr, cm^{-1}):

3406, 2932, 2860, 1743, 1665, 1536, 1461, 1386, 1341, 1260, 1203, 1149, 1128, 1092, 765, 747

High Resolution FAB-MS(m/e , $(C_{36}H_{53}N_5O_7 + H)^+$):

50 Calcd: 668.4024

Found: 668.4004

1H -NMR(300MHz, $CDCl_3$, δ ppm):

55 0.70-1.92(25H,m), 0.81(3H,t,J = 7.1Hz), 1.14(3H,d, J = 7.0Hz), 1.15(3H,d,J = 7.0Hz), 3.23(1H,dd,J = 5.3Hz,
15.1Hz), 3.36(1H,dd,J = 7.0Hz, 15.1Hz), 3.85-4.20(3H,m), 4.02(3H,s), 4.22-4.38(1H,m), 4.73-4.90(1H,m), 5.02
(1H,brs), 6.63(1H,d,J = 6.2Hz), 7.26(1H,t,J = 7.3Hz), 7.35(1H,t,J = 7.3Hz), 7.49(1H,s), 7.59(1H,d,J = 7.3Hz), 7.65-
(1H,d,J = 7.0Hz), 8.16(1H,d,J = 7.3Hz)

Example 43

Compound 43

- 5 m.p.: 96-104 °C
 IR(KBr, cm⁻¹):
 3328, 2962, 2872, 1743, 1656, 1530, 1461, 1386, 1341, 1311, 1260, 1227, 1092, 765, 747
 High Resolution FAB-MS(m/e, (C₃₂H₄₇N₅O₇ + H)⁺):
 Calcd: 614.3554
 10 Found: 614.3558
¹H-NMR(300MHz, CDCl₃, δ ppm):
 0.79(3H, d, J = 6.7Hz), 0.82(3H, t, J = 7.0Hz), 0.90(3H, d, J = 6.7Hz), 1.00-1.40(6H, m), 1.17(3H, d, J = 7.0Hz), 1.18
 (3H, d, J = 7.0Hz), 1.42-2.10(7H, m), 3.26(1H, dd, J = 6.0Hz, 14.3Hz), 3.31(1H, dd, J = 6.0Hz, 14.3Hz), 3.73(1H, dd,
 J = 7.2Hz, 7.5Hz), 3.94-4.05(1H, m), 4.01(3H, s), 4.06-4.22 (1H, m), 4.34(1H, dt, J = 5.8Hz, 8.0Hz), 4.86(1H, dt,
 15 J = 6.0Hz, 7.8Hz), 5.09(1H, d, J = 7.2Hz), 6.74(1H, d, J = 7.8Hz), 7.26(1H, t, J = 7.3Hz), 7.34(1H, t, J = 7.3Hz), 7.49-
 (1H, s), 7.50(1H, d, J = 5.8Hz), 7.61(1H, d, J = 7.3Hz), 8.15(1H, d, J = 7.3Hz)

Example 44

20 Synthesis of Compound 44

Compound 44 was prepared using 1-adamantanamine instead of diisopropylamine in the same manner described in Example 33-(2).

- m.p.: 129-137 °C
 25 IR(KBr, cm⁻¹):
 3316, 2962, 2872, 1743, 1635, 1539, 1461, 1386, 1260, 1092, 747
 High Resolution FAB-MS(m/e, (C₃₆H₅₁N₅O₇ + H)⁺):
 Calcd: 666.3867
 Found: 666.3879
 30 ¹H-NMR(300MHz, DMSO-d₆, δ ppm):
 0.61(3H, d, J = 3.9Hz), 0.63(3H, d, J = 4.4Hz), 0.87(3H, t, J = 6.8Hz), 0.90-2.00(24H, m), 2.85(1H, dd, J = 11.2Hz,
 14.4Hz), 3.11-3.25(1H, m), 3.83-3.95(1H, m), 3.96(3H, s), 4.10-4.20(1H, m), 4.56-4.67(1H, m), 5.63(1H, s), 5.75
 (1H, d, J = 7.0Hz), 7.25(1H, t, J = 7.5Hz), 7.32(1H, t, J = 7.5Hz), 7.49(1H, s), 7.70(1H, d, J = 7.5Hz), 8.04
 (1H, d, J = 7.5Hz), 8.18(1H, d, J = 7.8Hz), 8.31(1H, d, J = 8.7Hz)

35 Example 45

Synthesis of Compound 45

- 40 Compound 45 was prepared using D-1-aminobutyric acid benzyl ester as a starting material in the same manner described in Example 1-(2) and (3).
 m.p.: 216-218 °C
 IR(KBr, cm⁻¹):
 3448, 2968, 1743, 1647, 1599, 1536, 1461, 1389, 1341, 1314, 1260, 1170, 1095, 1050, 765, 747
 45 High Resolution FAB-MS(m/e, (C₂₈H₄₀N₄O₈ + H)⁺):
 Calcd: 561.2925
 Found: 561.2911
¹H-NMR(300MHz, CD₃OD, δ ppm):
 0.77(3H, d, J = 6.4Hz), 0.78(3H, d, J = 6.4Hz), 0.87(3H, t, J = 7.4Hz), 1.23-1.35(2H, m), 1.37(9H, s), 1.68-1.93
 50 (3H, m), 3.07(1H, dd, J = 5.1Hz, 10.7Hz), 3.14-3.17(1H, m), 3.95-4.02(1H, m), 4.00(3H, s), 4.18(1H, dd, J = 5.1Hz,
 6.5Hz), 4.65-4.75(1H, m), 7.24(1H, t, J = 7.7Hz), 7.29 (1H, t, J = 7.7Hz), 7.51(1H, s), 7.68(1H, d, J = 7.7Hz), 8.10-
 (1H, d, J = 7.7Hz)

Example 46

Synthesis of Compound 46

- 5 Compound 46 was prepared using t-butylisocyanate in the same manner described in Example 21-(3).
 m.p.: 116-121 °C
 IR(KBr, cm⁻¹):
 3400, 2962, 1743, 1650, 1557, 1461, 1389, 1260, 1092, 765, 747
 High Resolution FAB-MS(m/e, (C₃₀H₄₅N₅O₇ + H)⁺):
 10 Calcd: 588.3397
 Found: 588.3369
¹H-NMR(300MHz, DMSO-d₆, δ ppm):
 0.61(6H, d, J = 5.1Hz), 0.87(3H, t, J = 7.0Hz), 1.15(9H, s), 0.82-1.89(9H, m), 2.86(1H, dd, J = 11.3Hz, 14.4Hz), 3.11-
 3.21(1H, m), 3.96(3H, s), 4.09-4.21(2H, m), 4.58-4.70 (1H, m), 5.73(1H, d, J = 9.8Hz), 5.73(1H, s), 7.25(1H, t, J = 7.5Hz)-
 15 , 7.32(1H, t, J = 7.5Hz), 7.50(1H, s), 7.72(1H, d, J = 7.5Hz), 8.04(1H, d, J = 7.8Hz), 8.21(1H, d, J = 7.5Hz), 8.31-
 (1H, d, J = 8.8Hz)

Example 47

20 Synthesis of Compound 47

- Compound 47 was prepared using 2-aminopyridine instead of hexamethyleneimine in the same manner described in Example 28.
 m.p.: 109-111 °C
 25 IR(KBr, cm⁻¹):
 3364, 3076, 2968, 2872, 1716, 1656, 1575, 1461, 1389, 1260, 1203, 1092, 765, 723
 High Resolution FAB-MS(m/e, (C₃₁H₄₀N₆O₇ + H)⁺):
 Calcd: 609.3036
 Found: 609.3030
 30 ¹H-NMR(300MHz, DMSO-d₆, δ ppm):
 0.64(3H, d, J = 6.1Hz), 0.67(3H, d, J = 6.1Hz), 0.82(3H, t, J = 7.2Hz), 1.05-1.38(7H, m), 1.56-1.80(2H, m), 2.89
 (1H, dd, J = 11.2Hz, 14.6Hz), 3.13(1H, dd, J = 1.3Hz, 14.6Hz), 3.96(3H, s), 4.13-4.32(2H, m), 4.63-4.76(1H, m), 6.95
 (1H, dd, J = 5.5Hz, 7.0Hz), 7.26(1H, t, J = 7.7Hz), 7.33 (1H, t, J = 7.7Hz), 7.38(1H, d, J = 8.7Hz), 7.54(1H, s), 7.65-7.75-
 (1H, m), 7.78(1H, d, J = 7.7Hz), 7.86-8.02(1H, m), 8.05 (1H, d, J = 7.8Hz), 8.14(1H, dd, J = 1.3Hz, 5.5Hz), 8.2
 35 (1H, d, J = 7.7Hz), 8.50(1H, d, J = 8.8Hz), 9.34(1H, brs)

Example 48

Synthesis of Compounds 48 and 49

- 40 Compound 48 (ester form) and Compound 49 (deprotected form) were prepared using heptamethyleneimine instead of hexamethyleneimine in the same manner described in Example 28.

Compound 48

- 45 (colorless amorphous)
 IR(KBr, cm⁻¹):
 3304, 2932, 2866, 1743, 1632, 1533, 1461, 1386, 1257, 1152, 1092, 765, 747
 High Resolution FAB-MS(m/e, (C₃₇H₅₇N₅O₇ + H)⁺):
 50 Calcd: 684.4337 Found: 684.4336
¹H-NMR(300MHz, CDCl₃, δ ppm):
 0.82(3H, t, J = 7.1Hz), 0.84(3H, d, J = 6.3Hz), 0.85(3H, d, J = 6.3Hz), 1.06-1.80(19H, m), 1.41(9H, s), 3.19(1H, dd,
 J = 5.9Hz, 14.7Hz), 3.33(1H, dd, J = 6.1Hz, 14.7Hz), 3.13-3.50(4H, m), 4.01(3H, s), 3.96-4.10(1H, m), 4.27-4.37 (1H, m)-
 , 4.59(1H, d, J = 6.8Hz), 4.80-4.90(1H, m), 6.53 (1H, d, J = 8.8Hz), 6.99(1H, d, J = 7.7Hz), 7.25(1H, t, J = 7.3Hz), 7.33-
 55 (1H, t, J = 7.3Hz), 7.47(1H, s), 7.62 (1H, d, J = 7.3Hz), 8.17(1H, d, J = 7.3Hz)

Compound 49

m.p.: 99-104 °C

IR(KBr, cm⁻¹):

5 3316, 2932, 2872, 1743, 1632, 1533, 1461, 1386, 1260, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₃₃H₄₉N₅O₇ + H)⁺):

Calcd: 628.3710

Found: 628.3693

¹H-NMR(300MHz, CDCl₃, δ ppm):

10 0.80(3H,d,J = 5.9Hz), 0.81(3H,d,J = 5.9Hz), 0.83(3H,t, J = 6.0Hz), 1.09-1.95(19H,m), 3.05-3.46(6H,m), 3.88-3.98 (1H,m), 4.02(3H,s), 4.31-4.41(1H,m), 4.77-4.86(1H,m), 4.98(1H,d,J = 7.0Hz), 6.63(1H,d,J = 7.4Hz), 7.25(1H,t, J = 7.5Hz), 7.34(1H,t,J = 7.5Hz), 7.48(1H,s), 7.58(1H,d, J = 7.5Hz), 7.68(1H,d,J = 7.8Hz), 8.16(1H,d,J = 7.5Hz)

Example 49

15

Synthesis of Compound 50Compound 50 was prepared using H-DNva-O^tBu as a starting material in the same manner described in Example 39.

20 m.p.: 112-116 °C

IR(KBr, cm⁻¹):

3382, 2962, 2872, 1743, 1656, 1620, 1530, 1461, 1386, 1260

High Resolution FAB-MS(m/e, (C₃₂H₄₇N₅O₇ + H)⁺):

Calcd: 614.3554

25 Found: 614.3539

¹H-NMR(300MHz, CDCl₃, δ ppm):

30 0.81(6H,d,J = 5.4Hz), 0.84(3H,t,J = 7.3Hz), 1.14(3H,d, J = 6.5Hz), 1.15(3H,d,J = 6.6Hz), 1.01-1.89(13H,m), 3.29(2H,d,J = 5.7Hz), 3.92-4.02(2H,m), 4.01(3H,s), 4.09-4.16 (1H,m), 4.28-4.35(1H,m), 4.83(1H,dt,J = 6.3Hz, 5.7Hz), 4.96(1H,d,J = 4.8Hz), 6.67-6.72(1H,brs), 7.25(1H,t, J = 7.8Hz), 7.34(1H,t,J = 7.8Hz), 7.48(1H,s), 7.58(1H,d, J = 7.8Hz), 7.65(1H,d,J = 6.3Hz), 8.16(1H,d,J = 7.8Hz)

Example 50Synthesis of Compound 51

35

Compound 51 was prepared using H-DNva-O^tBu as a starting material in the same manner described in Example 40.

m.p.: 115-119 °C

IR(KBr, cm⁻¹):

40 3358, 2962, 2872, 1746, 1659, 1620, 1533, 1461, 1386, 1260

High Resolution FAB-MS(m/e, (C₃₃H₄₉N₅O₇ + H)⁺):

Calcd: 628.3732

Found: 628.3683

¹H-NMR(300MHz, CDCl₃, δ ppm):

45 0.84(9H,s), 0.84(3H,t,J = 7.2Hz), 1.14(6H,d,J = 7.1Hz), 1.01-1.89(12H,m), 3.28(1H,dd,J = 5.1Hz, 14.1Hz), 3.32(1H,dd,J = 6.0Hz, 14.1Hz), 3.87-4.01(2H,m), 4.01(3H,s), 4.10-4.20(1H,m), 4.32(1H,dt,J = 8.1Hz, 4.5Hz), 4.76(1H,ddd,J = 5.1Hz, 6.9Hz, 8.1Hz), 4.98(1H,d,J = 5.4Hz), 6.57-6.60(1H,brs), 7.24(1H,t,J = 7.8Hz), 7.34(1H,t, J = 7.8Hz), 7.48(1H,s), 7.58(1H,d,J = 7.8Hz), 7.69(1H,d, J = 8.1Hz), 8.16(1H,d,J = 7.8Hz)

Example 51Synthesis of Compound 52

55 Compound 52 was prepared using H-Ile-OBzl•TsOH as a starting material in the same manner described in Example 39.

m.p.: 85-95 °C

IR(KBr, cm⁻¹):

3328, 2968, 2872, 1743, 1671, 1539, 1461, 1386, 1344, 1311, 1260, 1203, 1149, 1092, 765, 747

High Resolution FAB-MS(m/e,(C₃₃H₄₉N₅O₇ + H)⁺):

Calcd: 628.3710

Found: 628.3689

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

5 0.42(3H,d,J = 6.8Hz),0.64(3H,t,J = 7.3Hz),0.81-0.98 (1H,m),0.86(3H,t,J = 7.1Hz),1.04(3H,d,J = 4.8Hz), 1.06-
(3H,d,J = 4.8Hz),1.20-1.80(14H,m),2.84(1H,dd, J = 11.4Hz,14.9Hz),3.24(1H,dd,J = 2.6Hz,14.9Hz), 3.78-
(1H,t,J = 7.8Hz),3.96(3H,s),4.07-4.20(3H,m), 4.58-4.68(1H,m),5.94(1H,d,J = 7.8Hz),7.25(1H,t, J = 7.6Hz),7.32-
(1H,t,J = 7.6Hz),7.52(1H,s),7.66(1H,d, J = 7.6Hz),8.04(1H,d,J = 7.6Hz),8.27(1H,d,J = 7.8Hz), 8.31-
(1H,d,J = 8.7Hz)

10 Each Compound 53 and 54 in the following Example 52 and 53 was prepared using each corresponding amino acid benzyl ester tosylate as a starting material in the same manner described in Example 51.

Example 52

15 Compound 53

m.p.: 85-94 °C

IR(KBr,cm⁻¹):

3328,2968,2872,1743,1671,1530,1461,1386,1344,1311, 1260,1200,1149,1092,765,747

20 High Resolution FAB-MS(m/e,(C₃₃H₄₉N₅O₇ + H)⁺):

Calcd: 628.3710

Found: 628.3698

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

25 0.51(3H,d,J = 6.5Hz),0.60(3H,t,J = 7.1Hz),0.80-0.92 (1H,m),0.86(3H,t,J = 7.0Hz),1.04(3H,d,J = 6.2Hz), 1.06-
(3H,d,J = 6.2Hz),1.20-1.80(14H,m),2.83(1H,dd, J = 11.7Hz,14.7Hz),3.23(1H,dd,J = 2.8Hz,14.7Hz), 3.88-
(1H,t,J = 7.7Hz),3.96(3H,s),4.02-4.18(3H,m), 4.59-4.69(1H,m),5.78(1H,d,J = 7.7Hz),7.25(1H,t, J = 7.6Hz),7.32-
(1H,t,J = 7.6Hz),7.53(1H,s),7.69(1H,d, J = 7.6Hz),8.04(1H,d,J = 7.6Hz),8.27(1H,d,J = 7.52Hz), 8.32-
(1H,d,J = 9.1Hz)

30 Example 53

Compound 54

m.p.: 108-111 °C

35 IR(KBr,cm⁻¹):

3340,2962,2872,1743,1650,1632,1518,1461,1386,1260, 1146,1092,765,747

High Resolution FAB-MS(m/e,(C₃₃H₄₉N₅O₇ + H)⁺):

Calcd: 628.3711

Found: 628.3705

40 ¹H-NMR(400MHz,CDCl₃, δ ppm):

0.81(3H,t,J = 6.8Hz),0.91(9H,s),1.19(6H,d,J = 6.8Hz), 1.12-1.36(4H,m),1.43-1.87(8H,m),3.25-
(1H,dd,J = 5.9Hz, 15.6Hz),3.34(1H,dd,J = 6.8Hz,15.6Hz),3.87-4.00(1H,m), 4.01(3H,s),4.01-4.06(1H,m),4.13-
4.18(1H,m),4.35-4.40(1H,m),4.81-4.87(1H,m),5.12(1H,d,J = 7.8Hz),6.76-6.82(1H,brs),7.27(1H,t,J = 7.8Hz),7.35-
(1H,t,J = 7.8Hz), 7.39-7.42(1H,brs),7.50(1H,s),7.63(1H,d,J = 7.8Hz), 8.17(1H,d,J = 7.8Hz)

45

Example 54

Synthesis of Compound 55

50 Compound 55 was prepared using 2-(2-chlorophenyl)carbamoyloxy-4-methylpentanoic acid, which was prepared using 2-hydroxy-5-methylpentanoic acid instead of H-Leu-OBzl in the same manner described in Example 18-(2), as a starting material in the same manner described in Example 33-(3).

m.p.: 161-163 °C

IR(KBr,cm⁻¹):

55 3310,2962,1740,1659,1533,1461,1449,1386,1260,1212, 1089,747

High Resolution FAB-MS(m/e,(C₃₂H₃₉ClN₄O₈ + H)⁺):

Calcd: 643.2535

Found: 643.2511

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.82(3H,t,J = 7.0Hz),0.91(6H,d,J = 4.5Hz),1.13-1.31 (4H,m),1.56-1.84(5H,m),3.16(1H,dd,J = 14.5Hz,7.2Hz),
3.29(1H,dd,J = 14.5Hz,6.7Hz),3.96(3H,s),4.41-4.48 (1H,m),4.78-4.86(1H,m),5.04-5.09(1H,m),6.60(1H,d,
J = 7.6Hz),6.78(1H,d,J = 7.7Hz),7.02(1H,t,J = 7.7Hz), 7.18-7.30(4H,m),7.35(1H,d,J = 7.7Hz),7.48(1H,s), 7.62-
5 (1H,d,J = 6.3Hz),7.99(1H,d,J = 8.0Hz),8.08(1H,d, J = 7.7Hz)

Example 55

Synthesis of Compound 56

Compound 56 was prepared using H-DVal-OBzl•TsOH instead of H-DNle-OBzl•TsOH in the same manner described in Example 1-(2) and (3).

m.p.: 203-207 °C

IR(KBr,cm⁻¹):

3328,2968,1722,1695,1530,1461,1389,1260,1170,1092, 765

High Resolution FAB-MS(m/e,(C₂₉H₄₂N₄O₈ + H)⁺):

Calcd: 575.3081

Found: 575.3066

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.65(3H,d,J = 6.4Hz),0.67(3H,d,J = 6.4Hz),0.85(3H,d, J = 6.9Hz),0.86(3H,d,J = 6.9Hz),0.90-1.48(3H,m),
1.31(9H,s),2.00-2.14(1H,m),2.91(1H,dd,J = 10.3Hz, 14.7Hz),3.12(1H,dd,J = 3.0Hz,14.7Hz),3.85-4.00(1H,m),
3.95(3H,s),4.04(1H,dd,J = 5.2Hz,7.9Hz),4.60-4.71 (1H,m),6.68(1H,d,J = 8.1Hz),7.23(1H,t,J = 7.5Hz), 7.31-
(1H,t,J = 7.5Hz),7.50(1H,s),7.69(1H,d,J = 7.5Hz), 7.80(1H,d,J = 7.9Hz),8.04(1H,d,J = 7.5Hz),8.16(1H,d,
J = 8.0Hz)

Example 56

Synthesis of Compound 57 and 58

(1) Preparation of Boc-DTrp(COOMe)_φ(CS-NH)-DNle-O^tBu

Boc-DTrp(COOMe)-DNle-O^tBu (300 mg, prepared in Example 21-(1)) and Lawesson's Reagent (140 mg) were suspended in toluene. The mixture was stirred at 100 °C under argon atmosphere for 5 h and concentrated under reduced pressure. The residue was purified by silica gel chromatography (Merck,
Kieselgel 60) with chloroform/diethyl ether = 99/1 for elution to give the product (172 mg).

FAB-MS(m/e, (C₂₈H₄₁N₃O₆S + H)⁺):547

(2) Preparation of Compound 57

A solution of Boc-DTrp(COOMe)_φ(CS-NH)-DNle-O^tBu (100 mg, obtained in (1)) in formic acid (3 ml) was stirred at room temperature for 2 h and concentrated under reduced pressure. The residue was dissolved in DMF (2 ml), and NMM (100 μl) was added. To the mixture were added 2,6-dimethylpiperidinocarbonyl-γMeLeu-OH (52 mg), HOBT•H₂O (34 mg) and EDCI•HCl (42 mg) under ice cooling. The mixture was stirred at the same temperature for 30 min and at room temperature for 14 h. The reaction mixture was diluted with ethyl acetate, washed with sat. aq NaHCO₃, 1N-hydrochloric acid and water successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 3/2 for development to give Compound 57 (28 mg).

m.p.: 75-78 °C

IR(KBr,cm⁻¹):

3256,2956,2872,1743,1626,1518,1461,1374,1341,1311, 1257,1227,1158,1089,1041,765,747

High Resolution FAB-MS(m/e,(C₃₈H₅₉N₅O₆S + H)⁺):

Calcd: 714.4264

Found: 714.4252

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.82(3H,t,J = 7.2Hz),0.90(9H,s),0.95-1.92(14H,m), 1.19(3H,d,J = 7.1Hz),1.20(3H,d,J = 7.1Hz),1.37(9H,s),
3.29(1H,dd,J = 7.2Hz,14.4Hz),3.47(1H,dd,J = 5.5Hz, 14.4Hz),4.00(3H,s),4.05-4.13(1H,m),4.11-4.22(1H,m),
4.19-4.26(1H,m),4.65(1H,d,J = 7.4Hz),4.72-4.79(1H,m), 4.97(1H,ddd,J = 5.5Hz,7.2Hz,8.0Hz),7.06(1H,d,

J = 8.0Hz), 7.26(1H, t, J = 7.4Hz), 7.33(1H, t, J = 7.4Hz), 7.49(1H, s), 7.72(1H, d, J = 7.4Hz), 8.15(2H, d, J = 7.4Hz)

(3) Preparation of Compound 58

5 Compound 57 (13 mg, prepared in (2)) was dissolved in formic acid (1 ml) under ice cooling, and the mixture was stirred at the same temperature for 1.5 h and at room temperature for 6 h. The mixture was diluted with ethyl acetate, washed with brine and water, dried over MgSO_4 and evaporated in vacuo. The residue was purified by silica gel chromatography (Wakogel C-200) with chloroform/methanol = 97/3 for elution to give Compound 58 (7.3 mg).

10 m.p.: 74-82 °C

IR(KBr, cm^{-1}):

3310, 2956, 2866, 1743, 1620, 1572, 1524, 1461, 1386, 1341, 1260, 1230, 1128, 1092, 765, 747

High Resolution FAB-MS(m/e, $(\text{C}_{34}\text{H}_{51}\text{N}_5\text{O}_6\text{S} + \text{H})^+$):

Calcd: 658.3638

15 Found: 658.3658

$^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

0.78(9H, s), 0.86(3H, t, J = 6.8Hz), 0.92-2.05(14H, m), 1.14(6H, d, J = 7.1Hz), 3.33(1H, dd, J = 7.9Hz, 14.7Hz),
3.54(1H, dd, J = 5.3Hz, 14.7Hz), 3.92-4.06(2H, m), 3.99 (3H, s), 4.09-4.16(1H, m), 4.94-5.07(2H, m), 5.17(1H, ddd,
J = 5.3Hz, 6.3Hz, 7.9Hz), 6.99(1H, d, J = 6.3Hz), 7.22(1H, t, J = 7.8Hz), 7.31(1H, t, J = 7.8Hz), 7.51(1H, s), 7.64(1H, d,
20 J = 7.8Hz), 8.11(1H, d, J = 7.8Hz), 8.89(1H, d, J = 7.5Hz)

Example 57

Synthesis of Compound 59

25

(1) Preparation of 2-hydroxy-3-(3-indolyl)propionyl-DNle-O^tBu

To a solution of DL-3-indolelactic acid (103 mg), H-DNle-O^tBu•HCl (100 mg) and NMM (58 μl) in dichloromethane (5.0 ml) were added HOBT•H₂O (81 mg) and EDCI•HCl (102 mg) under ice cooling. After
30 being stirred at room temperature over night, the mixture was diluted with dichloromethane (20 ml), washed with sat. aq. NaHCO_3 , 10% aq. citric acid and brine successively, dried over MgSO_4 and evaporated in vacuo. The residue was purified by MPLC (Merck, LiChroprep Si 60) with hexane/ethyl acetate = 3/2 for elution to give the product (80 mg).

FAB-MS(m/e, $(\text{C}_{21}\text{H}_{30}\text{N}_2\text{O}_4 + \text{H})^+$): 375

35

(2) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu- ϕ (COO)-DTrp-DNle-O^tBu

To a solution of 2-hydroxy-3-(3-indolyl)propionyl-DNle-O^tBu (62 mg, prepared in (1)) and 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-OH (72 mg) in THF (1.5 ml) were added EDCI•HCl (48 mg) and DMAP
40 (10 mg) under ice cooling. The mixture was stirred at room temperature for 3 d, and concentrated under reduced pressure. To the residue was added water (20 ml), and the mixture was extracted with ethyl acetate (10 ml $\times$ 3). The combined organic layers were washed with 1N-hydrochloric acid (10 ml), sat. aq. NaHCO_3 and brine successively, dried over MgSO_4 and evaporated in vacuo. The residue was purified by MPLC (Merck, LiChroprep Si 60) with hexane/ethyl acetate = 2/1 for elution to give the product (91 mg).

45 FAB-MS(m/e, $(\text{C}_{36}\text{H}_{56}\text{N}_4\text{O}_6 + \text{H})^+$): 641

(3) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu- ϕ (COO)-DTrp(COOMe)-DNle-O^tBu

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu- ϕ (COO)-DTrp-DNle-O^tBu (76 mg, prepared in
50 (2)) in dichloromethane were added methyl chloroformate (14 μl), pulverized NaOH (12 mg) and TBAHS (1 mg) under ice cooling. After being stirred at room temperature for 6 h, the mixture was diluted with dichloromethane (30 ml).

The solution was washed with water (20 ml), dried over MgSO_4 and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 2/1 for develop-
55 ment to give the compound (41 mg).

FAB-MS(m/e, $(\text{C}_{38}\text{H}_{58}\text{N}_4\text{O}_8 + \text{H})^+$): 699

(4) Preparation of Compound 59

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu- ϕ (COO)-DTrp(COOMe)-DNle-O^tBu (36 mg, prepared in (3)) in TFA (3 ml) was stirred at room temperature for 1 h, and concentrated under reduced pressure. The residue was triturated with water to give Compound 59 (32 mg).

m.p.: 74-79 °C

IR(KBr, cm⁻¹):

3406, 2962, 1746, 1671, 1629, 1539, 1461, 1386, 1260, 1200, 1164, 1092, 765, 747

FAB-MS(m/e, (C₃₄H₅₀N₄O₈ + H)⁺): 643

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.83(3H, t, J = 7.2Hz), 0.89(9H, s), 1.00-1.90(14H, m), 1.17(3H, d, J = 7.5Hz), 1.20(3H, d, J = 7.5Hz), 3.31(1H, dd, J = 7.5Hz, 14.9Hz), 3.40(1H, dd, J = 3.8Hz, 14.9Hz), 3.70-4.10(2H, m), 4.02(3H, s), 4.07-4.19(1H, m), 4.30-4.40 (1H, m), 5.48(1H, dd, J = 3.8Hz, 7.5Hz), 7.25(1H, t, J = 7.5Hz), 7.34(1H, t, J = 7.5Hz), 7.48(1H, s), 7.59(1H, d, J = 7.5Hz), 7.20-8.00(2H, brs), 8.16(1H, d, J = 7.5Hz)

Example 58

Synthesis of Compound 60

Compound 60 was prepared in the same manner described in Example 50.

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.82(3H, t, J = 7.2Hz), 0.89(9H, s), 1.15(3H, d, J = 7.1Hz), 1.05-1.75(14H, m), 1.16(3H, d, J = 7.1Hz), 1.41(9H, s), 3.17(1H, dd, J = 6.2Hz, 15.0Hz), 3.37(1H, dd, J = 5.9Hz, 15.0Hz), 3.93-4.10(2H, m), 4.00(3H, s), 4.10-4.23(1H, m), 4.24-4.36(1H, m), 4.65(1H, d, J = 7.1Hz), 4.81(1H, ddd, J = 5.9Hz, 6.2Hz, 8.5Hz), 6.48(1H, d, J = 8.5Hz), 7.03(1H, d, J = 7.6Hz), 7.24(1H, t, J = 7.6Hz), 7.33(1H, t, J = 7.6Hz), 7.47(1H, s), 7.62(1H, d, J = 7.6Hz), 8.16(1H, d, J = 7.8Hz)

Example 59

Synthesis of Compounds 61, 62, 63 and 64

(1) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DLmTyr-DNle-OBzl

The title compound was prepared from 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-OH, Boc-DLmTyr-OH and H-DNle-OBzl according to the same procedure described in Example 14.

FAB-MS(m/e, (C₃₇H₅₄N₄O₆ + H)⁺): 651

(2) Preparation of Compound 61

Compound 61 was prepared by deprotection of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DLmTyr-DNle-OBzl in a similar manner to that of Example 1-(3).

IR(KBr, cm⁻¹):

3328, 2956, 2872, 1602, 1533, 1464, 1404, 1341, 1251

FAB-MS(m/e, (C₃₀H₄₈N₄O₆ + H)⁺): 561

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.81(3H, t, J = 6.1Hz), 0.84(9H, s), 1.13(6H, d, J = 7.0Hz), 1.03-1.85(14H, m), 2.80-2.99(1H, m), 2.99-3.13(1H, m), 3.95-4.30(4H, m), 4.53-4.71(1H, m), 6.63(1H, d, J = 7.6Hz), 6.68(1H, d, J = 7.6Hz), 6.79(1H, brs), 7.05(1H, t, J = 7.6Hz), 7.46(1H, brs)

(3) Preparation of Compound 62

2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DLmTyr-DNle-OBzl (prepared in (1)) was acetylated by acetic anhydride and DMAP in THF. The product was deprotected in a similar manner to that of Example 1-(3) to afford Compound 63.

FAB-MS(m/e, (C₃₂H₅₀N₄O₇ + H)⁺): 603

(4) Preparation of Compound 63

2,6-dimethylpiperidinocarbonyl- γ MeLeu-DLmTyr-DNle-OBzl (prepared in (1)) was methylated by etherial diazomethane. The product was deprotected in a similar manner to that of Example 1-(3) to afford Compound 64.

FAB-MS(m/e, (C₃₁H₅₀N₄O₆ + H)⁺):575

(5) Preparation of Compound 64

2,6-dimethylpiperidinocarbonyl- γ MeLeu-DLmTyr-DNle-OBzl (prepared in (1)) was methoxycarbonylated by methyl chloroformate and DMAP in THF. The product was deprotected in a similar manner to that of Example 1-(3) to afford Compound 63.

FAB-MS(m/e, (C₃₂H₅₀N₄O₈ + H)⁺):619

Example 60

Synthesis of Compound 65

Compound 65 was prepared using DSer as a starting material in the same manner described in Example 6.

IR(KBr, cm⁻¹):

3328,2962,1743,1659,1563,1461,1386,1260,1167,1092, 1050,762

High Resolution FAB-MS(m/e, (C₂₇H₃₈N₄O₉ + H)⁺):

Calcd: 563.2717

Found: 563.2694

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.67(3H,d,J = 6.6Hz), 0.69(3H,d,J = 6.6Hz), 0.98-1.38 (3H,m), 1.31(9H,s), 2.93(1H,dd,J = 9.7Hz, 14.6Hz), 3.13- (1H,dd,J = 4.2Hz, 14.6Hz), 3.26-3.36(1H,m), 3.51-3.60(1H,m), 3.73-3.85(1H,m), 3.89-4.01(1H,m), 3.94 (3H,s), 4.55-4.68(1H,m), 6.68(1H,d,J = 8.0Hz), 7.22 (1H,t,J = 7.5Hz), 7.30(1H,t,J = 7.5Hz), 7.48(1H,s), 7.69(1H,d,J = 7.5Hz)-, 7.69(1H,d,J = 7.5Hz), 8.03(1H,d, J = 7.5Hz), 8.19(1H,d,J = 7.5Hz)

Example 61

Synthesis of Compound 66(1) Preparation of N-(2-thienylcarbamoyl)- γ MeLeu-OBzl

To a solution of 2-thiophenecarboxylic acid (30 mg) and diphenylphosphoryl aside (50.5 μ l) in toluene (10 ml) was added TEA (32.6 μ l) under ice cooling. The mixture was stirred at the same temperature for 1 h, at room temperature for 10 h and refluxed for 6 h . The mixture was cooled to room temperature. To the mixture was added a solution of γ MeLeu-OBzl•TsOH (61 mg) and TEA (21 μ l) in DMF (1 ml). After being stirred at room temperature for 15 h, the reaction mixture was diluted with ethyl acetate, washed with sat. aq. NaHCO₃, 1N-hydrochloric acid and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 4/1 for development to give the product.

FAB-MS(m/e, (C₁₉H₂₄N₂O₃S + H)⁺):361

(2) Preparation of Compound 66

N-(2-Thienylcarbamoyl)- γ MeLeu-OBzl (27.0 mg, prepared in (1)) was hydrolyzed with aqueous sodium hydroxide in methanol to afford N-(2-thienylcarbamoyl)- γ MeLeu-OH (22.0 mg), which was condensed with H-DTrp(COOMe)-DNle-O^tBu•HCl (35.0 mg, prepared in Example 21-(1)) in a similar manner to that of Example 33-(3) to afford Compound 66 (5.4 mg).

High Resolution FAB-MS(m/e, (C₃₁H₄₁N₅O₇S + H)⁺):

Calcd: 628.2805

Found: 628.2822

Each compound 67-70 in the following Examples 62-65 was prepared using each corresponding amine and amino acid benzyl ester tosylate in the same manner described in Example 36.

Example 62

Compound 67

5 m.p.: 98-102 °C

IR(KBr, cm⁻¹):

3316, 2962, 2872, 1743, 1632, 1521, 1461, 1386, 1260, 1092, 765, 747

High Resolution FAB-MS(m/e, (C₃₁H₄₅N₅O₇ + H)⁺):

Calcd: 600.3397

10 Found: 600.3371

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.50(3H,d,J = 6.7Hz), 0.57(3H,d,J = 6.7Hz), 0.86(3H,t, J = 6.8Hz), 1.10(6H,d,J = 5.6Hz), 0.92-2.10(11H,m),
 2.84(1H,dd,J = 11.2Hz, 14.4Hz), 3.21(1H,dd, 3.1Hz, 14.4Hz), 3.67-3.90(3H,m), 3.96(3H,s), 4.08-4.20(1H,m), 4.57-
 4.72(1H,m), 5.47(1H,d,J = 7.8Hz), 7.25(1H,t, J = 7.6Hz), 7.32(1H,t,J = 7.6Hz), 7.52(1H,s), 7.69(1H,d, J = 7.6Hz)-
 15 , 8.03(1H,d,J = 7.6Hz), 8.31(1H,d,J = 8.1Hz), 8.35(1H,d,J = 9.0Hz)

Example 63

Compound 68

20

IR(KBr, cm⁻¹):

2954, 2859, 1737, 1616, 1525, 1442, 1340, 1309, 1226, 1199, 1126, 765, 748

High Resolution FAB-MS(m/e, (C₃₃H₄₇N₅O₇ + H)⁺):

Calcd: 626.3554

25 Found: 626.3572

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

-0.80--0.95(2H,m), 0.01-0.39(3H,m), 0.86(3H,t, J = 7.0Hz), 1.03(3H,d,J = 7.4Hz), 1.05(3H,d,J = 7.4Hz), 0.98-
 1.82(14H,m), 2.85(1H,dd,J = 10.9Hz, 14.7Hz), 3.22(1H,dd, 3.1Hz, 14.7Hz), 3.95(3H,s), 3.90-4.20 (4H,m), 4.53-4.65-
 (1H,m), 6.05(1H,d,J = 6.8Hz), 7.24 (1H,t,J = 7.4Hz), 7.31(1H,t,J = 7.4Hz), 7.49(1H,s), 7.68(1H,d,J = 7.4Hz), 8.02-
 30 (1H,d,J = 7.4Hz), 8.24(1H,d, J = 6.9Hz), 8.25(1H,d,J = 8.8Hz)

Example 64

Compound 69

35

m.p.: 103-113 °C

IR(KBr, cm⁻¹):

2958, 2937, 1737, 1654, 1618, 1527, 1477, 1444, 1382, 1340, 1309, 1259, 1226, 1201, 1145, 1126, 765

High Resolution FAB-MS(m/e, (C₃₂H₄₅N₅O₇ + H)⁺):

Calcd: 612.3397

40 Found: 612.3384

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.02-0.39(4H,m), 0.79-0.97(1H,m), 0.84(3H,t,J = 7.0Hz), 1.04(3H,d,J = 6.7Hz), 1.06(3H,d,J = 6.7Hz), 1.12-1.78
 (12H,m), 2.85(1H,dd,J = 10.9Hz, 14.7Hz), 3.21-3.37 (2H,m), 3.96(3H,s), 4.02-4.21(3H,m), 4.50-4.62(1H,m), 6.27-
 45 (1H,d,J = 5.6Hz), 7.26(1H,t,J = 7.4Hz), 7.33(1H,t, J = 7.4Hz), 7.52(1H,s), 7.65(1H,d,J = 7.4Hz), 8.05(1H,d,
 J = 7.4Hz), 8.20(1H,d,J = 6.8Hz), 8.21(1H,d,J = 9.3Hz)

Example 65

50 Compound 70

IR(KBr, cm⁻¹):

2954, 2861, 1737, 1650, 1531, 1444, 1413, 1342, 1299, 1224, 1091, 765, 746

High Resolution FAB-MS(m/e, (C₃₃H₄₇N₅O₇ + H)⁺):

Calcd: 626.3554

55 Found: 626.3538

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.76-0.96(3H,m), 0.96-2.30(23H,m), 2.73-2.90(1H,m), 3.00-3.61(5H,m), 3.61-3.75(1H,m), 3.96(3H,s), 4.03-

4.20(1H,m),4.49-4.63(1H,m),5.97(1H,d,J = 4.9Hz),7.16-7.38(2H,m),7.53(1H,s),7.65(1H,d,J = 7.2Hz),8.05(1H,d,J = 7.2Hz),8.24(1H,d,J = 6.5Hz),8.32(1H,d,J = 7.7Hz)

Example 66

5

Synthesis of Compound 71

Compound 71 was prepared using DSer(Me) as a starting material in the same manner described in Example 6.

10 m.p.: 103-106 °C

IR(KBr,cm⁻¹):

3328,2962,1743,1527,1461,1386,1344,1260,1167,1092, 1047,1023,765

High Resolution FAB-MS(m/e,(C₂₈H₄₀N₄O₉ + H)⁺):

Calcd: 577.2874

15 Found: 577.2885

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.65(3H,d,J = 6.3Hz),0.67(3H,d,J = 6.3Hz),1.31(9H,s), 0.90-1.60(3H,m),2.80-3.20(2H,m),3.27(3H,s),3.43-3.75(2H,m),3.83-4.02(1H,m),3.95(3H,s),4.39-4.51 (1H,m),4.65-4.81(1H,m),6.67(1H,d,J = 7.1Hz),7.23 (1H,dt,J = 1.3Hz,7.6Hz),7.31(1H,dt,J = 1.3Hz,7.6Hz), 7.51(1H,s),7.73(1H,dd,J = 1.3Hz,7.6Hz),8.03(1H,dd, J = 1.3Hz,7.6Hz),8.17(1H,d,J = 8.3Hz),8.37(1H,d, J = 8.3Hz)

20

Example 67

Synthesis of Compounds 72, 73, 74 and 75

25

(1) Preparation of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NO₂)-DNle-O^tBu

The title compound was prepared from 2,6-dimethylpiperidinocarbonyl- γMeLeu-OH, Boc-DPhe(m-NO₂)-OH and H-DNle-O^tBu•HCl according to the same manner described in Example 14.

30 FAB-MS(m/e, (C₃₄H₅₅N₅O₇ + H)⁺):646

(2) Preparation of Compound 72

A solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NO₂)-DNle-O^tBu (34 mg, prepared in (1)) in 4N-hydrogen chloride/1,4-dioxane (3 ml) was stirred at room temperature for 2 h. The mixture was evaporated in vacuo and the residue was triturated with diethyl ether to give Compound 72 (24 mg).

35

IR(KBr,cm⁻¹):

3376,2956,2872,1725,1668,1536,1473,1452,1356,1248, 1200,1140

High Resolution FAB-MS(m/e,(C₃₀H₄₇N₅O₇ + H)⁺):

40

Calcd: 590.3554

Found: 590.3580

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.80-1.8(17H,m),0.87(9H,s),1.15(6H,d,J = 7.0Hz), 3.12-3.42(2H,m),4.0-4.85(6H,m),6.88(1H,d,J = 7.7Hz), 7.06(1H,d,J = 7.7Hz),7.46(1H,t,J = 7.8Hz),7.61-7.64 (1H,m),8.04-8.11(2H,m)

45

(3) Preparation of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NH₂)-DNle-O^tBu

A solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NO₂)-DNle-O^tBu (285 mg, prepared in (1)) in 95% ethanol (10 ml) was hydrogenated over 10% Pd/C at an atmospheric pressure of hydrogen for 3 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo to give the product (274 mg). FAB-MS(m/e, C₃₄H₅₇N₅O₅ + H)⁺:616

50

(4) Preparation of Compound 73 hydrochloride

Deprotection of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NH₂)-DNle-O^tBu (31 mg, prepared in (3)) in a similar manner to that of Example 67-(2) gave Compound 73 hydrochloride (28 mg).

55

m.p.: 144-148 °C

IR(KBr,cm⁻¹):

3400,2954,1774,1720,1655,1531,1466,1389,1246,1205, 1140,689

High Resolution FAB-MS(m/e,(C₃₀H₄₉N₅O₅ + H)⁺):

Calcd: 560.3812

Found: 560.3842

5 ¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.80(9H,s),0.86(3H,t,J = 7.0Hz),1.04(3H,d,J = 7.2Hz), 1.07(3H,d,J = 7.2Hz),1.12-1.81(14H,m),2.75(1H,dd,
J = 10.5Hz,13.6Hz),3.18(1H,dd,J = 3.5Hz,13.6Hz), 4.01-4.25(4H,m),4.36-4.49(1H,m),6.14(1H,d,J = 6.6Hz), 7.14-
7.34(4H,m),8.14(1H,d,J = 8.8Hz),8.23(1H,d, J = 7.6Hz),10.5(3H,brs)

10 (5) Preparation of Compound 74

To a solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NH₂)-DNle-O^tBu (31 mg, prepared in
(3)) in dichloromethane (1 ml) were added DMAP (10 mg) and formic pivalic anhydride (20 μl), and the
mixture was stirred at room temperature for 20 h. The mixture was diluted with dichloromethane (30 ml),
15 washed with 1N-aqueous sodium hydroxide (20 ml), 1N-hydrochloric acid (20 ml) and brine (20 ml)
successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC
(Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 1/2 for development to give ^tBu ester of Compound
74 (23 mg). The product (20 mg) was deprotected in a similar manner to that of Example 67-(2) to give
Compound 74 (16 mg).

20 m.p.: 123-125 °C

IR(KBr,cm⁻¹):

3300,2953,2800,1774,1670,1614,1535,1446,1396,1369, 1248,1142,789,690

High Resolution FAB-MS(m/e,(C₃₁H₄₉N₅O₆ + H)⁺):

Calcd: 588.3761

25 Found: 588.3759

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.79(9H,s),0.86(3H,t,J = 7.0Hz),0.96-1.80(23H,m), 2.62-2.75(1H,m),3.05-3.17(1H,m),3.83-4.49(3H,m), 6.05-
(1H,brs),6.95(1H,d,J = 7.6Hz),7.06-7.23(2H,m), 7.42(1H,s),8.15(1H,d,J = 7.3Hz),8.22(1H,d,J = 1.7Hz), 10.07-
(1H,brs)

30

(6) Preparation of Compound 75

To a solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DPhe(m-NH₂)-DNle-O^tBu (31 mg, prepared in
(3)) in dichloromethane (1 ml) were added DMAP (10 mg) and methyl chloroformate (12 μl), and the mixture
was stirred at room temperature for 12 h. The mixture was diluted with dichloromethane (30 ml), washed
35 with 1N-aqueous sodium hydroxide (20 ml), 1N-hydrochloric acid (20 ml) and brine (20 ml) successively,
dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel
60 F₂₅₄) with hexane/ethyl acetate = 1/1 for development to give ^tBu ester of Compound 75 (26 mg). The
product (23 mg) was deprotected in a similar manner to that of Example 67-(2) to give Compound 75 (16
40 mg).

m.p.: 116-118 °C

IR(KBr,cm⁻¹):

3300,2954,2858,1774,1716,1668,1614,1547,1448,1232, 1074,773

High Resolution FAB-MS(m/e,(C₃₂H₅₁N₅O₇ + H)⁺):

45 Calcd: 618.3867

Found: 618.3865

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.79(9H,s),0.86(3H,t,J = 7.0Hz),1.04(3H,d,J = 6.3Hz), 1.06(3H,d,J = 6.3Hz),0.98-1.78(14H,m),2.64(1H,dd,
J = 11.1Hz,13.4Hz),3.08(1H,dd,J = 3.3Hz,13.4Hz),3.63 (3H,s),3.99-4.43(5H,m),6.05(1H,d,J = 7.4Hz),6.86
50 (1H,d,J = 7.8Hz),7.08(1H,t,J = 7.8Hz),7.24(1H,d, J = 7.8Hz),7.30(1H,s),8.06-8.18(2H,m),9.49(1H,s)

Example 68

Synthesis of Compounds 76 and 77

55

Compounds 76 and 77 were prepared using 2,6-dimethylpiperidinocarbonyl- γMeLeu-OH as a starting
material in the same manner described in Example 18.

Compound 76

High Resolution FAB-MS(m/e, (C₃₆H₅₆N₄O₅S + H)⁺):

Calcd: 657.4050

5 Found: 657.4042

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.84(3H,t,J = 7.4Hz), 0.89(9H,s), 1.10-1.75(14H,m), 1.168(3H,d,J = 7.1Hz), 1.173(3H,d,J = 7.1Hz), 1.41(9H,s),
 3.37(1H,dd,J = 6.4Hz, 14.6Hz), 3.48(1H,dd,J = 6.6Hz, 14.6Hz), 3.96-4.08(2H,m), 4.13-4.23(1H,m), 4.29
 (1H,ddd,J = 5.7Hz, 7.8Hz, 7.8Hz), 4.66(1H,d,J = 6.8Hz), 4.83(1H,ddd,J = 6.4Hz, 6.6Hz, 8.5Hz), 6.53(1H,d,
 10 J = 8.5Hz), 6.95(1H,d,J = 7.8Hz), 7.26(1H,s), 7.30-7.40 (2H,m), 7.81-7.88(2H,m)

Compound 77

IR(KBr, cm⁻¹):

15 2775, 1712, 1666, 1531, 1429, 1203, 1138, 800, 762, 721

High Resolution FAB-MS(m/e, (C₃₂H₄₈N₄O₅S + H)⁺):

Calcd: 601.3423

Found: 601.3406

¹H-NMR(300MHz, CD₃OD, δ ppm):

20 0.84(9H,s), 0.8-1.9(17H,m), 1.16(6H,d,J = 7.1Hz), 3.2-3.5(2H,m), 4.08-4.38(4H,m), 4.8-4.9(1H,m), 7.28-7.41
 (2H,m), 7.31(1H,s), 7.83-7.90(2H,m)

Example 69

25 Synthesis of Compound 78 and 79

(1) Preparation of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(OH)-DNle-O^tBu

To a solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp-DNle-O^tBu (128 mg) in acetic acid (1.0
 30 ml) was added sodium cyanoborohydride (126 mg). The mixture was stirred at room temperature for 17 h,
 and water (10 ml) was added to the mixture. The mixture was extracted with ethyl acetate (10 ml × 3). The
 combined organic layers were washed with sat. aq. NaHCO₃ (3 ml), dried over MgSO₄ and evaporated in
 vacuo. The residue was purified by MPLC (Merck, LiChroprep Si 60) with hexane/ethyl acetate = 1/1 for
 35 elution to give a colorless powder (58.2 mg). To a solution of the product (32.1 mg) and sodium tungstate
 dihydrate (3.3 mg) in methanol (1.0 ml) and water (0.3 ml) was added 30% hydrogen peroxide (51 μl). After
 being stirred at room temperature for 1 h, the mixture was evaporated in vacuo. The residue was purified by
 silica gel chromatography (Wakogel, C-200) with hexane/ethyl acetate = 1/1 for elution to give the product
 (13.7 mg).

FAB-MS(m/e, (C₃₆H₅₇N₅O₆ + H)⁺): 656

40

(2) Preparation of Compound 78

A solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(OH)-DNle-O^tBu (11 mg, prepared in (1)) in
 formic acid (0.4 ml) was stirred at room temperature for 13 h. The mixture was concentrated under reduced
 45 pressure, and the residue was purified by preparative TLC (3M, Empore Silica Sheet) with
 chloroform/methanol/acetic acid = 20/1/1 for development to give the product (3.4 mg).

m.p.: 165 °C(dec.)

High Resolution FAB-MS(m/e, (C₃₂H₄₉N₅O₆ + H)⁺):

Calcd: 600.3761

50 Found: 600.3774

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.80(9H,s), 1.05(6H,d,J = 6.8Hz), 0.80-1.80(17H,m), 2.84(1H,dd,J = 9.7Hz, 14.4Hz), 3.10-3.20(1H,m), 4.00-
 4.25(4H,m), 4.35-4.45(1H,m), 6.10(1H,d,J = 7.1Hz), 6.95(1H,t,J = 7.5Hz), 7.09(1H,t,J = 7.5Hz), 7.17(1H,s), 7.28-
 (1H,d,J = 7.5Hz), 7.57(1H,d,J = 7.5Hz), 8.00-8.20 (2H,m)

55

(3) Preparation of Compound 79

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(OH)-DNle-O^tBu (11 mg, prepared in (1)) in methanol (3 ml) was treated with ethereal diazomethane. The mixture was evaporated in vacuo, and the residue was purified by silica gel chromatography (Wakogel, C-200) with hexane/ethyl acetate = 3/2 for elution to give a pale yellow powder (8.2 mg). The product (7.9 mg) was dissolved in formic acid (0.4 ml) and the mixture was stirred at room temperature for 14 h. The reaction mixture was evaporated in vacuo and the residue was purified by preparative TLC (3M, Empore Silica Sheet) with chloroform/methanol = 5/1 for development to give the product (4.1 mg).

m.p.: 115-120 °C

High Resolution FAB-MS(m/e, (C₃₃H₅₁N₅O₆ + H)⁺):

Calcd: 614.3917

Found: 614.3881

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.79(9H,s), 1.05(6H,d, J = 6.6Hz), 0.80-1.80(17H,m), 2.88(1H,dd, J = 9.1Hz, 14.6Hz), 3.13(1H,dd, J = 4.2Hz, 14.6Hz), 3.85-3.95(1H,m), 4.10-4.25(3H,m), 4.40-4.50 (1H,m), 3.98(3H,s), 6.11(1H,d, J = 7.9Hz), 7.01(1H,t, J = 7.5Hz), 7.15(1H,t, J = 7.5Hz), 7.29(1H,s), 7.35(1H,d, J = 7.5Hz), 7.59(1H,d, J = 7.5Hz), 7.80-7.90(1H,m), 7.90-8.05(1H,m)

Example 70

Synthesis of Compound 80,81,82 and 83(1) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-DNle-O^tBu

To a solution of 2,6-Dimethylpiperidinocarbonyl- γ -MeLeu-OH (64 mg), H-DTrp(7-OBzl)-OMe (61 mg) and HOBT·H₂O (35 mg) in dichloromethane (2 ml) was added EDCI·HCl (44 mg) under ice cooling. The mixture was stirred at the same temperature for 3 h and at room temperature over night. The mixture was diluted with ethyl acetate (30 ml), washed with sat. aq. NaHCO₃, 1N-hydrochloric acid and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 1/1 for elution to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-OMe (75 mg). The compound (73 mg) was dissolved in methanol (2.0 ml) and 1N-aqueous sodium hydroxide solution (1.0 ml) was added under ice cooling. The mixture was stirred at the same temperature for 6 h and water (30 ml) was added. The mixture was acidified with 1N-hydrochloric acid and extracted with dichloromethane (30 ml $\times$ 3). The combined organic layers were dried over MgSO₄ and evaporated to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-OH (56 mg). To a mixture of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-OH (55 mg), H-DNle-O^tBu·HCl (26 mg) and HOBT·H₂O (18 mg) in DMF (2 ml) were added NMM (13 μ l) and EDCI·HCl (23 mg) under ice cooling. The mixture was stirred at the same temperature for 2 h and at room temperature for 3 h. Water (50 ml) was added to the mixture and the mixture was extracted with ethyl acetate (20 ml $\times$ 3). The combined organic layers were washed with 10% aq. citric acid (20 ml), sat. aq. NaHCO₃ (20 ml) and brine (20 ml) successively, dried over MgSO₄ and evaporated in vacuo to give the product (72 mg).

FAB-MS(m/e, (C₄₈H₆₃N₅O₆ + H)⁺):746

(2) Preparation of Compound 80

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-DNle-O^tBu (9.8 mg, prepared in (1)) in dichloromethane (0.5 ml) were added methyl chloroformate (10 μ l), pulverized sodium hydroxide (5 mg) and TBAHS (2 mg), and the mixture was stirred at room temperature for 3 h. The mixture was diluted with dichloromethane, washed with water, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 1/1 for development to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp-(1-COOMe-7-OBzl)-DNle-O^tBu (5.4 mg). A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp-(1-COOMe-7-OBzl)-DNle-O^tBu (5.1 mg, obtained above) in methanol (1 ml) was hydrogenated over 10% Pd/C (5 mg) at an atmospheric pressure of hydrogen for 1 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 2/1 for elution. The product was dissolved in formic acid (1.0 ml) and the mixture was stirred at room temperature for 7 h. The reaction mixture was evaporated to give Compound 80 (1.1 mg).

High Resolution FAB-MS(m/e , $(C_{34}H_{51}N_5O_8 + H)^+$):

Calcd: 658.3816

Found: 658.3829

1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.85(3H,t,J = 7.0Hz), 0.88(9H,s), 1.14(3H,d,J = 7.0Hz), 1.15(3H,d,J = 7.0Hz), 1.07-2.13(14H,m), 3.19(1H,dd, J = 5.3Hz, 14.8Hz), 3.31(1H,dd, J = 6.6Hz, 14.8Hz), 3.82-4.00(2H,m), 4.05(3H,s), 4.07-4.18(1H,m), 4.23-4.33 (1H,m)-, 4.73-4.83(1H,m), 4.87(1H,d,J = 6.2Hz), 6.45 (1H,d,J = 8.6Hz), 6.89(1H,d,J = 7.3Hz), 7.03(1H,d, J = 7.3Hz), 7.18-(1H,t,J = 7.3Hz), 7.36(1H,s), 7.78(1H,d, J = 7.4Hz), 10.59(1H,s)

10 (3) Preparation of Compound 81

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OBzl)-DNle-O^tBu (60.6 mg, obtained in (1)) in methanol (5 ml) was hydrogenated over 10% Pd/C (20 mg) at an atmospheric pressure of hydrogen for 1 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OH)-DNle-O^tBu (50.2 mg). A mixture of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OH)-DNle-O^tBu (14.4 mg, obtained above) in formic acid (1.0 ml) was stirred at room temperature for 9 h. The mixture was evaporated and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 30/1/1 for development to give Compound 81 (7.3 mg).

20 High Resolution FAB-MS(m/e , $(C_{32}H_{49}N_5O_6 + H)^+$):

Calcd: 600.3761

Found: 600.3779

1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.79(3H,brs), 0.90(9H,s), 1.00-1.84(20H,m), 3.03-3.20 (1H,m), 3.27-3.46(1H,m), 3.90-4.30(4H,m), 4.70-4.90 (1H,m), 5.11(1H,brs), 6.55-7.20(7H,m), 9.44(1H,brs)

(4) Preparation of Compound 82

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OH)-DNle-O^tBu (33.9 mg, prepared in (3)) in diethyl ether (3 ml) and methanol (1 ml) was treated with ethereal diazomethane. The reaction mixture was evaporated and the residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 1/1 for elution to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OMe)-DNle-O^tBu (27.5 mg). A mixture of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OMe)-DNle-O^tBu (4.8 mg) in formic acid (1.0 ml) was stirred at room temperature for 7 h. The mixture was evaporated and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/ acetic acid = 30/1/1 for development to give Compound 82 (3.4 mg).

High Resolution FAB-MS(m/e , $(C_{33}H_{51}N_5O_6 + H)^+$):

Calcd: 614.3918

Found: 614.3904

40 1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.83(3H,t,J = 7.2Hz), 0.88(9H,s), 1.12(6H,d,J = 6.8Hz), 1.00-1.92(14H,m), 3.20(1H,dd,J = 5.7Hz, 14.7Hz), 3.45 (1H,dd,J = 5.7Hz, 14.7Hz), 3.80-4.00(2H,m), 3.94(3H,s), 4.05-4.20(1H,m), 4.20-4.32(1H,m), 4.68-4.78(1H,m), 4.99-(1H,brs), 6.56(1H,brs), 6.63(1H,d,J = 7.8Hz), 7.00(1H,t,J = 7.8Hz), 7.07(1H,s), 7.21(1H,d,J = 7.8Hz), 7.55(1H,brs)-, 8.54(1H,brs)

45

(5) Preparation of Compound 83

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OMe)-DNle-O^tBu (17.9 mg, prepared in (3)) in dichloromethane (1.0 ml) were added methyl chloroformate (20 μ l), pulverized sodium hydroxide (20 mg) and TBAHS (5 mg), and the mixture was stirred at room temperature for 3 h. The mixture was diluted with dichloromethane, washed with water, dried over $MgSO_4$ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 1/1 for development to give 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(1-COOMe-7-OMe)-DNle-O^tBu (12.4 mg). A mixture of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(7-OMe)-DNle-O^tBu (9.2 mg) in TFA (1 ml) was stirred at room temperature for 1 h and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 30/1/1 for development to give Compound 83 (8.1 mg).

High Resolution FAB-MS(m/e , $(C_{35}H_{53}N_5O_8 + H)^+$):

Calcd: 672.3972

Found: 672.4007

 $^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

0.84(3H,t,J = 7.0Hz), 0.84(9H,s), 1.15(6H,d,J = 6.8Hz), 1.09-1.94(14H,m), 3.25(2H,d,J = 6.4Hz), 3.89-4.04
 5 (2H,m), 3.95(3H,s), 3.97(3H,s), 4.11-4.22(1H,m), 4.23-4.33(1H,m), 4.66-4.77(1H,m), 5.06(1H,d,J = 6.1Hz), 6.66-
 (1H,d,J = 6.5Hz), 6.86(1H,t,J = 4.7Hz), 7.19(2H,d, J = 4.7Hz), 7.44(1H,s), 7.63(1H,d,J = 7.8Hz)

Example 71

10 Synthesis of Compound 84

To a solution of Compound 40 (27 mg, prepared in Example 40) in methanol (0.5 ml) was added a
 solution of sodium bicarbonate (3.5 mg) in water (0.1 ml). The mixture was evaporated in vacuo and the
 residue was dissolved in DMF (0.5 ml). Iodomethane (0.2 ml) was added to the mixture, and the mixture
 15 was stirred at room temperature for 20 h. The mixture was evaporated in vacuo and the residue was purified
 by preparative TLC (Merck, Kieselgel 60 F_{254}) with hexane/ ethyl acetate = 1/1 for development to give the
 product (11 mg).

m.p.: 80-88 °C

IR(KBr, cm^{-1}):

20 3412, 2956, 1746, 1656, 1623, 1560, 1521, 1461, 1386, 1341, 1260, 1092

High Resolution FAB-MS(m/e , $(\text{C}_{35}\text{H}_{53}\text{N}_5\text{O}_7 + \text{H})^+$):

Calcd: 656.4023

Found: 656.3996

 $^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

25 0.82(3H,t,J = 7.2Hz), 0.89(9H,s), 1.05-1.85(14H,m), 1.15(3H,d,J = 7.1Hz), 1.16(3H,d,J = 6.8Hz), 3.17(1H,dd,
 J = 5.7Hz, 15.0Hz), 3.39(1H,dd,J = 6.2Hz, 15.0Hz), 3.65 (3H,s), 3.90-4.06(2H,m), 4.06-4.20(1H,m), 4.36-4.47
 (1H,m), 4.64(1H,d,J = 6.8Hz), 4.78-4.90(1H,m), 6.38 (1H,d,J = 9.2Hz), 7.27(1H,brs), 7.25(1H,t,J = 7.3Hz), 7.34-
 (1H,t,J = 7.3Hz), 7.47(1H,s), 7.61(1H,d,J = 7.3Hz), 8.17(1H,d,J = 7.3Hz)

30 Example 72

Synthesis of Compound 85

Compound 85 was prepared using iodoethane instead of iodomethane in the same manner described in
 35 Example 71.

m.p.: 80-85 °C

IR(KBr, cm^{-1}):

3412, 2950, 1746, 1653, 1626, 1533, 1461, 1386, 1257, 1194, 1089, 465

High Resolution FAB-MS(m/e , $(\text{C}_{36}\text{H}_{55}\text{N}_5\text{O}_7 + \text{H})^+$):

40 Calcd: 670.4180

Found: 670.4161

 $^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

0.82(3H,t,J = 7.2Hz), 0.89(9H,s), 1.05-1.85(14H,m), 1.15(3H,d,J = 7.1Hz), 1.16(3H,d,J = 7.1Hz), 1.22(3H,t,
 J = 7.3Hz), 3.17(1H,dd,J = 5.8Hz, 14.7Hz), 3.38(1H,dd, J = 6.1Hz, 14.7Hz), 3.90-4.05(2H,m), 4.00(3H,s), 4.09-
 45 (2H,dq,J = 1.3Hz, 7.2Hz), 4.05-4.20(1H,m), 4.35-4.45 (1H,m), 4.66(1H,d,J = 7.3Hz), 4.75-4.88(1H,m), 6.49
 (1H,d,J = 8.6Hz), 7.18(1H,d,J = 7.3Hz), 7.24(1H,t, J = 7.3Hz), 7.33(1H,t,J = 7.3Hz), 7.48(H,s), 7.61(1H,d, J = 7.3Hz)-
 , 8.17(1H,d,J = 8.0Hz)

Example 73

50

Synthesis of Compound 86

Compound 86 was prepared using 2-iodopropane instead of iodomethane in the same manner
 described in Example 71.

55 m.p.: 75-80 °C

IR(KBr, cm^{-1}):

3400, 2950, 2872, 1745, 1656, 1623, 1527, 1461, 1386, 1341, 1311, 1260, 1197, 1146, 1107, 1092, 765

High Resolution FAB-MS(m/e , $(\text{C}_{37}\text{H}_{57}\text{N}_5\text{O}_7 + \text{H})^+$):

Calcd: 684.4337

Found: 684.4334

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.82(3H,t,J = 7.2Hz),0.88(9H,s),1.05-1.85(14H,m), 1.15(3H,d,J = 6.8Hz),1.16(3H,d,J = 7.1Hz),1.19(3H,d,
 5 J = 6.3Hz),1.21(3H,d,J = 6.3Hz),3.17(1H,dd,J = 5.9Hz, 14.9Hz),3.37(1H,dd,6.4Hz,14.9Hz),3.90-4.05(2H,m),
 4.00(3H,s),4.05-4.20(1H,m),4.31-4.42(1H,m),4.67 (1H,d,J = 6.8Hz),4.75-4.86(1H,m),4.94(1H,sept, J = 6.3Hz)-
 ,6.49(1H,d,J = 8.4Hz),7.12(1H,d,J = 7.6Hz), 7.24(1H,t,J = 7.8Hz),7.33(1H,t,J = 7.8Hz),7.48(1H,s), 7.62-
 (1H,d,J = 7.8Hz),8.17(1H,d,J = 7.8Hz)

10 Example 74

Synthesis of Compounds 87 and 88

To a solution of Compound 84 (94 mg, prepared in Example 71) in dry THF (1 ml) were added lithium
 15 chloride (12.3 mg), sodium borohydride (11 mg) and dry ethanol (2 ml), and the mixture was stirred at room
 temperature over night. The mixture was cooled with ice-water, adjusted to pH 4 by the gradual addition of
 10% aq. citric acid, and concentrated in vacuo. Water (20 ml) was added to the residue, and the mixture
 was extracted with dichloromethane (10 ml × 3). Combined organic layers were washed with brine, dried
 over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60
 20 F₂₅₄) with ethyl acetate for development to give Compound 87 (21 mg) and Compound 88 (11 mg).

Compound 87

IR(KBr,cm⁻¹):

25 3400,2950,2866,1743,1653,1620,1536,1461,1386,1260, 1092,765

High Resolution FAB-MS(m/e,(C₃₄H₅₃N₅O₆ + H)⁺):

Calcd: 628.4074

Found: 628.4075

¹H-NMR(300MHz,CDCl₃, δ ppm):

30 0.86(3H,t,J = 7.1Hz),0.87(9H,s),1.15(3H,d,J = 6.9Hz), 1.16(3H,d,J = 6.9Hz),1.05-1.85(14H,m),3.22(1H,dd,
 J = 4.9Hz,14.8Hz),3.45(1H,dd,J = 6.1Hz,14.8Hz),3.45-3.65(2H,m),3.68-3.79(1H,m),3.79-3.98(2H,m),4.01 (3H,s)-
 ,4.04-4.17(1H,m),4.65(1H,d,J = 6.1Hz),4.75-4.86 (1H,m),5.95(1H,d,J = 8.8Hz),7.16(1H,d,J = 8.1Hz),7.26
 (1H,dt,J = 1.3Hz,7.8Hz),7.36(1H,dt,J = 1.3Hz,7.8Hz), 7.48(1H,s),7.59(1H,dd,J = 1.3Hz,7.8Hz),8.19(1H,dd,
 J = 1.3Hz,7.8Hz)

35

Compound 88

IR(KBr,cm⁻¹):

3304,2950,2866,1743,1653,1611,1539,1461,1386,1254, 1089,1044,756

40 High Resolution FAB-MS(m/e,(C₃₅H₅₅N₅O₆ + H)⁺):

Calcd: 642.4231

Found: 642.4227

¹H-NMR(300MHz,CDCl₃, δ ppm):

45 0.85(3H,t,J = 7.0Hz),0.87(9H,s),1.15(3H,d,J = 7.1Hz), 1.16(3H,d,J = 7.1Hz),1.05-1.83(14H,m),1.46(3H,t,
 J = 7.0Hz),3.23(1H,dd,J = 5.3Hz,14.8Hz),3.44(1H,dd, J = 6.2Hz,14.8Hz),3.45-3.65(2H,m),3.68-3.78(1H,m), 3.78-
 3.98(2H,m),4.05-4.16(1H,m),4.40-4.53(2H,m), 4.64(1H,d,J = 6.1Hz),4.73-4.85(1H,m),5.95(1H,d, J = 8.8Hz),7.15-
 (1H,d,J = 8.1Hz),7.25(1H,dt,J = 1.2Hz, 7.5Hz),7.35(1H,dt,J = 1.2Hz,7.5Hz),7.50(1H,s), 7.59-
 (1H,dd,J = 1.2Hz,7.5Hz),8.19(1H,dd,J = 1.2Hz,7.5Hz)

50 Example 75

Synthesis of Compounds 89 and 90

Compound 89 (ester form) and Compound 90 (deprotected form) were prepared using 2,6-dimethyl-
 55 piperidine and H-Cpeg-OBzl•HCl instead of cyclopentylpropylamine and H-Leu-OBzl-TsOH in the same
 manner described in Example 36.

Compound 89

High Resolution FAB-MS(m/e , $(C_{38}H_{57}N_5O_7 + H)^+$):

Calcd: 696.4337

Found: 696.4319

 1H -NMR(300MHz, $CDCl_3$, δ ppm):

0.82(3H,t,J = 7.1Hz), 0.96-1.81(20H,m), 1.17(6H,d, J = 7.4Hz), 1.41(9H,s), 2.09-2.20(1H,m), 3.21(1H,dd, J = 5.7Hz, 14.7Hz), 3.33(1H,dd, J = 6.5Hz, 14.7Hz), 3.64-3.75(1H,m), 3.90-4.05(1H,m), 4.12-4.24(1H,m), 4.01 (3H,s)-, 4.25-4.35(1H,m), 4.82(1H,d, J = 6.9Hz), 4.80-4.93 (1H,m), 6.33(1H,d, J = 9.1Hz), 7.16(1H,d, J = 7.8Hz), 7.25 (1H,dt, J = 1.4Hz, 7.7Hz), 7.33(1H,dt, J = 1.4Hz, 7.7Hz), 7.47(1H,s), 7.63(1H,dd, J = 1.4Hz, 7.7Hz), 8.16(1H,dd, J = 1.4Hz, 7.7Hz)

Compound 90

High Resolution FAB-MS(m/e , $(C_{34}H_{49}N_5O_7 + H)^+$):

Calcd: 640.3710

Found: 640.3607

 1H -NMR(300MHz, $DMSO-d_6$, δ ppm):

0.86(3H,t,J = 6.9Hz), 1.03(3H,d, J = 6.8Hz), 1.05(3H,d, J = 6.8Hz), 0.92-1.80(20H,m), 1.80-1.91(1H,m), 2.81 (1H,dd, J = 11.6Hz, 14.6Hz), 3.30(1H,dd, J = 2.6Hz, 14.6Hz), 3.50-3.84(2H,m), 3.96(3H,s), 4.03-4.20(2H,m), 4.50-4.62(1H,m), 6.05(1H,d, J = 6.8Hz), 7.25(1H,dt, J = 1.3Hz, 7.3Hz), 7.32(1H,dt, J = 1.3Hz, 7.3Hz), 7.53(1H,s), 7.64 (1H,dd, J = 1.3Hz, 7.3Hz), 8.05(1H,dd, J = 1.3Hz, 7.3Hz), 8.23(1H,d, J = 7.6Hz), 8.33(1H,d, J = 8.8Hz)

Example 76

Synthesis of Compound 91

Compound 91 was prepared using H-D β Abu-O^tBu instead of H-DNle-O^tBu in the same manner described in Example 64.

m.p.: 122-126 °C

High Resolution FAB-MS(m/e , $(C_{30}H_{41}N_5O_7 + H)^+$):

Calcd: 584.3085

Found: 584.3079

 1H -NMR(300MHz, $DMSO-d_6$, δ ppm):

0.03-0.42(4H,m), 0.83-1.00(1H,m), 1.07(3H,d, J = 6.8Hz), 1.09(3H,d, J = 7.0Hz), 1.13(3H,d, J = 6.6Hz), 1.33-1.82 (6H,m), 2.15(1H,dd, J = 8.9Hz, 15.0Hz), 2.33-2.41(1H,m), 2.75-2.90(1H,m), 3.17-3.24(4H,m), 3.96(3H,s), 4.05-4.23- (1H,m), 4.34-4.46(1H,m), 6.37(1H,d, J = 5.6Hz), 7.26(1H,t, J = 7.1Hz), 7.34(1H,t, J = 7.1Hz), 7.50(1H,s), 7.63- (1H,d, J = 7.1Hz), 7.92(1H,d, J = 7.1Hz), 8.05(1H,d, J = 8.1Hz), 8.60(1H,d, J = 8.6Hz)

Example 77

Synthesis of Compound 92

(1) Preparation of 2,6-dimethylpiperidinocarbonyl- γ MeLeu-DTrp(COOMe)-OH

To a solution of Boc-DTrp-OBzl (12.19 g) in dichloromethane (60 ml) were added methyl chloroformate (4.8 ml), pulverized sodium hydroxide (4.1 g) and TBAHS (9.2 g), and the mixture was stirred at room temperature for 4 h. The reaction mixture was washed with water, dried over $MgSO_4$ and evaporated in vacuo. The residue was crystallized from dichloromethane and hexane to give Boc-DTrp(COOMe)-OBzl (12.57 g). Boc-DTrp(COOMe)-OBzl (4.52 g) was dissolved in 4N-hydrogen chloride/1,4-dioxane (50 ml) and the mixture was stirred at room temperature for 45 min. Ethyl ether (50 ml) was added to the mixture to complete the precipitation. The precipitate was collected by filtration and dried in vacuo to give H-DTrp(COOMe)-OBzl $\cdot$ HCl (3.71 g). To a suspension of H-DTrp(COOMe)-OBzl $\cdot$ HCl (1.94 g) in dichloromethane (50 ml) were added NMM (0.55 ml), 2,6-dimethylpiperidinocarbonyl- γ MeLeu-OH (1.42 g), HOBT $\cdot$ H₂O (0.84 g) and EDCI $\cdot$ HCl (1.05 g) under ice cooling. The mixture was stirred at the same temperature for 2 h and at room temperature over night. The reaction mixture was washed with sat. aq $NaHCO_3$ (50 ml), 10% aq. citric acid (50 ml) and brine (50 ml) successively, dried over $MgSO_4$ and evaporated. The residue was purified by MPLC (Merck, LiChroprep Si 60) with hexane/ethyl acetate = 2/1 for elution to give 2,6-

dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(COOMe)-OBzl (1.17 g). A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(COOMe)-OBzl (1.10 g) in methanol (30 ml) was hydrogenated over 10% Pd/C (50 mg) at an atmospheric pressure of hydrogen for 1.5 h. After removal of the catalyst by filtration, the filtrate was evaporated in vacuo to give the product (0.93 g).

(2) Preparation of Compound 92

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(COOMe)-OH (53 mg, prepared in (1)) and HOSu (17 mg) in DMF (1.0 ml) was added EDCI•HCl (29 mg) under ice cooling, and the mixture was stirred at the same temperature for 3.5 h. To the mixture was added a solution of glycine (11 mg) and NMM (16 μ l) in water (1.0 ml). The mixture was stirred at the same temperature for 2 h and at room temperature over night.

To the mixture was added 1N-hydrochloric acid (20 ml), and the mixture was extracted with dichloromethane (10 ml $\times$ 3). The combined organic layers were washed with brine (10 ml), dried over MgSO₄ and evaporated in vacuo.

The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 20/1/1 for development to give Compound 92 (23 mg).

m.p.: 124-127 °C

IR(KBr, cm⁻¹):

2960, 1738, 1659, 1612, 1531, 1458, 1444, 1383, 1259, 1090, 766, 748

High Resolution FAB-MS(m/e, (C₃₀H₄₃N₅O₇ + H)⁺):

Calcd: 586.3241

Found: 586.3232

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.74(9H,s), 1.16(3H,d, J = 6.8Hz), 1.19(3H,d, J = 6.8Hz), 1.31-1.82(8H,m), 3.14(1H,dd, J = 10.0Hz, 14.8Hz), 3.49(1H,dd, 4.4Hz, 14.8Hz), 3.74(1H,dd, J = 6.3Hz, 16.7Hz), 3.90-4.14(3H,m), 3.97(3H,s), 4.22(1H,dd, J = 6.3Hz, 16.7Hz), 4.77-4.93(2H,m), 7.13-7.36(3H,m), 7.51(1H,s), 7.57(1H,d, J = 7.7Hz), 7.87(1H,t, J = 6.3Hz), 8.13(1H,d, J = 7.7Hz)

Each Compound 93-95 in the following Examples 78-80 was prepared using each corresponding amino acid instead of glycine in the same manner described in Example 77.

Example 78

Compound 93

m.p.: 113-116 °C

IR(KBr, cm⁻¹):

3300, 2958, 1738, 1653, 1539, 1456, 1385, 1259, 1092, 766, 748

High Resolution FAB-MS(m/e, (C₃₁H₄₅N₅O₇ + H)⁺):

Calcd: 600.3397

Found: 600.3408

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.73(9H,s), 1.17(3H,d, J = 6.9Hz), 1.21(3H,d, J = 6.9Hz), 1.10-2.10(8H,m), 2.49-2.70(2H,m), 3.14(1H,dd, J = 9.1Hz, 14.7Hz), 3.38(1H,dd, 4.8Hz, 14.7Hz), 3.08-3.48(2H,m), 3.73-3.89(1H,m), 4.01(3H,s), 3.94-4.18(2H,m), 4.21-4.84(2H,m), 7.13(1H,d, J = 9.3Hz), 7.24(1H,t, J = 7.7Hz), 7.31(1H,t, J = 7.7Hz), 7.46(1H,t, J = 5.3Hz), 7.52(1H,s), 7.62(1H,d, J = 7.7Hz), 8.13(1H,d, J = 7.7Hz)

Example 79

Compound 94

m.p.: 137-141 °C

IR(KBr, cm⁻¹):

3400, 2956, 2576, 1734, 1653, 1522, 1456, 1385, 1259, 1092, 744

High Resolution FAB-MS(m/e, (C₃₃H₅₀N₆O₇ + H)⁺):

Calcd: 715.3819

Found: 715.3815

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.74(9H,s), 1.03(3H,d,J = 6.7Hz), 1.05(3H,d,J = 6.7Hz), 1.00-1.78(8H,m), 2.75-2.92(1H,m), 3.07-3.32(3H,m), 3.94(3H,s), 4.02-4.26(3H,m), 4.42-4.60(2H,m), 6.05 (1H,d,J = 7.9Hz), 7.00(1H,t,J = 7.8Hz), 7.06(1H,t, J = 7.8Hz)-, 7.20(1H,d,J = 2.4Hz), 7.23(1H,t,J = 7.8Hz), 7.31(1H,t,J = 7.8Hz), 7.33(1H,d,J = 7.8Hz), 7.51(1H,d, J = 7.8Hz), 7.49-(1H,s), 7.65(1H,d,J = 7.8Hz), 8.04(1H,d, J = 7.8Hz), 8.19(1H,d,J = 10.5Hz), 8.37(1H,d,J = 7.6Hz), 10.83-(1H,d,J = 2.4Hz)

Example 80

Compound 95

m.p.: 103-108 °C

IR(KBr, cm⁻¹):

3300, 2958, 2939, 1738, 1653, 1533, 1456, 1444, 1383, 1259, 1092, 766, 748

High Resolution FAB-MS(m/e, (C₃₅H₅₃N₅O₇ + H)⁺):

Calcd: 656.4023

Found: 656.4044

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.82(3H,t,J = 7.0Hz), 0.83(9H,s), 1.18(3H,d,J = 6.8Hz), 1.19(3H,d,J = 6.8Hz), 1.08-1.82(14H,m), 2.40(1H,dd, J = 5.2Hz, 15.4Hz), 2.53(1H,dd, J = 6.7Hz, 15.4Hz), 3.22 (1H,dd, J = 6.8Hz, 14.9Hz), 3.29(1H,dd, J = 6.8Hz, 14.9Hz), 3.86-4.24(4H,m), 4.02(3H,s), 4.67-4.79(1H,m), 6.48(1H,d,J = 6.9Hz), 7.16(1H,brs), 7.21-7.40(2H,m), 7.32-(1H,t,J = 7.8Hz), 7.48(1H,s), 7.60(1H,d,J = 7.8Hz), 8.16(1H,d,J = 7.8Hz)

Example 81

Synthesis of Compound 96

To a solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(COOMe)-OH (53 mg, prepared in Example 77-(1)) and HOSu (17 mg) in DMF (1.0 ml) was added EDCI•HCl (29 mg) under ice cooling, and the mixture was stirred at the same temperature for 3.5 h. To the mixture was added a solution of aminomethanesulfonic acid (17 mg) and NMM (16 μl) in water (1.0 ml). The mixture was stirred at the same temperature for 2 h and at room temperature over night. The mixture was purified by reverse phase MPLC (Merck, LiChroprep RP-18) with methanol/water (containing 0.1% TFA) = 3/2 for elution to give the product (26 mg).

m.p.: 135-140 °C

IR(KBr, cm⁻¹):

1782, 1737, 1666, 1547, 1458, 1385, 1259, 1207, 1169, 1092, 1041, 768

High Resolution FAB-MS(m/e, (C₂₉H₄₃N₅O₈S + H)⁺):

Calcd: 622.2911

Found: 622.2953

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.73(9H,s), 1.02(3H,d,J = 6.9Hz), 1.03(3H,d,J = 6.9Hz), 0.98-1.73(8H,m), 2.88(1H,dd, J = 10.2Hz, 15.1Hz), 3.09(1H,dd, 4.3Hz, 15.1Hz), 3.93(3H,s), 3.82-4.40(5H,m), 4.59-4.69(1H,m), 5.93(1H,brs), 7.22(1H,t, J = 7.7Hz), 7.30-(1H,t, J = 7.7Hz), 7.52(1H,s), 7.74(1H,d, J = 7.7Hz), 8.03(1H,d, J = 7.7Hz), 8.03(1H,d, J = 8.0Hz), 8.44(1H,t, J = 6.1Hz)

Each Compound 97-99 in the following Examples 82-84 was prepared using each corresponding aminosulfonic acid instead of aminomethanesulfonic acid in the same manner described in Example 81.

Example 82

Compound 97

m.p.: 115-120 °C

IR(KBr, cm⁻¹):

1780, 1738, 1666, 1549, 1456, 1385, 1259, 1209, 1168, 1041, 768, 748

High Resolution FAB-MS(m/e, (C₃₀H₄₅N₅O₈S + H)⁺):

Calcd: 674.2626

Found: 674.2632

¹H-NMR(300MHz, DMSO-d₆, δ ppm):

0.73(9H,s),1.03(3H,d,J = 6.2Hz),1.06(3H,d,J = 6.2Hz), 1.18-1.80(8H,m),2.42-2.66(2H,m),2.88-
(1H,dd,J = 4.4Hz, 10.6Hz),3.13-3.43(3H,m),3.94(3H,s),4.00-4.21(2H,m), 4.30-4.80(2H,m),6.10(1H,brs),7.23-
(1H,t,J = 7.6Hz), 7.32(3H,t,J = 7.6Hz),7.47(1H,s),7.84(1H,d,J = 7.6Hz), 8.04(1H,d,J = 7.6Hz),8.08(1H,t,J = 5.5Hz)-
,8.28(1H,d, J = 7.9Hz)

5

Example 83

Compound 98

10 m.p.: 141-144 °C

IR(KBr,cm⁻¹):

3400,2958,1778,1738,1660,1539,1456,1383,1259,1153, 1092,1036,768

High Resolution FAB-MS(m/e,(C₃₄H₅₃N₅O₈S + H)⁺):

Calcd: 692.3693

15

Found: 692.3713

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.73(9H,s),0.83(3H,t,J = 6.7Hz),1.04(3H,d,J = 6.7Hz), 1.05(3H,d,J = 6.7Hz),0.97-2.00(14H,m),2.42-2.53
(2H,m),2.87(1H,dd,J = 10.6Hz,15.0Hz),3.21(1H,dd, J = 3.2Hz,15.0Hz),3.94(3H,s),3.90-4.37(5H,m),6.10
(1H,d,J = 6.6Hz),7.24(1H,t,J = 7.4Hz),7.31(1H,t, J = 7.4Hz),7.51(1H,s),7.66(1H,d,J = 7.4Hz),7.75(1H,d,
20 J = 8.5Hz),8.04(1H,d,J = 7.4Hz),8.21(1H,d,J = 7.7Hz)

Example 84

Compound 99

25

IR(KBr,cm⁻¹):

2954,2866,1778,1738,1666,1539,1477,1456,1383,1257, 1169,1092,1041,768,748,627

High Resolution FAB-MS(m/e,(C₃₃H₅₁N₅O₈S + H)⁺):

Calcd: 678.3537

30

Found: 678.3507

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.72 + 0.75(9H,s),0.80-1.96(23H,m),2.80-2.93(1H,m), 3.05-3.18(1H,m),3.93(3H,s),3.99-4.71(5H,m),5.88 +
6.01(1H,brs),7.21(1H,t,J = 8.0Hz),7.30(1H,t,J = 8.0Hz), 7.72 + 7.75(1H,d×2,J = 8.0Hz), 7.56(1H,s),7.83-8.12
(3H,m)

35

Example 85

Synthesis of Compound 100

40

To a solution of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(COOMe)-OH (106 mg, prepared in Exam-
ple 77-(1)), diethyl (R)-1-aminopentylphosphonate (67 mg) and HOBt·H₂O (51 mg) in dichloromethane (10
ml) was added EDCI·HCl (73 mg) under ice cooling, and the mixture was stirred at room temperature for 16
h. The mixture was diluted with ethyl acetate, washed with 10% aq. citric acid, water, sat. aq. NaHCO₃ and
brine successively, dried over MgSO₄ and evaporated in vacuo.

45

The residue was purified by silica gel chromatography (Merck,Kieselgel 60) with hexane/ethyl acetate
= 1/1 for elution to give the diethyl ester of Compound 100 (61 mg). To a solution of the diethyl ester (41
mg) in dichloromethane (0.8 ml) was added bromotrimethylsilane (0.8 ml) under ice cooling. The mixture
was stirred at room temperature for 4 h, and concentrated in vacuo. The residue was dissolved in 90%
acetone/water (5 ml). The mixture was stirred at 45 min and evaporated in vacuo. The residue was triturated
50 with water to give Compound 100 (25 mg).

m.p.: 205-210 °C

IR(KBr,cm⁻¹):

2954,2949,1736,1649,1524,1458,1383,1259,1092,768

High Resolution FAB-MS(m/e,(C₃₃H₅₂N₅O₈P + H)⁺):

55

Calcd: 678.3632

Found: 678.3674

¹H-NMR(300MHz,CD₃OD, δ ppm):

0.82(9H,s),0.92(3H,t,J = 6.8Hz),1.13(3H,d,J = 6.8Hz), 1.14(3H,d,J = 6.8Hz),1.20-2.00(14H,m),3.0-3.1(1H,m),

3.3-3.4(1H,m),4.01(3H,s),4.05-4.30(4H,m),4.7-4.8 (1H,m),7.25(1H,t,J = 7.3Hz),7.31(1H,t,J = 7.3Hz), 7.53(1H,s)-
 ,7.67(1H,d,J = 7.3Hz),8.12(1H,d,J = 7.3Hz)

Example 86

5

Synthesis of Compound 101

To a solution of Compound 40 (32 mg, prepared in Example 40), 2-aminoethanol (4 mg) and
 HOBT·H₂O (10 mg) was added EDCI·HCl (13 mg) under ice cooling. The mixture was stirred at the same
 10 temperature for 1 h and at room temperature over night. The mixture was diluted with ethyl acetate (20 ml),
 washed with sat. aq. NaHCO₃ (20 ml), 10% aq. citric acid (20 ml) and brine successively, dried over MgSO₄
 and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with
 chloroform/methanol = 30/1 for development to give the product (23.5 mg).

m.p.: 106-109 °C

15 IR(KBr,cm⁻¹):

1736,1655,1610,1524,1458,1383,1257,1088,1257,1088, 766

High Resolution FAB-MS(m/e,(C₃₆H₅₆N₆O₇ + H)⁺):

Calcd: 685.4289

Found: 685.4280

20 ¹H-NMR(300MHz,CDCl₃, δ ppm):

0.78(9H,s),0.84(3H,t,J = 7.1Hz),1.13(6H,d,J = 7.1Hz), 1.04-1.79(13H,m),2.00-2.15(1H,m),3.22-3.43(4H,m),
 3.55-3.80(3H,m),3.82-4.17(2H,m),4.04(3H,s),4.44-4.56(2H,m),4.59-4.71(1H,m),4.68(1H,d,J = 4.7Hz), 6.31-
 (1H,d,J = 6.8Hz),6.75(1H,t,J = 4.3Hz),7.28(1H,t, J = 7.8Hz),7.38(1H,t,J = 7.8Hz),7.52(1H,s),7.59(1H,d, J = 7.8Hz)-
 ,7.73(1H,d,J = 9.0Hz),8.20(1H,d,J = 7.8Hz)

25

Example 87

Synthesis of Compound 102

30 Compound 102 was prepared by deprotection of 2,6-dimethylpiperidinocarbonyl- γMeLeu-DTrp-DNle-
 O^tBu with formic acid.

m.p.: 149-153 °C

IR(KBr,cm⁻¹):

3309,2947,1653,1522,1387,743

35 High Resolution FAB-MS(m/e,(C₃₂H₄₉N₅O₅ + H)⁺):

Calcd: 584.3812

Found: 584.3788

¹H-NMR(300MHz,CD₃OD, δ ppm):

40 0.8-1.9(17H,m),0.85(9H,s),1.13(3H,d,J = 7.1Hz), 1.15(3H,d,J = 7.1Hz),3.2-3.4(2H,m),4.0-4.4(4H,m), 4.7-4.8-
 (1H,m),6.97(1H,t,J = 7.9Hz),7.05(1H,t, J = 7.9Hz),7.08(1H,s),7.29(1H,d,J = 7.9Hz),7.59(1H,d, J = 7.9Hz)

Example 88

Synthesis of Compound 103

45

Dry hydrogen chloride gas was bubbled into a solution of Compound 102 (70 mg) in formic acid (3 ml),
 and the mixture was stirred at room temperature for 2.5 h. The mixture was evaporated in vacuo and the
 residue was purified by preparative TLC (Merck,Kieselgel 60 F₂₅₄) with chloroform/methanol = 10/1 for
 elution to give the product (45 mg).

50 m.p.: 132-135 °C

IR(KBr,cm⁻¹):

2954,1716,1653,1608,1522,1458,1387,795,750

High Resolution FAB-MS(m/e,(C₃₃H₄₉N₅O₆ + H)⁺):

Calcd: 612.3761

55 Found: 612.3746

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

0.75(9H,s),0.78-0.89(3H,m),1.01(3H,d,J = 6.8Hz), 1.02(3H,d,J = 6.8Hz),1.05-1.8(14H,m),2.85-3.00(1H,m),
 3.2-3.5(2H,m),3.95-4.20(4H,m),4.5-4.6(1H,m),6.03-6.09(1H,m),7.26-7.36(2H,m),7.51-7.62(1H,m),7.69

(1H,d,J = 7.6Hz), 7.95-8.25(2.5H,m), 9.18(0.5H,brs)

Example 89

5 Synthesis of Compound 104

To a solution of Compound 81 (7.3 mg, prepared in Example 70) in dichloromethane (0.5 ml) were added acetyl chloride (1.3 μ l) and DMAP (3.7 mg) under ice cooling. The mixture was stirred at the same temperature for 4 h and at room temperature for 2 h. Ethyl acetate (20 ml) was added to the mixture, and
 10 the mixture was washed with 1N-hydrochloric acid (20 ml) and brine (20 ml), dried over MgSO_4 and evaporated in vacuo.

The residue was purified by preparative TLC (Merck, Kieselgel 60 F_{254}) with chloroform/methanol/acetic acid = 20/1/1 for elution to give the product (2.2 mg).

High Resolution FAB-MS(m/e , $(\text{C}_{34}\text{H}_{51}\text{N}_5\text{O}_7 + \text{H})^+$):

15 Calcd: 642.3867

Found: 642.3851

$^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

0.83(3H,t,J = 7.0Hz), 0.88(9H,s), 1.14(6H,d,J = 5.8Hz), 1.07-1.87(13H,m), 2.12-2.42(1H,m), 2.38(3H,s), 3.18
 (1H,dd,J = 6.0Hz, 14.4Hz), 3.40(1H,dd,J = 6.0Hz, 14.4Hz), 3.93-4.30(4H,m), 4.68-4.80(1H,m), 4.95-
 20 (1H,d,J = 6.0Hz), 6.70(1H,d,J = 6.5Hz), 6.96(1H,d,J = 7.6Hz), 7.06(1H,t, J = 7.6Hz), 7.06(1H,s), 7.31-
 (1H,d,J = 6.5Hz), 7.49(1H,d, J = 7.6Hz), 8.62(1H,s)

Example 90

25 Synthesis of Compound 105

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(COOMe)-OH (20 mg) and NMM (4.2 μ l) in dry THF (2.0 ml) was added isopropyl chloroformate (5.3 μ l) at -25°C over 5 min. To the mixture was added a solution of (R)-5-(1-aminopentyl)-1H-tetrazole hydrochloride (10.9 mg) and NMM (6.3 μ l) in dry
 30 THF (2.0 ml) at -20°C . The mixture was stirred at -20°C - -15°C for 2 h and at 10°C for 10 h, then concentrated under reduced pressure. The residue was partitioned between ethyl acetate (10 ml) and water (10 ml). The organic layer was washed with 1N-hydrochloric acid (10 ml $\times$ 2) and water (10 ml), dried over anhydrous sodium sulfate and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F_{254}) with chloroform/methanol = 20/1 for development to give the product (13.5 mg).

35 m.p.: $111-116^\circ\text{C}$

IR(KBr, cm^{-1}):

3400, 3300, 2942, 2870, 1659, 1612, 1529, 1456, 1383, 1259, 1091, 766, 748

High Resolution FAB-MS(m/e , $(\text{C}_{34}\text{H}_{51}\text{N}_9\text{O}_5 + \text{H})^+$):

Calcd: 666.4092

40 Found: 666.4113

$^1\text{H-NMR}$ (300MHz, CDCl_3 , δ ppm):

0.83(9H,s), 0.88(3H,d,J = 6.7Hz), 0.98(3H,t,J = 6.7Hz), 0.99(3H,d,J = 6.7Hz), 1.08-1.90(13H,m), 2.16-2.30
 (1H,m), 3.30-3.48(2H,m), 3.54(1H,quint,J = 6.7Hz), 3.61-3.82(2H,m), 4.05(3H,s), 4.62-4.73(1H,m), 4.66
 (1H,d,J = 6.0Hz), 5.44(1H,dt,J = 3.9Hz, 9.7Hz), 6.32(1H,d, J = 6.6Hz), 7.30(1H,t,J = 7.4Hz), 7.40(1H,t,J = 7.4Hz),
 45 7.54(1H,s), 7.61(1H,d,J = 7.4Hz), 7.94(1H,d,J = 9.7Hz), 8.20(1H,d,J = 7.4Hz)

Each Compound 106 and 107 in the following Examples 91 and 92 was prepared using 2,6-dimethylpiperidinocarbonyl-Val-DTrp(COOMe)-OH and each corresponding aminosulfonic acid as starting materials in the same manner described in Example 81.

50 Example 91

Compound 106

IR(KBr, cm^{-1}):

55 3290, 2960, 2935, 1738, 1668, 1610, 1522, 1458, 1383, 1259, 1201, 1092, 768, 748

FAB-MS(m/e , $(\text{C}_{31}\text{H}_{47}\text{N}_5\text{O}_8\text{S} + \text{H})^+$): 650

$^1\text{H-NMR}$ (300MHz, CD_3OD , δ ppm):

0.52(3H,d,J = 6.8Hz), 0.77(3H,d,J = 6.8Hz), 0.80-1.00 (3H,m), 1.17 + 1.19(3H $\times$ 2,d $\times$ 2,J = 6.6Hz), 1.10-1.85

(13H,m),2.96(1H,dd,J = 11.2Hz,14.6Hz),3.30-4.50 (5H,m),4.01(3H,s),4.75(1H,dd,J = 3.9Hz,11.2Hz), 7.22-7.34-(2H,m),7.54(1H,s),7.67(1H,d,J = 7.2Hz), 8.10(1H,d,J = 8.1Hz)

Example 92

5

Compound 107

m.p.: 128-131 °C

IR(KBr,cm⁻¹):

10 3273,2960,1778,1736,1662,1541,1456,1382,1259,1201, 1039,768,748

High Resolution FAB-MS(m/e,(C₃₂H₄₉N₅O₈S + H)⁺):

Calcd: 664.3380

Found: 664.3437

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

15 0.51(3H,d,J = 6.4Hz),0.65(3H,d,J = 6.4Hz),0.83(3H,t, J = 7.0Hz),1.07(6H,d,J = 6.5Hz),1.01-2.00(13H,m)-
2.40-2.55(2H,m),2.83(1H,dd,J = 11.0Hz,14.5Hz),2.23(1H,dd, J = 3.0Hz,14.5Hz),3.69-3.83(1H,m),3.95(3H,s)-
3.97-4.23(3H,m),4.35-4.48(1H,m),5.95(1H,brs),7.25(1H,t, J = 7.5Hz),7.32(1H,t,J = 7.5Hz),7.51(1H,s),7.66(1H,d,
J = 7.5Hz),7.81(1H,d,J = 8.3Hz),8.03(1H,d,J = 7.5Hz), 8.34(1H,d,J = 8.1Hz)

20 Each Compound 108 and 109 in the following Examples 93 and 94 was prepared using 2,6-dimethylpiperidinocarbonyl-Leu-DTrp(COOMe)-OH and each corresponding aminosulfonic acid as starting materials in the same manner described in Example 81.

Example 93

25 Compound 108

m.p.: 139-142 °C

IR(KBr,cm⁻¹):

3271,2956,2871,1774,1736,1655,1543,1458,1383,1259, 1092,1039,768

30 High Resolution FAB-MS(m/e,(C₃₃H₅₁N₅O₈S + H)⁺):

Calcd: 678.3537

Found: 678.3585

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

35 0.67(3H,d,J = 5.6Hz),0.71(3H,d,J = 5.6Hz),0.83(3H,t, J = 6.8Hz),1.04(3H,d,J = 6.7Hz),1.05(3H,d,J = 6.7Hz),
1.10-2.00(15H,m),2.45-2.65(2H,m),2.85(1H,dd, J = 10.8Hz,14.5Hz),3.22(1H,dd,J = 3.4Hz,14.5Hz), 3.95(3H,s)-
3.95-4.45(5H,m),6.07(1H,brs), 7.24(1H,t,J = 7.6Hz),7.32(1H,t,J = 7.6Hz),7.47(1H,s), 7.66(1H,d,J = 7.6Hz),7.78-
(1H,d,J = 8.5Hz),8.04(1H,d, J = 7.6Hz),8.21(1H,d,J = 7.8Hz)

Example 94

40

Compound 109

IR(KBr,cm⁻¹):

3271,2956,2872,1738,1668,1610,1539,1456,1385,1259, 1203,1092,1041,768

45 High Resolution FAB-MS(m/e,(C₃₂H₄₉N₅O₈S + Na)⁺):

Calcd: 686.3187

Found: 686.3200

¹H-NMR(300MHz,DMSO-d₆, δ ppm):

50 0.57-0.89(9H,m),0.95-1.10(6H,m),1.10-1.97(15H,m), 2.78-2.93(1H,m),3.09-3.31(1H,m),3.94 + 3.95-
(3H,s×2), 4.00-4.75(5H,m),5.85 + 6.10(1H,brs×2),7.18 + 7.54 (1H,brs×2),7.20-7.36(2H,m),7.46 + 7.55(1H,s×2),
7.64 + 7.76(1H,d×2,J = 7.3Hz),8.03 + 8.05(1H,d×2, J = 7.3Hz),7.95 + 8.33(1H,d×2,J = 9.5Hz,8.8Hz)

55

Example 95

Synthesis of Compound 110

5 (1) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(2-Br)-DNle-O^tBu

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp-DNle-O^tBu (320 mg) in anhydrous acetic acid (1.0 ml) was added dropwise over 10 min a solution of N-bromosuccinimide (107 mg) in anhydrous acetic acid (3.0 ml). After being stirred at room temperature for 20 h, the mixture was diluted with
 10 dichloromethane (50 ml) and the solution was washed with water (25 ml $\times$ 2), sat. aq. NaHCO₃ (25 ml) and sat. brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 30/1 for development to give the product (15 mg).

FAB-MS (m/e, (C₃₆H₅₆BrN₅O₅ + H)⁺) : 718 and 720

15

(2) Preparation of Compound 110

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(2-Br)-DNle-O^tBu (14 mg, prepared in (1)) in TFA (1.0 ml) was stirred under ice cooling for 3 h. The mixture was concentrated under reduced pressure
 20 and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 30/1/1 for development to give Compound 110 (8.4 mg).

m.p.: 145-148 °C

IR(KBr, cm⁻¹):

2954, 2868, 1724, 1659, 1620, 1514, 1247, 1136, 743

25 High Resolution FAB-MS(m/e, (C₃₂H₄₈BrN₅O₅ + H)⁺):

Calcd: 662.2917

Found: 662.2944

¹H-NMR(300MHz, CDCl₃, δ ppm):

0.78(3H,t,J = 7.1Hz), 0.91(9H,s), 1.13(3H,d,J = 7.1Hz), 1.16(3H,d,J = 7.1Hz), 0.98-1.86(14H,m), 3.19(1H,dd,
 30 J = 6.2Hz, 14.6Hz), 3.39(1H,dd, 5.7Hz, 14.6Hz), 3.93-4.20 (3H,m), 4.20-4.35(1H,m), 4.50-4.66(1H,m), 4.90-5.07
 (1H,m), 6.54-6.66(1H,m), 7.09(1H,t, J = 7.6Hz), 7.15 (1H,t, J = 7.6Hz), 7.25(1H,d, J = 7.6Hz), 7.62(1H,d, J = 7.6Hz)-
 , 7.30-7.44(1H,m), 8.74(1H,brs)

Example 96

35

Synthesis of Compound 111

(1) Preparation of α -N,1-bis-t-butoxycarbonyl- α -N-trifluoroacetyl-2-chloro-D-tryptophan methyl ester

40 To a solution of α -N-trifluoroacetyl-2-chloro-D-tryptophan methyl ester (283 mg) in acetonitrile (5 ml) were added di-tert-butyl dicarbonate and DMAP (20 mg) and the mixture was stirred at room temperature for 23 h. The mixture was evaporated in vacuo and the residue was purified by MPLC (Merck, LicChrorep Si 60) with hexane/ethyl acetate = 5/1 for elution to give the product (265 mg).

FAB-MS (m/e, (C₂₄H₂₈F₃ClN₂O₇)⁺) : 548 and 550

45

(2) Preparation of α -N,1-bis-t-butoxycarbonyl-2-chloro-D-tryptophan

To a solution of α -N,1-bis-t-butoxycarbonyl- α -N-trifluoroacetyl-2-chloro-D-tryptophan methyl ester (255 mg, prepared in (1)) in methanol (5 ml) was added 1N NaOH (1.03 ml) and the mixture was stirred at room
 50 temperature for 7 h. The mixture was diluted with water (50 ml), acidified with 10% citric acid aq. solution and the mixture was extracted with ethyl acetate (30 ml $\times$ 3). Combined organic extracts were washed with sat. brine (30 ml), dried over MgSO₄ and evaporated in vacuo to give the product (206 mg).

FAB-MS (m/e, (C₂₁H₂₇ClN₂O₆)⁺) : 438 and 440

55 (3) Preparation of Boc-DTrp(1-Boc,2-Cl)-DNle-O^tBu

To a solution of α -N,1-bis-t-butoxycarbonyl-2-chloro-D-tryptophan (206 mg, prepared in (2)) and H-DNle-O^tBu•HCl (125 mg) in dichloromethane (10 ml) were added NMM (62 μ l), HOBT•H₂O (86 mg) and

EDCI•HCl (107 mg) under ice cooling. The mixture was stirred at the same temperature for 1 h and at room temperature for 3 h. The mixture was diluted with dichloromethane (20 ml), washed with sat. aq. NaHCO₃ - (10 ml), 10% citric acid aq. solution (10 ml) and sat. brine (10 ml) successively, dried over MgSO₄ and evaporated in vacuo.

5 The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 5/1 for elution to give the product (235 mg).

FAB-MS (m/e, (C₃₁H₄₆ClN₃O₇ + H)⁺) : 608 and 610

(4) Preparation of H-DTrp(1-Boc,2-Cl)-DNle-O^tBu

10

A solution of Boc-DTrp(1-Boc,2-Cl)-DNle-O^tBu (252 mg, prepared in (3)) in formic acid (10 ml) was stirred at room temperature for 1 h. The mixture was concentrated under reduced pressure and the residue was dissolved in ethyl acetate (50 ml). The solution was washed with sat. aq. NaHCO₃ (50 ml × 2) and sat. brine (50 ml) successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 30/1 for development to give the product (98 mg).

15

FAB-MS (m/e, (C₂₆H₃₈ClN₃O₅ + H)⁺) : 508 and 510

(5) Preparation of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(1-Boc,2-Cl)-DNle-O^tBu

20

To a solution of H-DTrp(1-Boc,2-Cl)-DNle-O^tBu (39 mg, prepared in (4)) and 2,6-dimethylpiperidinocarbonyl-γMeLeu-OH (34 mg) in dichloromethane (3 ml) were added HOBT•H₂O (18 mg) and EDCI•HCl (23 mg) under ice cooling.

25

The mixture was stirred at the same temperature for 1 h and at room temperature for 3 h. The mixture was diluted with dichloromethane (30 ml), washed with sat. aq. NaHCO₃ (20 ml), 10% citric acid aq. solution (20 ml) and sat. brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 2/1 for development to give the product (51 mg).

FAB-MS (m/e, (C₄₁H₆₄ClN₅O₇ + H)⁺) : 774 and 776

30

(6) Preparation of Compound 111

35

A mixture of 2,6-dimethylpiperidinocarbonyl-γMeLeu-DTrp(1-Boc,2-Cl)-DNle-O^tBu (12.3 mg, prepared in (5)) in TFA (1.0 ml) was stirred at room temperature for 1 h and concentrated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 30/1/1 for development to give Compound 111 (7.4 mg).

m.p.: 145-149 °C

IR(KBr,cm⁻¹):

3290,2956,2360,1646,1618,1533,1247,1128,743

40

High Resolution FAB-MS(m/e,(C₃₂H₄₈ClN₅O₅ + H)⁺):

Calcd: 618.3422

Found: 618.3431

¹H-NMR(300MHz,CDCl₃, δ ppm):

45

0.78(3H,t,J = 7.2Hz),0.91(9H,s),1.13(3H,d,J = 6.9Hz), 1.16(3H,d,J = 6.9Hz),0.96-1.88(14H,m),3.20(1H,dd, J = 5.7Hz,15.0Hz),3.39(1H,dd,5.7Hz,15.0Hz),3.90-4.08 (2H,m),4.08-4.20(1H,m),4.20-4.33(1H,m),4.48-4.64 (1H,m),4.90-5.05(1H,m),6.49-6.63(1H,m),7.10(1H,t, J = 8.0Hz),7.17(1H,t,J = 8.0Hz),7.25(1H,d,J = 8.0Hz), 7.59-(1H,d,J = 8.0Hz),7.33-7.48(1H,m),8.61(1H,brs)

Each Compound 112 and 113 in the following Examples 97 and 98 was prepared using 2,6-dimethylpiperidinocarbonyl-Leu-OH or hexahydroazepinylcarbonyl-Leu-OH respectively instead of 2,6-dimethylpiperidinocarbonyl-γMeLeu-OH in Example 96-(5) in the same manner described in Example 96-(5) and (6).

50

Example 97

55

Compound 112

m.p.: 142-147 °C

IR(KBr,cm⁻¹):

3402,2956,2870,2360,1653,1622,1522,1387,1340,1238, 1128,743

High Resolution FAB-MS(m/e, (C₃₁H₄₆ClN₅O₅ + H)⁺):

Calcd: 604.3266

Found: 604.3260

5 ¹H-NMR(300MHz,CDCl₃, δ ppm):

0.79(3H,t,J = 7.2Hz),0.86(3H,d,J = 4.9Hz),0.88(3H,d, J = 4.9Hz),1.15(3H,d,J = 6.9Hz),1.16(3H,d,J = 6.9Hz),
0.93-1.87(15H,m),3.21(1H,dd,J = 6.3Hz,14.4Hz), 3.38(1H,dd,6.3Hz,14.4Hz),3.91-4.07(2H,m),4.07-4.20 (1H,m)-
,4.20-4.34(1H,m),4.52-4.73(1H,m),4.82-5.02 (1H,m),6.52-6.70(1H,m),7.11(1H,t,J = 7.1Hz),7.17 (1H,t,J = 7.1Hz)-
,7.26(1H,d,J = 7.1Hz),7.59(1H,d, J = 7.1Hz),7.31-7.49(1H,m),8.63(1H,brs)

10

Example 98

Compound 113

15 m.p.: 127-130 °C

IR(KBr,cm⁻¹):

3300,2929,2864,1647,1529,1452,1342,1217,743

High Resolution FAB-MS(m/e, (C₃₀H₄₄ClN₅O₅ + H)⁺):

Calcd: 590.3109

20 Found: 590.3067

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.78(3H,t,J = 7.2Hz),0.86(3H,d,J = 4.4Hz),0.88(3H,d,J = 4.4Hz),0.92-1.87(17H,m),3.05-3.47(6H,m),3.92-
4.06 (1H,m),4.23-4.38(1H,m),4.51-4.70(1H,m),4.92-5.08 (1H,m),6.50-6.70(1H,m),7.11(1H,t,J = 7.6Hz),7.17
(1H,t,J = 7.6Hz),7.26(1H,d,J = 7.6Hz),7.59(1H,d,J = 7.6Hz),7.37-7.52(1H,m),8.60(1H,brs)

25

Example 99

Synthesis of Compound 114

30 (1) Preparation of α-N-trifluoroacetyl-2-chloro-1-methyl-D-tryptophan

To a solution of α-N-trifluoroacetyl-2-chloro-1-methyl-D-tryptophan methyl ester (525 mg) in methanol
(10 ml) was added 1N NaOH (1.70 ml) and the mixture was stirred at room temperature for 6 h. The mixture
was diluted with water (50 ml), acidified with 1N HCl and the mixture was extracted with diethyl ether (30 ml
× 3). Combined organic extracts were washed with sat. brine (30 ml), dried over MgSO₄ and evaporated in
35 vacuo. The residue was recrystallized from ethyl acetate-hexane to give the product (344 mg).

FAB-MS (m/e, (C₁₄H₁₂ClF₃N₂O₃ + H)⁺) : 449 and 451

(2) Preparation of α-N-trifluoroacetyl-DTrp(1-Me,2-Cl)DNle-O^tBu

40

To a solution of α-N-trifluoroacetyl-2-chloro-1-methyl-D-tryptophan (340 mg, prepared in (1)) and H-
DNle-O^tBu•HCl (230 mg) in dichloromethane (20 ml) were added NMM (0.12 ml), HOBT•H₂O (180 mg)
and EDCI•HCl (225 mg) under ice cooling. The mixture was stirred at the same temperature for 1 h and at
room temperature over night. The mixture was diluted with ethyl acetate (100 ml), washed with sat. aq.
45 NaHCO₃ (100 ml), 1N HCl (100 ml) and sat. brine (100 ml) successively, dried over MgSO₄ and evaporated
in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl
acetate = 4/1 for elution to give the product (571 mg).

(3) Preparation of H-DTrp(1-Me,2-Cl)-DNle-O^tBu

50

To a solution of α-N-trifluoroacetyl-DTrp(1-Me, 2-Cl)-DNle-O^tBu (84 mg, prepared in (2)) in 1,4-dioxane
(10 ml) was added 1M Na₂CO₃ aq. solution (3.0 ml) and the mixture was stirred at 50 °C over night. The
mixture was diluted with water (30 ml) and extracted with ethyl acetate (10 ml × 3). Combined organic
extracts were washed with sat. brine, dried over MgSO₄ and evaporated in vacuo. The residue was purified
55 by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 3/7 for development to give the
product (47 mg).

FAB-MS (m/e, (C₂₂H₃₂ClN₃O₃ + H)⁺) : 422 and 424

(4) Preparation of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(1-Me,2-Cl)-DNle-O^tBu

To a solution of H-DTrp(1-Me,2-Cl)-DNle-O^tBu (47 mg, prepared in (3)) and 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-OH (38 mg) in DMF (2.0 ml) were added HOBT•H₂O (22 mg) and EDCl•HCl (27 mg) under ice cooling. The mixture was stirred at the same temperature for 30 min and at room temperature for 1.5 h. The mixture was concentrated in vacuo and the residue was dissolved in ethyl acetate. The solution was washed with sat. aq. NaHCO₃, 1N HCl and sat. brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with hexane/ethyl acetate = 1/1 for development to give the product (73 mg).

FAB-MS (m/e, (C₃₇H₅₈ClN₅O₅ + H)⁺) : 688 and 690

(5) Preparation of Compound 114

A solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp(1-Me,2-Cl)-DNle-O^tBu (28 mg, prepared in (4)) in TFA (3 ml) was stirred at room temperature for 1 h and concentrated in vacuo. The residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 9/1 for development to give Compound 114 (19 mg).

m.p.: 103-108 °C

IR(KBr,cm⁻¹):

3300,2954,2937,1652,1647,1541,1533,1522,741

High Resolution FAB-MS(m/e,(C₃₃H₅₀ClN₅O₅ + H)⁺):

Calcd: 632.3578

Found: 632.3564

¹H-NMR(300MHz,CDCl₃, δ ppm):

0.78(3H,t,J = 7.4Hz),0.86(9H,s),1.14(3H,d,J = 5.8Hz), 1.15(3H,d,J = 5.8Hz),0.91-1.92(14H,m),3.34-
(1H,dd,J = 5.7Hz,14.6Hz),3.44(1H,dd,6.2Hz,14.6Hz),3.73(3H,s), 3.84-4.19(3H,m),4.30-
(1H,ddd,J = 5.7Hz,6.2Hz,7.6Hz), 4.66-4.74(1H,m),4.90(1H,d,J = 5.5Hz),6.41(1H,d,J = 7.6 Hz),7.11-
(1H,t,J = 7.8Hz),7.2(1H,t,J = 7.8Hz),7.25(1H, d,J = 7.8Hz),7.58(1H,d,J = 7.8Hz),7.62(1H,d,J = 6.9Hz)

Example 100

Synthesis of Compound 115

To a solution of 2,6-dimethylpiperidinocarbonyl- γ -MeLeu-DTrp-DNle-O^tBu (199 mg) in acetic acid (2 ml) was added methanesulfonyl chloride and the mixture was stirred at room temperature for 2 days. The mixture was concentrated under reduced pressure and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 10/1 for development to give Compound 115 (65 mg).

m.p.: 158-161 °C

IR(KBr,cm⁻¹):

3305,2954,2869,1652,1618,1521,1448,1386,1247,744

High Resolution FAB-MS(m/e,(C₃₃H₅₁N₅O₅S + H)⁺):

Calcd: 630.3689

Found: 630.3663

¹H-NMR(300MHz,acetone-d₆, δ ppm):

0.80-1.90(23H,m),0.83(9H,s),2.47(3H,s),3.27(1H,dd, J = 8.9Hz,14.2Hz),3.51(1H,dd,4.4Hz,14.2Hz),4.00-4.15
(1H,m),4.15-4.35(3H,m),4.69-4.77(1H,m),5.65(1H,d, J = 8.4Hz),7.02(1H,t,J = 7.6Hz),7.10(1H,t,J = 7.6Hz), 7.31-
(1H,d,J = 7.6Hz),7.40-7.43(1H,m),7.69(1H,d,J = 7.6 Hz),7.75(1H,d,J = 6.6Hz),10.37(1H,s)

Example 101

Synthesis of Compound 116

To a solution of Compound 115 (50 mg, prepared in Example 100) in acetic acid (2 ml) was added 30% hydrogen peroxide (13.6 μ l) and the mixture was stirred at room temperature for 2 h. The mixture was concentrated under reduced pressure and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol/acetic acid = 10/1/1 for development to give Compound 116 (18.9 mg).

m.p.: 161-168 °C

IR(KBr,cm⁻¹):

3390,2956,1652,1603,1538,1405,1128,746

High Resolution FAB-MS(m/e, (C₃₃H₅₂N₅O₆S + H)⁺):

Calcd: 646.3638

Found: 646.3663

5 ¹H-NMR(300MHz,acetone-d₆-CD₃OD,dppm):

0.80-1.90(23H,m),0.82 + 0.87(9H,s×2),3.06(3H,s), 3.15-3.65(2H,m),4.05-4.40(4H,m),4.62-4.78(1H,m), 7.12-(1H,t,J = 7.5Hz),7.26(1H,t,J = 7.5Hz),7.46(1H,d, J = 7.5Hz),7.81-7.89(1H,m)

Example 102

10

Synthesis of Compound 117

To a solution of Compound 116 (23 mg, prepared in Example 101) in acetic acid (1 ml) was added 30% hydrogen peroxide (41 μl) and the mixture was stirred at room temperature for 4 h. The mixture was concentrated under reduced pressure and the residue was purified by preparative TLC (Merck, Kieselgel 60 F₂₅₄) with chloroform/methanol = 10/1 for development to give Compound 117 (4 mg).

FAB-MS (m/e, (C₃₃H₅₁N₅O₇S-H)⁻) : 660

¹H-NMR(300MHz,acetone-d₆, δ ppm):

0.80(9H,s)0.84-1.86(23H,m),3.34(3H,s)3.50-3.65 (1H,m),3.78(1H,dd,J = 4.3Hz,14.3Hz),3.90-4.04(1H,m),
20 4.15-4.32(3H,m),4.65-4.75(1H,m),5.64(1H,d,J = 6.5Hz),7.19(1H,t,J = 7.4Hz),7.36(1H,t,J = 7.4Hz), 7.55-(1H,d,J = 7.4Hz),7.71(1H,brs),7.87(1H,d,J = 7.4Hz),7.97(1H,d,J = 8.0Hz),10.95(1H,brs)

Referential Example 1

25 Synthesis of (R)-2-Aminohexanesulfonic acid

(1) Preparation of (R)-2-t-butoxycarbonylamino-1-hexanol

To a solution of Boc-DNle-OMe (4.23 g) in dry THF (30 ml) were added lithium chloride (0.85 g), sodium borohydride (0.76 g) and dry ethanol (60 ml), and the mixture was stirred at the same temperature for 2 h and at room temperature over night. The mixture was cooled with ice-water, adjusted to pH 3 by the gradual addition of 10% aq. citric acid, and concentrated in vacuo. Water (200 ml) was added to the residue, and the mixture was extracted with dichloromethane (50 ml × 3).

The combined organic layers were washed with brine, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 1/1 for elution to give the product (2.95 mg).

(2) Preparation of (R)-2-t-butoxycarbonylamino-1-hexyl methanesulfonate

To a solution of (R)-2-t-butoxycarbonylamino-1-hexanol (2.17 g, prepared in (1)) in dichloromethane (20 ml) were added triethylamine (1.67 ml) and methanesulfonyl chloride (0.93 ml) under ice cooling. After being stirred at the same temperature for 1 h, the mixture was washed with water, 10% aq. citric acid, sat. aq. NaHCO₃ and brine successively, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 2/1 for elution to give the product (1.32 g).

(3) Preparation of (R)-2-aminohexanesulfonic acid

A mixture of (R)-2-t-butoxycarbonylamino-1-hexyl methanesulfonate (428 mg, prepared in (1)) in 4N-hydrogen chloride/1,4-dioxane (10 ml) was stirred at room temperature for 1 h, and evaporated in vacuo. The residue was dissolved in water (3.0 ml) and sodium sulfite (378 mg) was added. After being stirred at room temperature for 4 d, the reaction mixture was passed through a Dowex 50W-X8 column (H⁺ form) and washed with excess water. The combined water solution was concentrated in vacuo to give the product (114 mg).

55 FAB-MS(m/e, (C₆H₁₅NO₃S + H)⁺):182

Referential Example 2

Synthesis of (R)-3-Aminoheptanoic acid

(1) Preparation of (R)-2-t-butoxycarbonylaminoheptanenitrile

A mixture of (R)-2-t-butoxycarbonylamino-1-hexyl methanesulfonate (561 mg, prepared in Referential Example 1-(2)) and sodium cyanide (117 mg) in DMF (3 ml) was stirred at 60 °C for 5 h. After cooling, water (50 ml) was added to the mixture. The mixture was extracted with dichloromethane (25 ml × 3). Combined organic layers were washed with brine, dried over MgSO₄ and evaporated in vacuo. The residue was purified by silica gel chromatography (Merck, Kieselgel 60) with hexane/ethyl acetate = 3/1 for elution to give the product (327 mg).

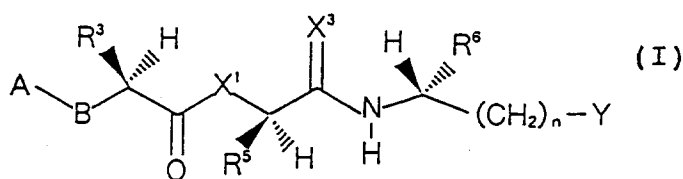
(2) Preparation of (R)-3-aminoheptanoic acid

A mixture of (R)-2-t-butoxycarbonylaminoheptanenitrile (165 mg, prepared in (1)) in 6N-hydrochloric acid (10 ml) was refluxed for 3 h. After cooling, the mixture was evaporated in vacuo. The residue was dissolved in methanol (2 ml) and propylene oxide (1 ml) was added to the mixture. After being stirred at room temperature over night, the mixture was evaporated in vacuo. The residue was crystallized from methanol and diethyl ether to give the product (52 mg).

FAB-MS(m/e, (C₇H₁₅NO₂ + H)⁺):146

Claims

1. A peptide derivative of the formula:



wherein A is a group of the formula R¹¹O-CO- (wherein R¹¹ is a lower alkyl group or a phenyl group), or a group of the formula R¹²R¹³N-C(=O)- (wherein R¹² is a lower alkyl group, a cycloalkyl group, a 1-adamantyl group, a phenyl group wherein one or two optional hydrogen atoms on the benzene ring may independently be replaced by a halogen atom, a trifluoromethyl group, a nitro group, an amino group or a formylamino group, a pyridyl group, or a thienyl group, R¹³ is a hydrogen atom, a lower alkyl group or a cycloalkyl group, or R¹² and R¹³ form, together with the adjacent nitrogen atom, a 5- to 9-membered nitrogen-containing saturated heterocyclic ring having 4 to 8 carbon atoms, wherein among methylene groups forming the ring, one optional methylene group not adjacent to the above nitrogen atom may be replaced by a thio group, and one to four optional hydrogen atoms on the carbon atoms of the heterocyclic ring may independently be replaced by a lower alkyl group, and further two adjacent carbon atoms in the heterocyclic ring may form a benzo-fused ring); B is an oxygen atom or a group of the formula -NR²- (wherein R² is a hydrogen atom or a lower alkyl group); R³ is a lower alkyl group, a cycloalkyl group, an aryl group, a heterocyclic group, a cycloalkyl lower alkyl group, an aryl lower alkyl group or a heterocyclic lower alkyl group; X¹ is an oxygen atom or a group of the formula -NR⁴- (wherein R⁴ is a hydrogen atom or a lower alkyl group); R⁵ is a 3-indolylmethyl, 3-benzothienylmethyl, 1-naphthylmethyl or benzyl group wherein one or two optional hydrogen atoms on the ring may be replaced by a hydroxyl group, a halogen atom, a formyl group, a lower alkyl group, a lower alkoxy group, a lower alkylthio group, a lower alkylsulfinyl group, a lower alkylsulfonyl group, a lower alkoxy carbonyl group, a nitro group or a group of the formula R⁵¹-CO-X² {wherein R⁵¹ is a lower alkyl group, a lower alkoxy group, or an amino group which may be substituted by a lower alkyl group, and X² is an oxygen atom or a group of the formula -NR⁵²- (wherein R⁵² is a hydrogen atom or a lower alkyl group)}; X³ is an oxygen atom or a sulfur atom; R⁶ is a hydrogen atom, or a lower alkyl or lower alkenyl group which may have a substituent selected from the group consisting of a hydroxyl group, a lower alkoxy group, a lower alkylthio group and a heterocyclic group; n is 0 or 1; Y is a hydroxymethyl group,

a group of the formula CO_2R^{71} (wherein R^{71} is a hydrogen atom or a lower alkyl group), a group of the formula CONHR^{72} (wherein R^{72} is a hydrogen atom or a lower alkyl group which may have a substituent selected from the group consisting of a hydroxyl group, a carboxyl group and a sulfo group), a 1H-tetrazol-5-yl group, a sulfo group and a phosphono group; or a pharmaceutically acceptable salt thereof.

2. The peptide derivative or the pharmaceutically acceptable salt thereof according to Claim 1, wherein R^5 is a 3-indolylmethyl group wherein the indole ring is substituted at the 1-position and/or 2-position by one or two substituents selected from the group consisting of a lower alkoxy group, a lower alkylthio group, a lower alkanoyloxy group, a lower alkoxy carbonyl group, a halogen atom or a lower alkyl group; and R^6 is a lower alkyl group or a lower alkenyl group which may have a substituent selected from the group consisting of a lower alkoxy group and a lower alkylthio group.

3. An endothelin antagonistic agent comprising a compound as defined in Claim 1 or a pharmaceutically acceptable salt thereof.

4. A drug for treating hypertension, pulmonary hypertension, Raynaud's disease, acute renal failure, myocardial infarction, angina pectoris, cerebral infarction, cerebral vasospasm, arteriosclerosis, asthma, gastric ulcer, diabetes, endotoxin shock, endotoxin-induced multiple organ failure or disseminated intravascular coagulation, and/or cyclosporin-induced renal failure or hypertension, which contains a peptide derivative of the formula (1) as defined in Claim 1 or a pharmaceutically acceptable salt thereof.

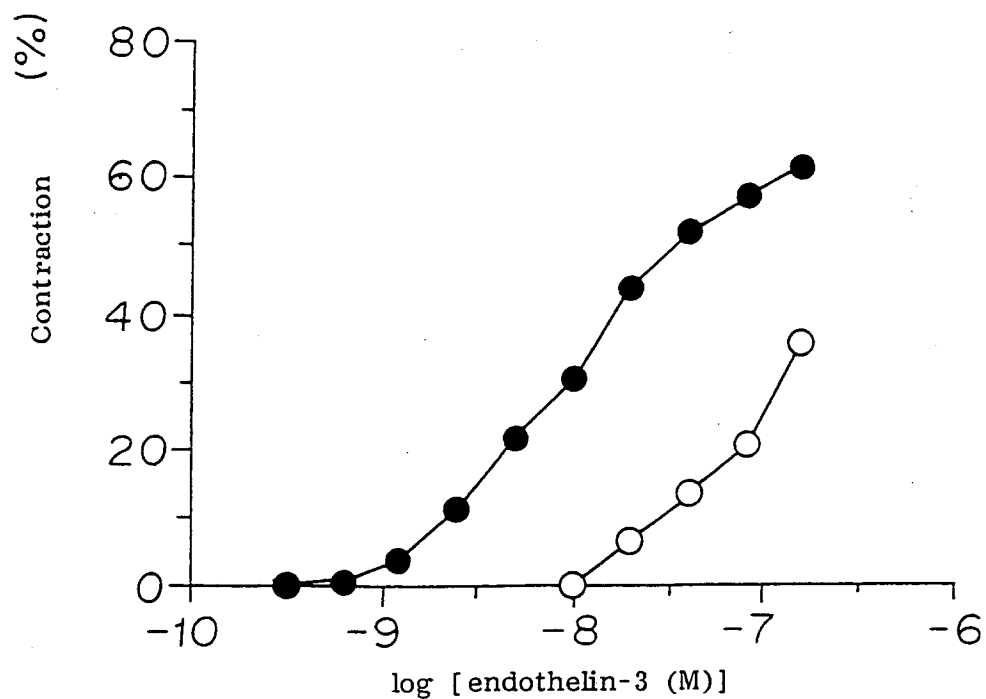
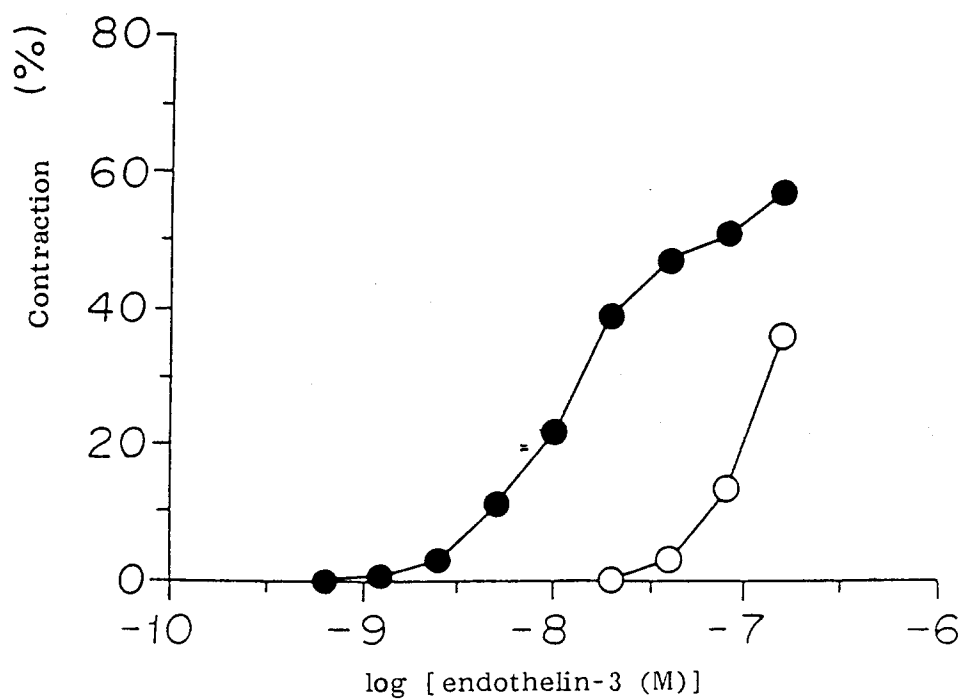
FIGURE 1**FIGURE 2**

FIGURE 3

